# AccDSL

## Company Histories, Accounting Interpretations, and Scenario Simulation

Product and method proposal

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This document specifies a system that translates company disclosures into traceable evidence, represents several compatible economic and accounting histories, and simulates their future journals and financial statements under explicit economic scenarios.

The proposed data contracts, methods and acceptance cases specify the intended implementation. Current source and domain limitations remain explicit; this candidate is not a completed product or a claim of full proposal readiness. Proposed libraries can be replaced while preserving those contracts. Sector rule packages require their own accounting review before they can be described as compliant with a reporting standard.

Research materials, source qualifications and review records accompany this document in `docs/design/research` and `docs/design/review`.

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# Chapter 1

# The product decision

A company's accounts describe the financial consequences of its activities, but they leave much of the activity itself unobserved. Revenue combines many sales, a borrowing balance combines contracts with different terms, and a provision combines estimates about uncertain future payments. Aggregation is necessary for reporting. It also makes the inverse task—recovering economic history from published reports—fundamentally different from entering known transactions into a ledger.

AccDSL will preserve this distinction. Its first product is an evidence-backed representation of a company's historical record. Its second is a set of compatible histories and current economic states. Its third is a collection of conditional simulations, each explaining how a particular history, accounting treatment and business response produces future journals and reports. These products share identifiers and provenance, so an analyst can move from a forecast balance to the assumption or disclosure responsible for it.

The recommended design is a small, typed accounting language over a shared intermediate representation, combined with an evidence database, contract calculators and replaceable inference engines. A language earns its place by making recurring accounting decisions explicit and checkable: which entity and period an amount belongs to, whether it is known or assumed, which rule applies, and how an event changes a balanced ledger. The language does not supply missing facts or determine a company's future behaviour.

## 1.1 Why journals are necessary but insufficient

Consider a company with a borrowing of 100 currency units and annual interest expense of 5. Assume that the entire principal was outstanding throughout the year, interest was paid at year-end, and there were no fees, defaults, hedges or foreign-exchange effects. Both a fixed-rate loan at 5% and a floating-rate loan whose rate happened to be 5% generate the same entries:

| Account | Debit | Credit |
|---|---|---|
| Cash, on initial borrowing | 100 | |
| Borrowing liability | | 100 |
| Interest expense, at year-end | 5 | |
| Cash | | 5 |

Even the full historical journal would not distinguish these two contracts. Now impose a scenario in which the floating rate resets immediately to 7% for the whole next year. The fixed

loan produces interest of 5; the floating loan produces interest of 7. If the disclosure is also compatible with any fixed-rate fraction $\alpha \in [0, 1]$, future interest is

$$I(\alpha) = 100[0.05\alpha + 0.07(1 - \alpha)] = 7 - 2\alpha. \tag{1.1}$$

The historical accounts are compatible with the whole interval $[5, 7]$ of future interest. This is a range of feasible amounts. It becomes a probability distribution only when a distribution for $\alpha$ is supplied. Neither the number of generated interpretations nor a language model's confidence provides that distribution.

Journals remain essential because simulated economic events must have coherent accounting consequences. However, the simulated state must also contain contract terms, maturities, customer or borrower cohorts, collateral, unrecognised commitments, and management decisions. These determine what the journals will be under a changed economy. Reconstructing every historical transaction is unnecessary when a smaller state preserves the responses needed for the intended scenarios. Establishing that smaller state is an explicit modelling task, examined in Chapter 6.

The analogy with competing legal interpretations is useful at the level of representation. Several readings may survive the evidence and require separate reasoning. In accounting, some alternatives concern the meaning of a disclosure; others concern undisclosed facts or future behaviour. The system must identify which kind of uncertainty is involved. A known contractual term must not become an arbitrary interpretive choice merely because another choice is convenient for a simulation.

## 1.2 Users, questions and deliverables

An accounting analyst needs to inspect a number in its original table, verify its entity, currency and period, and see how it reconciles with the notes. A financial modeller needs to distinguish a reported balance from assumptions about its composition. A language engineer needs deterministic semantics and tests. A researcher needs to compare explanations under the same scenario, including explanations that fit history equally well.

The first release must answer the following complete workflow. An analyst loads a set of reports and specifies an information cutoff. AccDSL preserves the original documents and extracts source-linked observations. It identifies reconciliation failures and constructs an explicitly bounded family of compatible histories. The analyst declares additional assumptions where simulation requires them. The system executes each requested scenario and returns statements, journal traces, explanation-specific distributions or bounds, and unresolved limitations. When a later filing arrives, the analyst can compare the revised interpretation with the earlier information set.

| Requirement | Required behaviour |
|---|---|
| **R1 Evidence** | Every observed amount or narrative assertion retains its document version, location, extraction method and original representation. |
| **R2 History** | Queries distinguish economic time, reporting period, publication time and database ingestion time; restatements remain recoverable. |
| **R3 Meaning** | Entity, perimeter, dimensions, currency, measurement basis, standard version and accounting concept accompany values. |

| Requirement | Required behaviour |
|---|---|
| **R4 Ambiguity** | Discrete alternatives and continuous unknowns retain their joint constraints across accounts and periods. Missing values never mean zero. |
| **R5 Accounting** | All committed journals balance in the book's functional currency, and balances reconcile to explicitly classified movements. |
| **R6 Economics** | Contracts, exposures and behavioural assumptions drive events. Accounting identities alone never determine scenario responses. |
| **R7 Results** | Conditional results survive by interpretation and scenario. Weighted results require declared, normalised probability measures. |
| **R8 Explanation** | A report cell can be traced to journals, events, rules, assumptions and evidence, including consolidation adjustments. |
| **R9 Extensibility** | Standard, industry and company packages extend shared types through versioned interfaces, with explicit unsupported cases. |
| **R10 Reproduction** | A saved run records source versions, model versions, parameters, calendars, seeds and numerical conventions. |

These are acceptance requirements. Python, SQL storage, a particular parser, and a particular constraint solver are implementation choices. Initial support will cover a non-financial company with cash, receivables, payables, inventory, fixed assets, ordinary debt and ordinary equity. Banks and insurers must be representable in the core data model from the outset, but their executable measurement and behavioural packages follow dedicated milestones. Until a package supports an instrument, the result is explicitly incomplete.

## 1.3 What constitutes a useful first implementation

The first demonstrator should reconstruct three reporting dates and the intervening annual and interim flows for one company, including at least one restatement or revised comparative. It must preserve a borrowing-composition ambiguity and execute the fixed/floating example through both journals and all affected statement lines. A synthetic company supplies known hidden history; a public company supplies realistic disclosure and extraction problems. Their roles differ: the synthetic case tests preservation and rejection of known alternatives, while the public case tests traceability and practical coverage.

Success is a reproducible chain of evidence and computation, with identified gaps. A convincing-looking complete ledger reconstructed from incomplete disclosures would fail this product contract if its assumptions were hidden. Forecast accuracy requires a later evaluation with historical information cutoffs and suitable baselines; it is not established by the first demonstrator.

# Chapter 2

# A complete company and a real disclosure

## 2.1 What a balance must remember

Northstar Components is a synthetic seller of one homogeneous good. Its disclosures, transactions and contract terms are specified together, so we can check whether reconstruction preserves a known possible history. All amounts in this example are currency units, not thousands. The company has no foreign currency, subsidiaries, leases, returns or other comprehensive income. These are declared features of the example, not assumptions to impose on an imported company. Its source package and expected results are supplied in `fixtures/northstar.json`.

At the beginning of 2024, Northstar holds cash of 60, receivables of 20, inventory of 30 and plant costing 100 with accumulated depreciation of 20. It owes suppliers 20 and lenders 100. Share capital is 50 and retained earnings are 20. Thus assets of 190 equal liabilities of 120 plus equity of 70. Inventory consists of six units costing 5 each. The existing plant depreciates by 10 a year, with no residual value. The loan remains outstanding throughout 2024; both possible contract types charge 5% during that year.

The six months ending June and December provide two successive observation periods. Here the chosen accrual convention assigns one half-year to each. Cash-flow dates within a half-year are not recovered from a period aggregate; the synthetic history supplies a feasible ordering in which inventory is purchased before delivery and customer receipts precede the payments they fund. More detailed liquidity questions require more detailed dates.

| Movement | First half | Second half |
|---|---:|---:|
| Delivered sales on credit | 110 | 150 |
| Customer collections | 110 | 135 |
| Inventory purchases on credit | 70 | 85 |
| Supplier payments | 65 | 80 |
| Cost of goods delivered | 75 | 90 |
| Payroll paid | 20 | 22 |
| Plant purchased at period end | 10 | 12 |
| Depreciation | 5 | 6 |
| Interest accrued and paid | 2.5 | 2.5 |
| Tax expense | 1.5 | 5.9 |
| Tax paid | 3.5 | 3.9 |
| Dividends paid | 4 | 5 |

Each row has an economic meaning before it has a posting. Delivery of the good transfers control and creates an unconditional receivable; invoicing is assumed simultaneous in this example. Collection settles that receivable. Purchases increase inventory and trade payables; paying the supplier settles the payable. Delivery separately removes inventory at cost. Plant purchases are capital expenditure, while depreciation consumes its service potential. The addition of 10 enters service after June and has a five-year life, producing one unit of second-half depreciation. The addition of 12 enters service after December and has a twelve-year life.

Tax is deliberately simple: 20% of positive profit before tax, with all these expenses deductible and no deferred tax. The first-half payment exceeds its tax expense by 2, leaving prepaid tax; that prepayment is used in the second half. A genuine corporate tax package would need jurisdiction, tax basis, losses, timing differences and payment schedules. The example's tax rule governs its own numbers only.

## 2.2 Recovering movements without inventing invoices

Suppose the annual report discloses sales of 260, cost of sales of 165 and the opening and closing working-capital balances, while the collection and purchase totals are hidden. Under Northstar's explicit no-write-off, no-acquisition and no-currency-effect scope, the three bridges are

$$35 = 20 + 260 - C, \tag{2.1}$$

$$20 = 30 + P - 165, \tag{2.2}$$

$$30 = 20 + P - S. \tag{2.3}$$

Here $C$ denotes collections, $P$ purchases and $S$ supplier payments, all in currency units over the same year. Substitution gives $C = 245$, $P = 155$ and $S = 145$. These are identifiable aggregate movements within this model. They do not identify which invoice was paid on which day. If acquisitions or write-offs are admitted, their movement variables enter the same bridges and these totals may cease to be unique.

The loan remains different. Write $F$ and $V$ for fixed and floating principal. The source imposes $F + V = 100$, $F, V \geq 0$ and $0.05F + 0.05V = 5$. The last equality adds no information about the split. Both endpoints and every intermediate allocation survive. Northstar therefore has identified working-capital totals and an unidentified financing exposure in the same joint historical family. A solver must not turn its convenient witness $F = 0$ into the company's recovered contract.

The model's historical slots are one delivery, collection, purchase, supplier settlement, payroll, plant addition, depreciation, interest, tax payment and dividend per half-year. Each slot represents a classified aggregate event. The interim filing supplies the information for allocating the annual movements between the two half-years; the annual bridges alone identify only their totals. The template for collection contributes $+C$ to cash and $-C$ to receivables; its state effect reduces the unpaid receivable pool by $C$. The purchase template adds inventory quantity $P/5$, inventory cost $P$ and a payable $P$. A cash-balance identity by itself cannot substitute for either template.

## 2.3 The complete historical statements

The corrected statements follow from those movements. Temporary income and expense accounts close to retained earnings at each half-year boundary; annual income is the sum of the two periods' classified movements, excluding closing entries.

| Balance | Opening | June | December |
|---|---|---|---|
| Cash | 60 | 65 | 74.6 |
| Receivables | 20 | 20 | 35 |
| Inventory | 30 | 25 | 20 |
| Plant, net | 80 | 85 | 91 |
| Prepaid tax | 0 | 2 | 0 |
| Total assets | 190 | 197 | 220.6 |
| Trade payables | 20 | 25 | 30 |
| Debt | 100 | 100 | 100 |
| Share capital | 50 | 50 | 50 |
| Retained earnings | 20 | 22 | 40.6 |
| Liabilities and equity | 190 | 197 | 220.6 |

Annual revenue of 260 less cost of sales 165, payroll 42, depreciation 11 and interest 5 gives profit before tax of 37. Tax expense is 7.4 and profit is 29.6. Other comprehensive income is zero under the declared scope, so total comprehensive income is also 29.6. Equity moves from 70 to $70 + 29.6 - 9 = 90.6$ after dividends.

The cash-flow statement supplies an independent reconciliation. Collections 245 less supplier payments 145, payroll 42, interest 5 and tax payments 7.4 produce operating cash of 45.6. Investing cash is $-22$ and financing cash is $-9$, giving a cash increase of 14.6. The indirect route gives the same operating cash: profit 29.6 plus depreciation 11, minus the receivables increase 15, plus the inventory decrease 10 and the payables increase 10. The prepaid-tax balance is zero at both annual endpoints. The two cash routes use different information and therefore provide a useful check on classification.

## 2.4 A correction changes what was knowable

The original interim filing reported revenue of 120 and receivables of 30. Ten units of revenue had been recorded before delivery. Its tax expense was 3.5, profit 14 and equity 80. The annual filing subsequently corrected the interim revenue and receivable by 10 and recognised the excess tax payment as prepaid tax of 2. Corrected interim profit is 6 and equity is 72, as above. The original statement remains an accurate record of what was published, even though its revenue recognition was wrong under the stated delivery rule.

For the temporal fixture, the original report becomes available on 1 August 2024 and is ingested on 3 August. The correction and annual report become available on 2 March 2025 and are ingested on 5 March. An investor-information query at 1 September 2024 00:00 UTC returns 120. For this synthetic example, take UTC as the reporting timezone and use an instant cutoff of 3 March 2025 at 00:00 UTC, after the correction's entire publication-date interval has ended. That query returns 110 when evaluated using the subsequently acquired archive. A query that additionally asks what the database contained on 4 March still returns 120. Publication and ingestion answer different questions.

The annual total 260 and corrected interim total 110 share revision group `v2`, so second-half revenue is $260 - 110 = 150$. Subtracting the original 120 would yield 140, but that result mixes incompatible revisions. The query must reject the subtraction unless a reviewed compatibility relationship supports it. A later date alone does not establish such a relationship.

## 2.5 One economic scenario, several complete futures

For 2025, assume demand for 30 units at price 10, unit purchase cost 5 and a target closing inventory of eight units. Opening inventory is four units, so the purchase decision is $30+8-4=34$ units, costing 170. Deliveries cost 150, leaving inventory of 40. Collect 90% of current sales and 10 of old receivables: $270+10=280$, leaving receivables of 55. Pay old suppliers 30 and 130 of the new purchases, leaving payables of 40. Payroll is 50, capital expenditure of 20 occurs at year-end and dividends are 8.

These decisions are inputs to a conditional forecast. They are not estimates of demand or optimal management responses. No new borrowing is available. Under the existing asset schedules, depreciation is $10+10/5+12/12=13$. The new year-end asset begins depreciation in the next period. Let $\alpha = F/100$. With the floating rate resetting to 7%, interest is $I = 7-2\alpha$. Profit before tax is $300-150-50-13-I = 80+2\alpha$, giving tax $16+0.4\alpha$ and profit $64+1.6\alpha$.

| 2025 statement amount | All fixed | All floating |
|---|---|---|
| Revenue | 300 | 300 |
| Interest expense | 5 | 7 |
| Tax expense | 16.4 | 16 |
| Profit and comprehensive income | 65.6 | 64 |
| Operating cash flow | 48.6 | 47 |
| Investing cash flow | −20 | −20 |
| Financing cash flow | −8 | −8 |
| Closing cash | 95.2 | 93.6 |
| Receivables / inventory / net plant | 55 / 40 / 98 | 55 / 40 / 98 |
| Assets | 288.2 | 286.6 |
| Payables / debt | 40 / 100 | 40 / 100 |
| Equity | 148.2 | 146.6 |

The joint symbolic result is cash $93.6+1.6\alpha$, profit $64+1.6\alpha$ and equity $146.6+1.6\alpha$, with the same $\alpha \in [0,1]$ throughout. Selecting the smallest cash and largest equity independently would describe no feasible company. A requested distribution requires a measure over that same historical family; the accounting equations do not supply one.

The later method schedules use the floating share $x$ rather than the fixed share $\alpha$ used here: $x = 1-\alpha$. Both parameterisations describe the same joint family. Its 2025 opening cash is 74.6 and equity is 90.6; the forecast spends 20 on period-end capital expenditure, distributes 8 and settles its stipulated tax in the same year. These teaching assumptions are retained when the later lease example introduces a different tax-payment schedule.

## 2.6 Apple: source labels do not identify exposure

The real reference package is Apple's 2024 Form 10-K and first-quarter 2025 Form 10-Q, for fiscal dates 28 September and 28 December 2024 respectively (Apple Inc., 2024, 2025). The annual balance sheet also supplies 30 September 2023. The annual filing is archived as the company's PDF from AnnualReports; the interim filing is archived from SEC EDGAR with its inline XBRL. Apple's Note 1 describes a 52/53-week year ending on the last Saturday of September. Calendar-year assumptions would therefore misstate the periods.

The package inventory preserves every annual PDF page and every interim top-level section, all thirteen annual notes and all ten interim notes. It records internal cross-references and externally

incorporated material, including the proxy statement, as distinct obligations. Statement and debt facts below are structured; policies, other notes and exhibits retain their source locations and specific modelling obligations. This is complete inventory coverage of the acquired files, with partial semantic formalisation. It does not claim to reconstruct every Apple contract or to have acquired every incorporated external document.

Apple's annual balance sheet reports assets 364,980, liabilities 308,030 and equity 56,950, in millions of US dollars. Note 9 reports term-debt principal 97,341, unamortised discount/premium and issuance costs of −321, and hedge accounting adjustments of −358. Thus its carrying amount is $97{,}341 - 321 - 358 = 96{,}662$, split into current 10,912 and non-current 85,750. Fair value is separately disclosed as 88.4 billion. None of these three measurements may replace another merely because a table calls it debt.

Note 9 identifies the notes as fixed-rate, then states that interest-rate swaps effectively convert a portion to floating rates. Note 4 reports interest-rate contract notional of 14,575 at the annual date. A notional is neither a liability carrying amount nor sufficient information for an exact interest sensitivity: reset timing, remaining periods, curve conventions and hedge coverage still matter. The cross-reference connects the derivative to the borrowing without erasing either contract.

At the interim date, total term-debt carrying amount is $10{,}848 + 83{,}956 = 94{,}804$. Cash is 30,299, total assets 344,085, liabilities 277,327 and equity 66,758. The interim debt note's rounded "94.8 billion" and the statement's 94,804 million can coexist; their precision differs. Treating them as conflicting exact numbers would create an artificial infeasibility.

These balances leave two practical financing questions open: how much cash can fund a payment, and when the payment falls due. The annual cash and cash equivalents balance of 29,943 includes 2.6 billion held in escrow and restricted from general use for the State Aid tax obligation. Note 4 states the inclusion explicitly, and Note 7 connects the escrow to that obligation (Apple Inc., 2024, Notes 4 and 7, printed pp. 37 and 39). Subtracting the reported restricted component gives $29{,}943 - 2{,}600 = 27{,}343$ million as an indicative remainder at the displayed amounts. The restricted component is rounded to tenths of a billion, so this subtraction does not establish an exact freely available balance. The earlier case annotation that described the reported cash balance as excluding restricted cash was wrong relative to this footnote. Correcting that interpretation changes funding calculations without changing the issuer's reported number. It also requires revisiting any liquidity result that relied on the exclusion.

The debt movement supplies a second reason to preserve a calculation's missing terms. The first-quarter cash flow statement reports repayments of term debt of 1,009 million, while carrying amount falls by $96{,}662 - 94{,}804 = 1{,}858$ million. At the displayed amounts, the reconciliation is

$$94{,}804 = 96{,}662 - 1{,}009 - 849.$$

The 849 million is the residual after the reported cash repayment; it does not identify an exchange loss, an income-statement charge or a particular journal. Explaining it requires the missing movement information, including applicable currency, discount/premium and hedge-accounting effects. The quarterly cash flow statement still reconciles independently: operating inflow 29,935 plus investing inflow 9,792 less financing outflow 39,371 gives the reported cash increase of 356, from 29,943 to 30,299 million (Apple Inc., 2025, statement of cash flows and Note 6). A balanced cash bridge therefore does not identify the debt history.

The annual filing's securities exhibit adds terms that a debt-total query cannot supply. Exhibit 4.1 describes two euro-denominated series bearing the year 2025 in their names. Its reported outstanding principal is dated 18 October 2024, after the annual balance-sheet date. The description concerns Apple Inc. as issuer, rather than every subsidiary in the consolidated group (Apple Inc., 2024, Exhibit 4.1, pp. 3–4).

| Series | Principal (EUR m) | Fiscal year | Scheduled maturity |
|---|---|---|---|
| 0.875% 2025 Notes | 1,250 | 2025 | 24 May 2025 |
| 0.000% 2025 Notes | 1,000 | 2026 | 15 November 2025 |

Apple's disclosed 52-week fiscal 2025 ends on 27 September 2025 under its last-Saturday convention. The November maturity therefore belongs to fiscal 2026 even though the bond's name says 2025. The annual Note 9 principal-payment schedule uses fiscal years, whereas the instrument's stated maturity is a calendar date. Joining those fields on their year number would put a billion euros of scheduled principal into the wrong fiscal bucket. Neither the later holding date nor the euro principal may be silently substituted for a September US-dollar consolidated balance.

Suppose the disclosed faces remain outstanding until scheduled maturity, with no early redemption, default or additional withholding-related payment, and euro remains the payment currency. The annual coupon on the 0.875% series is then the reported face multiplied by the stated rate:

$$1{,}250 \times 0.00875 = 10.9375 \quad \text{million EUR.}$$

Its final scheduled principal plus annual coupon is consequently 1,260.9375 million EUR; the zero-coupon series has 1,000 million EUR of scheduled principal and zero stated coupon. Both scheduled maturity dates fall on a Saturday. Exact settlement requires the applicable payment-adjustment provision and business calendar. Exhibit 4.1 is expressly a summary qualified by the 2013 and 2018 indentures; its payment-currency exceptions and redemption rights also matter. The retained 2021 indenture cannot replace those earlier contracts. The calculation establishes conditional scheduled amounts, while their exact settlement dates and subsequent holdings remain unresolved.

Those missing facts have different consequences for the financial statements. Principal and coupon payments reduce cash, but settling principal differs from recognising interest expense. The latter also depends on carrying amount, effective-interest accrual and the treatment of discounts and hedges; Note 9 expressly includes applicable amortisation and hedging adjustments in effective rates. Exchange rates, hedge designations and tax treatment can affect profit, other comprehensive income, retained earnings and closing debt differently. The cash flow statement additionally needs the payment classification and any associated hedge settlements. A usable answer thus carries the conditional euro schedule together with these unresolved dependencies. A dollar company forecast must supply consistent holdings, exchange, hedge and tax assumptions across all affected statements.

To demonstrate a source-to-forecast calculation, freeze the interim balance sheet and apply a *marginal* financing stress for a following 13-week period: a 200-basis-point rate increase, no principal change, no offsetting hedge beyond the explicitly selected exposure, immediate cash settlement and no tax effect. An assumed effective floating exposure $E = 14{,}575$ million produces $0.02 \times (13/52)E = 72.875$ million additional interest. This exposure uses the annual notional as a conditional input; it is not asserted to be the interim net sensitivity. Under that condition, debit interest expense and credit cash by 72.875. The resulting marginal statements have cash 30,226.125, assets 344,012.125, liabilities 277,327, equity 66,685.125, profit and comprehensive income of $-72.875$, and operating cash flow of $-72.875$ relative to the frozen baseline. The balance-sheet difference equals the equity difference exactly.

If exposure is instead only restricted to $0 \le E \le 14{,}575$, the conditional added interest lies in $[0, 72.875]$ before monetary rounding. That restriction is a stress assumption and is not claimed to bound every real Apple exposure. This real case verifies evidence, context, mappings and a coherent marginal propagation. Northstar supplies the complete-company forecast; a complete Apple forecast remains dependent on its unmodelled operations and undisclosed terms.

## 2.7 Apple: revenue recognition and cash collection

An iPhone sale can create revenue today and an obligation to provide services later. Apple's annual revenue note separates delivered hardware and bundled software from the right to receive product-related services such as iCloud, Siri and Maps. It allocates the price and related discounts by relative stand-alone selling prices. For the bundled services those prices are estimates, because observable separate prices are unavailable. The hardware portion is recognised when control transfers, generally on shipment; the service portion is deferred and recognised straight-line over the estimated period of provision (Apple Inc., 2024, Note 2, printed p. 35). A historical model consequently needs the allocation and service cohort as well as the sale date and amount. Neither the product label nor the customer payment supplies those missing terms.

The note describes collection shortly after control transfer or the start of service delivery. It does not specify one collection lag for every contract. Products revenue also reflects expected returns, price protection and customer incentives. Revenue excludes taxes collected for remittance to governments (Apple Inc., 2024, Notes 1–2, printed pp. 34–35). Together these qualifications prevent a simple conversion of reported revenue into gross customer receipts.

Presentation creates a second distinction. Where Apple obtains control of a third-party product before sale, it reports the gross amount billed. For third-party applications sold through the App Store, its stated treatment records only the retained commission in Services revenue. Services also already includes recognition of the deferred value of services bundled with products (Apple Inc., 2024, Note 2, printed p. 35). Adding that recognition to reported Services a second time would double count it; treating Services as gross App Store customer spending would change the measured quantity.

The product and geographic tables describe different divisions of the same sales. The five product/service rows sum to 391,035 million dollars in fiscal 2024 and 124,300 million in the quarter ended 28 December 2024. The geographic segment sales sum to those same totals. They must not be added together. Apple says the product mix is generally similar across segments, except that iPhone represents a moderately higher share in Greater China. That qualitative statement does not identify a product-by-region matrix (Apple Inc., 2024, Notes 2 and 13, printed pp. 35–36 and 46–47) (Apple Inc., 2025, Notes 2 and 10, printed pp. 6 and 11).

### 2.7.1 The movement that the note leaves unidentified

At 28 September 2024 Apple reports total deferred revenue of 12.8 billion dollars. At 28 December it reports 13.2 billion, and says that quarter's net sales include 3.7 billion recognised from the deferred balance at the September date (Apple Inc., 2025, Note 2, printed p. 6). The balance sheet's December current deferred-revenue line is 8,461 million, with 8,249 million at September (Apple Inc., 2025, printed p. 3). Those are current liabilities; the note's larger amounts are totals. The precision differs too: the total is displayed to tenths of a billion, whereas the current line is displayed in millions.

Let $D_0$ and $D_1$ denote opening and closing total deferred revenue. During the quarter, let $A$ be additions to deferred revenue, $R_o$ revenue released from the opening balance, and $R_n$ revenue released from amounts added during the quarter. Let $M$ collect other signed movements, positive when they increase the liability. These symbols describe a balance reconciliation; $M$ is not an assertion that a particular adjustment occurred. In billions of dollars,

$$D_1 = D_0 + A - R_o - R_n + M, \qquad A - R_n + M = D_1 - D_0 + R_o \simeq 13.2 - 12.8 + 3.7 = 4.1. \tag{2.4}$$

The source identifies $R_o$, not the whole release $R_o + R_n$. The 4.1 billion residual therefore

measures net additions after new-origin releases and other movements at the displayed amounts. Calling it gross new deferrals, billings or cash collected would be wrong relative to what has been calculated.

Rounding also matters. If each of the three displayed inputs is rounded to the nearest 0.1 billion, a closed outer envelope for that residual is $[3.95, 4.25]$ billion: the lower endpoint uses $13.15 - 12.85 + 3.65$, and the upper uses $13.25 - 12.75 + 3.75$. This is a conditional precision calculation, not a verified issuer rounding rule or a bound on gross collections. Its endpoints leave the treatment of exact rounding ties unspecified.

Two constructed histories make the missing movement visible. For this illustration alone, use the displayed amounts as exact inputs and set $M = 0$. Keep the reported quarter's total revenue at 124.3 billion. The table supplies two different choices for $A$ and $R_n$; other recognised revenue is the remainder needed to match that total.

| Quarterly amount (USD billion) | History 1 | History 2 |
|---|---|---|
| New deferrals, $A$ | 4.1 | 5.1 |
| Release from those new deferrals, $R_n$ | 0 | 1.0 |
| Release from the opening balance, $R_o$ | 3.7 | 3.7 |
| Other recognised revenue | 120.6 | 119.6 |
| Closing deferred revenue | 13.2 | 13.2 |
| Total recognised revenue | 124.3 | 124.3 |

Both histories satisfy the selected note totals, yet their new deferrals differ by one billion. They are witnesses within this aggregate model, not reconstructed Apple contracts or a fit to every financial-statement constraint. Additional contract, cohort, return and movement information can rule out or refine them. No probability attaches to either merely because two examples were written down.

The corresponding recognition entry debits deferred revenue and credits revenue when service is provided. Receipt or billing of consideration that creates the deferral instead credits deferred revenue and debits cash or, where an unconditional right to payment exists, a receivable. Collection of that receivable changes cash and receivables without creating revenue a second time. For example, hold History 1's other revenue cash-settled and omit costs, taxes, refunds and other movements solely to isolate this subsystem. Receiving its 4.1 billion of new consideration gives revenue-side cash inflow 124.7 billion. If that new consideration is receivable at quarter end, cash inflow is 120.6 billion and receivables increase 4.1 billion. In both cases the asset increase is 124.7, matched by a 0.4 increase in deferred revenue and 124.3 of revenue before omitted expenses. These conditional entries explain the identification problem; neither inflow is an estimate of Apple's actual cash collections.

### 2.7.2 A dated recognition expectation is not a cash forecast

At both September and December, Apple expects 64% of the then-reported total deferred revenue to be realised in less than a year, 25% within one to two years, 9% within two to three years and 2% after three years (Apple Inc., 2024, Note 2, printed p. 36) (Apple Inc., 2025, Note 2, printed p. 6). These are two dated expectations about two balances. Multiplication at the displayed December values gives $13.2 \times 0.64 = 8.448$ billion in the first bucket. The corresponding September calculation gives 8.192 billion. Their difference does not measure revenue recognised during the intervening quarter: the dates, remaining horizons and contract populations have changed.

The percentages divide an expected amount by recognition horizon. They do not assign probabilities to alternative histories, fix collection dates or forecast all future sales. The 8.448 billion calculation also cannot replace the reported 8.461 billion current liability. Rounded amount and percentage disclosures do not justify an exact equality between those figures, and the query must retain the source's distinct description of each quantity. A future calculation that requires monthly service or receipts needs finer timing and payment evidence.

The note leaves another part of future service unquantified. For certain long-term arrangements, Apple has not yet delivered the service, has no right to bill for it and attributes the unbilled consideration entirely to that undelivered service. It reports neither revenue nor an amount for those obligations in this passage (Apple Inc., 2024, Note 2, printed p. 35). A query must retain an unknown amount; setting it to zero or equating it with recorded deferred revenue would erase the distinction the company makes.

These disclosures therefore yield several different results. The total and current deferred balances are reported observations. The 4.1 residual is a derived net movement with unresolved components. The recognition buckets are management's dated expectations, and the excluded consideration is unquantified. Contract-level allocations, service lives, new-origin releases and collection facts remain necessary for a particular future journal path. The annual and interim periods also stay separate: the December 2024 quarter follows the September 2024 fiscal year, so subtracting it from that annual revenue cannot produce the year's final nine months. The complete revenue-note reading makes these consequences explicit while the wider Apple disclosure package remains only partly formalised.

## 2.8 Apple: commitments and the cash needed to fulfil them

The same distinction between recognition and cash arises when Apple buys capacity and components. A lease liability measures one part of an existing obligation; a purchase commitment describes goods or services still to be acquired. A funding projection needs the payments arising from both, together with new orders and operating decisions. Adding every disclosed amount would mix measurements and could count the same requirement twice.

### 2.8.1 From a lease liability to future payments

At 28 September 2024, Apple's lease note reports the following amounts in millions of dollars (Apple Inc., 2024, Note 8, printed p. 42). The maturity table first totals undiscounted payments and then deducts imputed interest to reach the recognised liability. The right-of-use assets are reported separately.

| Amount (USD million) | Operating | Finance | Total |
|---|---|---|---|
| Undiscounted lease payments | 13,518 | 1,272 | 14,790 |
| Less: imputed interest | 1,984 | 376 | 2,360 |
| Recognised lease liabilities | 11,534 | 896 | 12,430 |
| Right-of-use assets | 10,234 | 1,069 | 11,303 |

Thus $14{,}790 - 2{,}360 = 12{,}430$ reconciles the maturity and liability measurements. The recognised liability is already included in the balance sheet; it is not an extra cash requirement to add to 14,790. Its current components total $1{,}488 + 144 = 1{,}632$ million, whereas the fiscal-2025 maturity bucket is 1,991 million. A current carrying amount and a gross payment bucket answer different questions. The right-of-use asset likewise need not equal the remaining liability; these

reported balances do not disclose the complete amortisation, prepayment and modification history.

Apple also reports 849 million of fixed payments under additional leases that had not commenced. It expects them to commence in 2025–2026, with terms from less than one year to twenty-one years. Adding this separately disclosed amount to the maturity total gives $14{,}790 + 849 = 15{,}639$ million. That sum is consistent, under a nearest-0.1-billion rounding assumption, with the 15.6 billion of fixed lease obligations in management's cash-requirements discussion (Apple Inc., 2024, printed pp. 25 and 42). Numerical compatibility does not establish a contract-by-contract reconciliation. It neither adds 849 to the recognised liability nor identifies its commencement and payment dates.

The remaining lease information cannot supply those dates mechanically. The note describes original terms typically no longer than ten years, with renewal options, some reasonably certain to be exercised. Its weighted-average remaining term of 10.3 years describes a different quantity. The reported 3.1% discount rate, generally based on estimated incremental borrowing rates, does not supply every contract's rate or the timing hidden in "thereafter". Applying that one rate to invented year-end payments would introduce new assumptions. The policy combining lease and nonlease components also has a stated scope: corporate, data-centre and retail facilities (Apple Inc., 2024, Notes 1 and 8, printed pp. 34 and 42).

Fixed and variable costs add another distinction. For 2024, Apple reports 2.0 billion of fixed operating-lease cost, 1.9 billion of fixed operating-lease cash payments and 1.0 billion of noncash right-of-use assets obtained for lease liabilities. Variable costs across its leases were 13.8 billion; payments of that kind primarily depend on purchases of output from the leased assets (Apple Inc., 2024, Note 8, printed p. 42). Historical expense is therefore not a fixed future cash schedule. Forecasting it requires output, price and settlement assumptions. Adding it to manufacturing commitments without identifying their relationship could double count procurement requirements.

### 2.8.2 Why a fall in commitments does not identify cash spent

The purchase note and management discussion disclose different views. Note 12 lists 11,226 million of future payments under unconditional purchase obligations whose remaining terms exceed one year. Its fiscal-2025 row is 3,206 million (Apple Inc., 2024, Note 12, printed p. 45). The term restriction selects contracts; it does not place every payment beyond the coming year. The note describes these as off-balance-sheet commitments, primarily involving supplier arrangements, intellectual property, content and distribution rights.

Management's discussion provides additional near-term evidence. At the same September date it reports manufacturing purchase obligations of 53.0 billion, of which 52.9 billion is payable within twelve months. Other purchase obligations total 12.0 billion, with 4.1 billion within twelve months; that description includes capital assets as well as supplier, content and rights arrangements (Apple Inc., 2024, printed p. 25). A model confined to Note 12 would miss those reported aggregates. Conversely, subtracting 11.226 from 12.0 and labelling the difference short-term contracts is unsupported: the category and term populations have not been reconciled, and the precision differs.

At 28 December 2024 manufacturing obligations are 39.8 billion, all payable within twelve months. The interim report says contractual cash requirements have not changed materially since the annual filing except for commercial paper, manufacturing obligations and the State Aid tax payable (Apple Inc., 2025, printed p. 16). That qualitative statement is not an exact December lease measurement or evidence that every lease remained unchanged. The two manufacturing totals are dated observations over changing orders and payment horizons. Their

13.2 billion difference does not identify purchases or cash paid during the quarter.

To see what is missing, consider an explicitly simplified model in billions of dollars. Let $O_0$ and $O_1$ denote a comparable population of undelivered fixed-price orders at the two dates. Let $N$ be new orders and $F$ the value removed when goods are delivered. Assume no cancellation, price or currency change. Treat the displayed 53.0 and 39.8 as exact inputs for this illustration. These assumptions define the example; the disclosure does not establish this roll-forward convention for every reported obligation. The balance relation is

$$O_1 = O_0 + N - F, \qquad F - N = 53.0 - 39.8 = 13.2. \tag{2.5}$$

With no new orders, delivery is 13.2. With 5.0 of new orders, delivery is 18.2. Both leave the same closing undelivered amount of 39.8. The source balances alone cannot choose between these conditional movements; actual contract changes and delivery evidence could do so.

Payment is a further choice. Suppose the first history's 13.2 of delivered goods all remain unsold inventory, with no opening payables, prepayments or tax. Paying on delivery debits inventory and credits cash by 13.2: total assets and profit do not change within this subsystem. Delivery on credit instead debits inventory and credits trade payables by 13.2: assets and liabilities rise together, with no immediate cash or profit effect. Settling that payable later debits the payable and credits cash; consumption, sale or impairment would require further entries. These examples isolate recognition and settlement. They are not estimates of Apple's actual purchases or payments, or a fit to all its financial statements. No probability follows from writing two histories.

Delivery assumptions also need the narrative that surrounds the purchase table. Apple describes single or limited sources for some components, industry-wide shortages and price fluctuations, and capacity or yield constraints for new technologies. Suppliers may favour common components over custom ones. Its named manufacturing locations describe outsourced production, not a product-by-revenue-region allocation or ownership map (Apple Inc., 2024, Note 12, printed p. 46). A scenario in which input availability changes may alter output, variable lease payments, inventory and receipts together. The passage motivates those dependencies without identifying their size or probabilities.

The same note, and the interim contingency note, preserve management's dated assessment of unresolved legal claims. At each date, management states that there was not at least a reasonable possibility that Apple had incurred a material loss, or a material loss greater than a recorded accrual (Apple Inc., 2024, Note 12, printed p. 46) (Apple Inc., 2025, Note 9, printed p. 11). It does not make litigation exposure zero. Contract terms, supply constraints, commencement, delivery and settlement must therefore remain in the historical record alongside reported balances. They determine which future journal paths can be supported when a funding question is asked.

# Chapter 3

# Representing a company's historical record

Northstar's corrected receivable, Apple's debt measurements and its deferred revenue show why the source record must retain more than a label and a number. Section 2.7 distinguishes a reported balance, a derived movement, a dated expectation and an unquantified obligation. Section 2.8 adds the contract and delivery facts needed to turn a commitment into a future payment. The procedures below determine which assertions enter a model and which remain unresolved.

## 3.1 The record is a collection of assertions

The original record includes financial statements, notes, accounting policies, auditor opinions, management discussion, segment tables, security terms, regulatory returns and subsequent announcements. A complete ingestion process accounts for every page and every extracted block. A complete financial model is a stronger claim: it would require economically relevant information that public reports may never disclose. AccDSL therefore reports document coverage, semantic coverage and simulation coverage separately.

Each source span receives a disposition: structured observation, narrative assertion, relationship, policy evidence, contextual material, duplicate, or unresolved extraction. "Unresolved" remains visible. A commitment buried in a note is not discarded because it has no current balance-sheet line. An auditor's qualification is retained as evidence about the record; it does not become an automatic numerical haircut. Management's forecast is a dated assertion with an author and modality, not an observed future outcome.

An observation is an immutable record with the fields below. Values inferred from observations are separate records, linked by a derivation. This avoids overwriting a company's statement with an analyst's preferred reconstruction.

| Field group | Minimum contents |
|---|---|
| Identity and source | Observation ID; document hash; filing identifier; PDF page and bounding box or HTML/XBRL locator; literal text; extraction version. |

| Field group | Minimum contents |
|---|---|
| Reporting context | Legal entity; reported group perimeter; statement or note; instant or interval; standard and taxonomy versions; dimensions such as segment, product, geography, seniority or measurement category. |
| Amount | Original string; parsed number; sign interpretation; unit; scale; currency; precision or rounding interval; missing-value category. |
| Meaning | Source label; taxonomy QName when present; one or more proposed concept mappings; whether the assertion is reported, estimated or conditional. |
| Timing | Economic validity when known; reporting period; publication and availability timestamps; ingestion timestamp; correction or supersession links. |
| Reliability and review | Extraction confidence, reviewer decision, source status and known contradictions, all distinct from a probability over histories. |

A narrative assertion adds actor, predicate, object, scope, quantifier and modality. For example, "substantially all debt is fixed rate" has a source span and a qualitative quantifier. It does not assert $\alpha = 1$. A reviewed interpretation can introduce a numerical restriction such as $\alpha \geq a$, but the threshold $a$ remains an analyst assumption unless the source defines it. Boilerplate policy language also does not prove that a company holds a particular instrument.

## 3.2 Different names, different measurements

An accounting concept is more than a row label. "Debt" can refer to legal principal, amortised carrying amount, undiscounted contractual cash flows or a fair-value disclosure. These amounts can all be correct at the same date. Likewise, "revenue" may be reported gross or net, a segment measure may differ from the consolidated measure, and an alternative performance measure may exclude selected expenses. A mapping must state its relationship: equivalent meaning, component of, broader than, presentation reclassification, or measurement transformation.

The common vocabulary will preserve source concepts as first-class identifiers. Company extensions may inherit a broad economic category without being declared identical to a standardised concept. For instance, an extension can be a component of borrowings while retaining an unresolved split between lease liabilities and other financing. Many-to-many mappings are allowed; each mapping has applicable dates, dimensions and evidence. An aggregation mapping does not silently become a measurement conversion.

Standards differ in recognition, measurement, presentation and disclosure. Consequently, conversion between IFRS, US GAAP and Chinese Accounting Standards for Business Enterprises (ASBE) is not generally a dictionary lookup. The original basis remains preserved, and any comparison basis is a separate derived book. The same principle applies to tax and prudential capital. A tax loss carryforward, accounting deferred tax asset and regulatory capital deduction can concern the same economic loss while belonging to different measurement systems.

## 3.3 Time, revisions and information availability

At least four times matter. A machine may enter service in November; its cost appears in the accounts for the year ending December; the annual report is published in March; and AccDSL ingests the report in April. A later report may revise the December comparative. These times answer different questions and must remain distinct.

The storage model uses validity intervals for economic assertions and transaction intervals for database versions, following the distinction in Snodgrass and Ahn (1986). Publication and availability timestamps are additional domain fields. Database transaction time records what AccDSL contained; it cannot substitute for what investors could have known. An "as known on" query applies an explicit availability rule and does not include a later restatement simply because its economic period is earlier.

AccDSL supports three named views: the original filing view; the latest restated view, with a stated information cutoff; and a harmonised research view, with explicit transformations. No view changes the stored source facts. Corrections have links to the replaced assertion and a reason. Where two sources conflict without a clear supersession relationship, both survive with a conflict record.

Instant values and interval flows have different types. A balance at 30 June is not a six-month flow. Calendar intervals use half-open boundaries $[t_0, t_1)$ internally; a filing period ending 30 June maps to the next boundary at 1 July in its reporting calendar. Quarter-only earnings may be derived from year-to-date values only if the periods, basis, perimeter, units and revision vintage are compatible. The derivation retains both source observations and their correlated rounding uncertainty. Fifty-two/fifty-three-week calendars, stub periods and reporting-date changes require explicit calendar objects.

Accounting-policy changes, estimate changes and error corrections also have different temporal effects. The applicable rule package decides whether a change is prospective, requires restatement, or adjusts opening balances. The engine never infers that effect from a generic "revision" flag.

## 3.4 The company boundary changes

Legal entity, reporting group and economic exposure are separate dimensions. Acquisitions, disposals, changes in control, joint arrangements, special-purpose vehicles and non-controlling interests can change the reporting perimeter. A group's increase in property need not be capital expenditure; it may reflect an acquisition, a currency translation or a revaluation. The bridge between opening and closing property must identify these movements before a scenario model can extrapolate investment.

The company graph therefore stores dated ownership, control and counterparty relationships. Control is an accounting conclusion with supporting evidence, not a rule that blindly consolidates every majority-owned entity. Group reporting combines entity books with a separate adjustment and elimination book. Intragroup receivables and payables require matching counterparties, currencies and dates. Unmatched eliminations remain unresolved; a balancing "plug" is not a recovered transaction.

Acquired opening balances, goodwill, purchase-price allocation, divested operations and non-controlling interests each have their own event types. Analyses can preserve reported scope or define a constant perimeter, but the latter is a derived modelling assumption, not the original record.

## 3.5 Further problems that the representation must handle

Several difficulties extend beyond differences in standards, industries and company labels. Recognition and settlement occur on different dates: a sale, invoice, cash receipt and impairment can belong to four periods. Net balances hide gross flows: repeated borrowing and repayment can leave the same closing debt. Netting can conceal offsetting exposures and liquidity needs. Different contractual maturities and repricing dates can share the same current carrying amount. Guarantees, undrawn credit facilities, litigation and covenants can be economically important before an asset or liability is recognised.

Foreign currency introduces transaction currency, functional currency and presentation currency, together with different translation conventions for stocks and flows. Currency translation effects must not be mistaken for cash receipts. Inventory and depreciation depend on vintages or lots; an aggregate stock and average age may be insufficient under a shock. Tax bases can evolve differently from carrying amounts. Deferred tax recognition requires assumptions about recoverability, rather than multiplication by a tax rate alone.

Public information is selectively disclosed and rounded. Duplicate XBRL facts may agree, overlap within rounding bounds or contradict one another. A dash can mean zero, immaterial, unavailable or inapplicable. Tables can mix units, span pages and embed signs in parentheses. Text and table references can be wrong in the source itself. AccDSL retains these distinctions and gives an analyst a local repair path without pretending that extraction confidence settles economic interpretation.

Finally, strategic behaviour matters. A firm may refinance, cut investment, raise equity, change prices, breach a covenant or fail. A bank may preserve a capital ratio by shrinking assets; an insurer may change reinsurance or asset allocation. Such responses cannot be deduced from the balance-sheet identity. They require explicitly calibrated or assumed policies, and alternative policies must remain separate when the available record does not distinguish them.

## 3.6 Account for a disclosure before translating it

The importer first freezes a filing package: documents, byte hashes, reporting dates, availability evidence and incorporated references. It compares the document's contents page and statement index with detected section and note headings. A missing heading, table continuation or referenced exhibit creates an obligation to resolve; successful text extraction cannot discharge it. Each PDF page and each HTML content unit then has a stable locator and a disposition. Child units such as a table or note belong to a parent page/section; the importer records overlap instead of counting the same content twice.

Three denominators remain separate. Acquired-document coverage counts source units with a disposition, including unresolved units. Semantic coverage counts identified assertions whose meaning and context have been reviewed. Simulation coverage counts the operations and exposures required by a specified query that have supported state and rules. The first can be complete while the other two are partial. An assertion inventory can itself be incomplete; until the page/section reconciliation is reviewed, semantic coverage is a lower-bound inventory rather than a percentage of all financial meaning.

Cross-reference traversal is a work-list algorithm. Start with the filing's statement and note references, resolve each target against the acquired manifest, add newly discovered targets, and stop when no unvisited target remains or a declared resource limit is reached. Document hash plus target locator prevents cycles from expanding forever. A target has one of four results: resolved, duplicate of a resolved source, unavailable, or ambiguous. The last two

remain obligations. Apple's proxy and prior-filed agreements show why closure over acquired files is weaker than closure over incorporated disclosures. A manifest may declare a bounded acquisition scope, but must show these outstanding references beside its completeness status.

A narrative translator records the actor, predicate, object, quantifier, modality, applicable entities/dates and qualifications before proposing any constraint. "All notes are fixed-rate" constrains contractual coupon type; it does not constrain the combined note-and-swap exposure. "May repurchase" creates an authorised action with limits, not an obligatory payment. "Substantially all" remains qualitative unless a separately reviewed assumption gives it a threshold. A conditional guarantee becomes an obligation with a trigger and exposure, not a present liability merely because its maximum amount is known. Unsupported predicates retain their literal text and links to affected balances. The reviewer accepts the translation separately from accepting that the text was extracted correctly.

## 3.7 Mappings must establish a financial relationship

A mapping record identifies source concepts, target concept, relationship, applicable entity/basis/period/dimensions, evidence, reviewer and any required transformation. Equivalence permits substitution only when all these contexts match. A component relationship permits inclusion in a sum only through a declared disjoint partition. A broader concept supplies containment, not an equality. Reclassification changes presentation while preserving the stated population and total; measurement transformation must supply an actual formula and inputs. These relationships cannot compose indiscriminately: component followed by equivalence remains component, while two measurement transforms compose only if their intermediate bases and information dates agree.

For example, Northstar's product revenue might split into 180 domestic and 80 overseas, while its customer channel split is 200 wholesale and 60 retail. Both classify the same 260. Adding all four numbers gives 520 and is invalid. The partition records therefore share a population ID but have different axes. To remove an axis, `sumOver` requires evidence that members are disjoint and exhaustive for that context. Where the source supplies only named segments, an explicit unallocated member carries the residual; it is not assumed zero. Geography-by-product cells cannot be recovered from the two marginals without further constraints or assumptions.

Apple supplies another counterexample: its vendor non-trade receivables arise from component sales to manufacturing vendors, whose gains reduce product cost of sales when the final products are sold (annual Note 4, p. 38). They cannot be relabelled as ordinary revenue-generating trade receivables without changing meaning. Likewise gross derivative assets and liabilities cannot be silently netted. Company extensions can define these distinctions and dated presentation changes; they cannot override mandatory recognition rules in a selected standards package.

## 3.8 Selecting original and revised assertions

The query procedure first filters assertions by the declared availability cutoff, then by an optional database-time cutoff. A day-only filing timestamp is conservatively available at the next local midnight in its named timezone; any more precise rule needs evidence. Within the eligible set, explicit supersession removes only the targeted assertions in matching contexts. The superseding assertion must itself be eligible. Follow the acyclic graph transitively; retain all maximal assertions if two eligible corrections conflict. File order and ingestion recency never decide which is true.

The original view selects the initial assertion chain in a named filing; the latest-available view selects eligible maximal versions; the harmonised view adds reviewed transformations to

those selections. A revision to revenue does not automatically revise every note or change the company's accounting basis. An estimate change takes effect in future measurements from its specified date. An error correction may change a comparative and an opening balance bridge, each as a new assertion with its own lineage. Northstar's explicit query fixture in Chapter 2 checks all these clocks without assuming that the most recently stored row is always the appropriate history.

## 3.9 What an accepted reading permits

Apple's inventory policy illustrates the distance between reading a sentence and reconstructing a business. Its reported use of FIFO supports a proposed cost-formula reading for the inventory and reporting contexts covered by that policy. It does not reveal the quantities and ages of the actual lots, establish that no write-down occurred, or supply the authority for every US accounting treatment. Those are separate inputs to a cost and impairment calculation. Similarly, a straight-line depreciation policy leaves useful lives, residual values, components and available-for-use dates to be established. Section 9.5 tests those limits against the reported asset balances, depreciation and cash payments, then constructs two different asset books compatible with the selected amounts.

The source preparation retains six located Apple policy passages: consolidation of Apple and wholly owned subsidiaries; comparative reclassification; FIFO; straight-line depreciation; the interim statement's estimates and annual-report reference; and its 52/53-week fiscal calendar. Each proposed reading carries the original document hash, extraction hash and exact character slice. These are company assertions. The consolidation sentence, for example, cannot exclude every partly owned entity without a control analysis. The calendar sentence does not make every future fiscal year 52 weeks or make a contractual interest year 365 days. The exact quotations and proposed predicates are retained in `../methods/source-translations-v1.json`; the meaning explained here is not an invented human acceptance of them.

A reading progresses through separate decisions. First, extraction review asks whether the located words and numbers were transcribed faithfully. Translation review asks whether the proposed predicate preserves their scope, quantifier, exceptions and reporting context. A model then admits that predicate, or an explicitly labelled assumption, for a particular purpose and information view. Accounting eligibility subsequently asks whether a method applies to the actual event. A faithful transcription of an erroneous company assertion is still faithful transcription. The four decisions therefore cannot share a single "approved" flag.

An explicit presumption in a standard also differs from filling an unknown with a convenient answer. Article 6 of the retained official ASBE 21 text requires both practical ability throughout use and economic benefit for a substantive substitution right. It specifies a presumption when that right is difficult to determine. Applying that conditional rule requires its stated circumstances; it does not transform missing lessor information into a reported fact. Nor may a translator assign arbitrary numerical thresholds to "substantive", "probable" or "highly probable".

## 3.10 Three meanings of disclosure coverage

The Apple annual/interim preparation inventories 125 quantities, 13 annual notes, 10 interim notes, 70 exhibit references and seven further package obligations. All 23 notes retain unresolved semantic content; none is recorded as fully formalised. Downloading their pages answers a useful but narrower question than constructing a complete company model.

Three populations make that distinction inspectable. Retained-unit coverage partitions acquired documents into nonoverlapping pages or smaller leaf units, including duplicates and unreadable material. Assertion-review coverage counts the separately enumerated material assertions and their review decisions. Query-exposure coverage identifies the sources, legal facts, methods and authority that a particular result requires. A fully downloaded annual report can have incomplete assertion review, and a reviewed debt total can leave a refinancing query unresolved because an incorporated agreement is missing. No weighted average of these populations establishes completeness.

Preparation begins with declared annual and interim roots and a policy for following incorporated references. Each edge is recorded as satisfied, duplicate or cyclic, outside the declared scope with a reason, inaccessible, or still unexpanded. A bounded traversal saves its unexplored frontier and budget so it can resume. An unknown external extent remains unknown. Acquired proxy and registration documents improve the Apple record, but do not prove that every executed amendment, issuance term and exhibit dependency has been closed. The present real package remains incomplete for that reason.

Container choice can itself create a false missing-source diagnosis. Apple's retained annual PDF contains securities and award exhibits absent from the retained HTML container; other incorporated award forms are present in earlier reports in the retained corpus. Locating those bodies resolves a source-access question. It still leaves their qualifications and outgoing references to be interpreted. Conversely, the cash restriction in Section 2.6 shows why a located note can change the meaning of an already extracted figure: correct arithmetic alone would not detect an erroneous "unrestricted" label.

## 3.11 Displayed precision does not determine a rounding rule

Suppose a statement displays 100 in millions. The scale establishes how the literal is read; it does not establish how the issuer rounded its underlying amount. Under an explicitly admitted nearest-rounding assumption, an integer display $m$ at scale $s$ permits the conservative compatibility enclosure

$$s(m - \tfrac{1}{2}) \le y \le s(m + \tfrac{1}{2}).$$

The closed endpoints accommodate an unknown tie rule within that nearest-rounding model. A specified tie rule can exclude an endpoint. Truncation of a positive amount instead permits $sm \le y < s(m + 1)$. Thus the illustrative display 100 corresponds to 99.5–100.5 million under the first assumption and 100–101 million, excluding 101, under the second. These are different restrictions on a history.

The retained Apple arithmetic uses the first enclosure as a conditional compatibility assumption; the issuer's rounding convention has not been established. Without that assumption or source support, the record states known display precision and unknown rounding convention and supplies no hard half-unit constraint. A one-unit difference between a displayed total and the sum of displayed components is consequently not enough to establish a contradiction. The source value is scaled once, while its literal and precision metadata remain available for inspection.

## 3.12 Public availability and system possession are different views

Northstar's correction explains why a result needs a reporting vintage. A historical forecast also needs to establish when the exact document bytes were available. A report downloaded today may carry an old date on its current web page; that does not prove that the same bytes

were public on that date. Public availability evidence and an ingestion receipt answer different questions.

An investor-information query supplies a cutoff $K$. Each admitted source must have supported public availability no later than $K$ and evidence linking the selected bytes to that availability. Later ingestion is allowed. Later extraction review can read eligible old bytes, but an interpretation that uses later information fails the cutoff on its own dependencies. This view establishes public availability, not actual investor awareness or the ability to trade.

A system-as-of query instead supplies an instant $S$ and requires an immutable receipt showing possession of those bytes by $S$. Publication cannot replace that receipt. An accepted-layer query also requires review decisions visible by $S$; a raw-layer query may return a then-unreviewed document. A query asking both what was public and what the system possessed must pass both filters. Unknown publication alone need not block a pure possession question.

Archival exploration permits inspection of retained sources with incomplete timing. With a cutoff, unknown publication yields an unresolved candidate, not a cutoff-qualified selection. Without a cutoff, the archive may display the retained lineage, explicitly without a historical-knowledge claim. A known later version is excluded from an earlier view and cannot suppress an eligible old assertion. An unresolved potentially relevant version keeps the affected answer unresolved even when an older assertion is displayed.

Date-only evidence requires particular care. For publication date $D$ in a named timezone, let $[a_D, b_D)$ be the interval from local midnight to the next local midnight. A cutoff before $a_D$ excludes the source; a cutoff inside the interval leaves availability unresolved; a cutoff at or after $b_D$ supports admission on this date evidence. No arbitrary noon or fixed 24-hour duration is inserted. Missing timezone leaves the instant comparison unresolved.

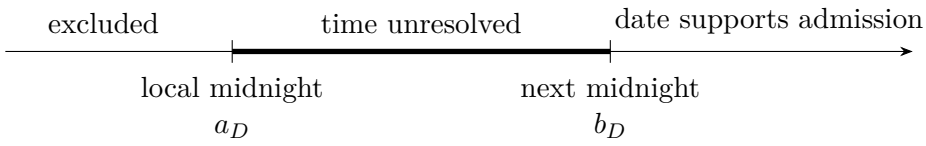



Figure 3.1: A dated publication supports historical admission only after its possible publication interval has ended. The line marks uncertainty about an instant, not a delay imposed on an already known timestamp.

Revision selection follows admission. Original selection returns eligible roots; latest-available selection returns eligible assertions without an eligible superseding descendant for the same context. Unrelated roots and multiple live leaves remain competing assertions. A harmonised view adds explicit transformations with their own admitted sources, method and opening bridges. File order never selects a winner.

The synthetic interim example has original first-half revenue 120, revised first-half revenue 110 and revised annual revenue 260. Compatible subtraction gives second-half revenue $260 - 110 = 150$. The arithmetic $260 - 120 = 140$ is easy but mixes incompatible vintages. An original-view display may show both source amounts while refusing that derived flow. A change in calendar or perimeter likewise requires a specific bridge.

The investigated real sources have not yet supplied a qualifying unresolved same-context contradiction. Known revisions, different reporting bases and synthetic conflicts remain useful cases of their own kinds. A genuine conflict case must first match entity, perimeter, concept, dimensions, unit, basis, period, versions and precision, and investigate extraction and supersession. The absence of a qualifying example in a bounded search is not evidence that the corpus contains no conflicts.

# Chapter 4

# What the literature contributes

The research problem crosses accounting, databases, programming languages, econometrics and document understanding. Each field resolves part of it. The review below distinguishes results that can be adopted directly, engineering ideas that need adaptation, and evidence that does not establish the intended product capability. Source versions and technical anchors are recorded in the accompanying research ledgers. The search includes work outside the original collection, with public-index searches and backward and forward citation searches through 2 October 2026. It is a scoped design survey, rather than a claim to enumerate every paper in these fields.

## 4.1 Accounting structure and company dynamics

Nehmer and Robinson (1997), especially Sections 2–5, supplies a useful algebraic starting point: balances occupy a structured state space, transactions change that state, and aggregation relates detailed accounts to reports. This supports a distinction between conservation constraints and permitted transactions. An entry can conserve the accounting total and still be an inadmissible transaction. AccDSL adopts that distinction in its posting validator. Its particular event vocabulary, evidence model and inverse-history semantics are new design decisions, not results established by the algebraic paper.

Doshi (2024) develops an accounting DSL with time, transaction logs and correction mechanisms. The relevant material is in Chapters 4 and 7–9 of the 2024 thesis; the original collection's filename misleadingly suggests 2026. It demonstrates why corrections and dates belong in the language rather than in informal comments. The thesis does not establish recovery of an entire company history from public reports or inference over economically distinct histories. Babcinețchi et al. (2022), pp. 342–345, offers a much smaller accounting language and grammar proposal. It is useful as a syntax precedent, but it provides insufficient semantics or evaluation to govern AccDSL by itself.

Shahnazarian (2004) connects accounting dynamics with behavioural and econometric equations. Sections 2.3–2.8 describe the accounting relationships; the optimisation and econometric discussion in Chapter 3 and the data and evaluation material in Chapters 4 and 8 show how the model is used with standardised Swedish corporate data. This is an important precedent for simulating companies under changed conditions. Its standardised tax-data starting point differs from heterogeneous public-report extraction, and its working-capital funds formulation should not be imported as a general modern cash-flow statement implementation. The transferable idea is the explicit separation of accounting evolution and behavioural assumptions.

Pierson (2020), Sections 3–4 and the reporting structures, makes that separation operational in system dynamics. Receivables, payables and wages accumulate differences between accrual and payment flows; reporting-period accumulators turn instantaneous flows into statements. AccDSL can use these structures as examples of coherent movement schedules. The model's payment delays, inventory approximations and simplified equity treatment remain modelling choices. They are not universal defaults for companies or evidence that an aggregate state is sufficient for all shocks.

Bank models demonstrate why journal generation needs economic state. Halaj (2016), Section 3, combines funding shocks, fire sales, loans, securities and risk constraints. The homogeneous portfolios and specified funding process make the model tractable, but limit instrument-level interpretation. Behn et al. (2019), Section 2, makes managers' choices over asset growth, debt, risk and payouts depend on capital and liquidity constraints. These models motivate interfaces for management policy and regulatory calculations. Their parameterisations must be calibrated and evaluated for the target application; passing their accounting identities would not validate an AccDSL forecast.

Two important neighbouring traditions remain explicit coverage qualifications. McCarthy's 1982 REA model is a foundational candidate for the relation between resources, events and agents; Caiani and co-authors' stock-flow-consistent agent-based benchmark is a candidate for simulation across interacting sectors. Verified primary PDFs could not be obtained in this retrieval campaign after institutional and publisher attempts. Their records are retained in the omission register, and no technical result from either is used to justify the design. AccDSL's event and counterparty records are specified and justified in Chapters 7 and 8. A later multi-company equilibrium engine would require expansion of the stock-flow-consistent literature before implementation.

## 4.2 Incomplete information and provenance

Financial disclosures are incomplete databases in a substantive sense. Multiple underlying records can yield the same published observations. Libkin (2014), Sections 2–5, explains the semantics of incomplete databases, labelled nulls, conditional tables and certain answers. Repeated occurrences of the same unknown must share a value. Conditional tables can represent dependence between alternatives that independent missing cells cannot express. This is directly relevant to a debt note and a cash-flow table that refer to the same unknown repayment.

AccDSL adopts shared symbolic unknowns and conditions. It extends the database idea with accounting constraints, continuous amounts, temporal transitions and economic event models. A "certain" answer means a statement that holds for every history in the declared feasible family. The family itself may omit unknown mechanisms; certainty is therefore always relative to a model scope. Imieliński and Lipski's original 1984 paper is a foundational backward lead whose publisher copy was inaccessible in this campaign. The inspected Libkin paper supplies the technical discussion actually used here.

Green et al. (2007), Definitions 3.1–3.2 and Section 4, treats provenance as annotations that compose through relational operations. Combining sources and alternative derivations can therefore have formal semantics, rather than being a list of unrelated citations. This is useful for report aggregation and explanation queries. Positive relational operations covered by provenance semirings do not by themselves specify every operation AccDSL needs: negation, nonlinear valuation, optimisation and probabilistic conditioning require their own derivation records. Accordingly, the implementation starts with a typed dependency graph and may use semiring annotations for supported relational subqueries. A dependency edge is not a posterior

probability.

Snodgrass and Ahn (1986) distinguishes historical validity from the time facts are stored. The archived paper is *Temporal Databases*, by Snodgrass and Ahn, not the later Jensen–Snodgrass paper initially returned by a search. Its snapshot, rollback, historical and temporal examples support the need to recover both old business facts and old database beliefs. AccDSL adds public availability time because a database insertion date cannot establish whether a historical investor could have seen a filing.

## 4.3 Inverse problems, identification and probability

Stuart (2010), Section 2.2, starts from an observation equation $y = G(u)$ and distinguishes inversion, regularisation and Bayesian inference. The discussion makes explicit that a regularising norm or prior is an additional assumption. AccDSL has a similar observation problem, although its unknowns include discrete accounting interpretations and event structures as well as continuous parameters. The infinite-dimensional well-posedness theorems in that paper are not automatically applicable to this mixed, potentially discontinuous model.

The relevant issue is often identification rather than computation alone. Giacomini and Kitagawa (2021), Sections 2.1–2.6 and Theorem 1 of the archived 2020 working version, shows how a class of priors can preserve uncertainty within an identified set. A likelihood that is flat over the unresolved part does not learn the conditional prior over that part. Posterior lower and upper probabilities summarise a class of posterior beliefs. The paper's regularity conditions and application to structural vector autoregressions do not prove that an arbitrary accounting program admits the same algorithms or guarantees. The design borrows the distinction between learnable quantities and assumption-dependent distributions; the simple accounting bounds in Chapter 6 are derived directly.

Hoeting et al. (1999), equations (1)–(3), gives Bayesian model averaging as a weighted sum of model-conditional distributions, with weights determined by model priors and marginal likelihoods. Sections 3 and 5 discuss computation and prior specification. This supports an optional weighted mode, provided AccDSL specifies the likelihood, priors and candidate model space. It does not justify giving equal mass to every syntactically different journal reconstruction. Two encodings of the same economic history must not gain extra weight by duplication. Model averaging also remains conditional on the considered model family, rather than repairing missing mechanisms.

Scenario semantics need similar care. Pearl (2009), Section 3.2.1 and equations (5)–(6), distinguishes an intervention that replaces a structural equation from ordinary conditioning. AccDSL will distinguish a supplied exogenous path, an observational conditional forecast and an intervention. A rise in rates imposed by a stress test is not automatically distributed as the collection of historical periods in which rates happened to be high. Identifying a causal effect requires assumptions beyond reproducing accounts.

## 4.4 Languages for rules and contracts

van Deursen et al. (2000) surveys DSL benefits, risks and implementation options. Mernik et al. (2005), Sections 1.1 and 2.2–2.5 of the archived CWI report, treats application libraries as serious competitors and distinguishes deciding, analysing, designing and implementing a DSL. Its implementation decision tree explicitly includes embedding in a host language. The literature therefore supports a conditional design decision: a DSL is attractive when domain

constraints and repeated notation justify it, but an ordinary typed library may be cheaper and equally expressive. AccDSL will implement one shared representation accessible from both a Python API and a textual language, so syntax does not become a prerequisite for testing the accounting model.

Merigoux et al. (2021), Sections 3–4 and Figures 8–9, provides Catala's scopes, exceptions, default calculus and translation into a conventional functional target. Its explicit empty and conflict outcomes are valuable precedents for accounting rules with exceptions. They are different from probabilistic uncertainty. AccDSL will preserve this distinction: conflicting applicable rules produce a diagnostic, while an evidence-supported alternative reading produces a conditional model branch. The paper's compiler correctness results do not establish correctness of our accounting rules or of an automatic translation from disclosures.

The inspected LegalMath code offers reusable ideas in its shared typed interpretation, partial observations, source references and validity/knowledge times. Its ordinary frontend execution selects a reading. AccDSL needs an additional execution mode that retains a family of readings, continuous unknowns and economic state. A bounded candidate generator is useful for search, but its candidate limit does not prove completeness.

Peyton Jones et al. (2000), Sections 3–5, builds complex financial contracts from combinators and separates their compositional meaning from a chosen valuation model. The ACTUS technical specification similarly separates contract states, scheduled events, state transitions and payoffs, with risk factors supplied through an interface (ACTUS Financial Research Foundation, 2026). These are strong reasons to represent contractual obligations independently of accounting classifications. Neither contract cash flows nor a derivative price determine the recognition, measurement and presentation rules of a complete IFRS or US GAAP book.

## 4.5 Extraction and financial reasoning

The document literature supports a staged ingestion process. PubTables-1M (Smock et al., 2022), in its canonicalisation procedure and evaluation section, distinguishes table detection, structure recognition and functional analysis. Its GriTS evaluation compares reconstructed table structure. TableBank (Li et al., 2020), Sections 3–4, uses document structure to obtain weak supervision for table detection and recognition. GTE and FinTabNet (Zheng et al., 2021), Sections 3–5, provide a financial-document setting and company-level dataset splits. These sources help select extraction components and evaluation units. Their table scores do not establish that a debt note has been interpreted correctly across years or that every off-balance-sheet exposure has been found.

The Docling report (Auer et al., 2024), Section 3, describes PDF parsing, page layout analysis and table structure recognition within a document-conversion pipeline. The locally archived version lists Christoph Auer first; this version difference is recorded in the catalogue. Docling is an ingestion candidate whose output must retain source locations and undergo accounting validation. Its code and model licences must be considered separately when packaging a deployment.

FinQA (Chen et al., 2021), Sections 3–6, provides questions with executable reasoning programs and supporting evidence. TAT-QA (Zhu et al., 2021), Sections 2–4, combines table cells and text spans with aggregation operations, and explicitly treats numerical signs in evaluation. DocFinQA (Reddy et al., 2024) extends financial questions to long documents and evaluates retrieval/context choices. FinBen (Xie et al., 2024) broadens the collection of financial tasks and model evaluations. These benchmarks can test retrieval and local numerical reasoning. Their question-answer format does not require a complete multi-period company representation,

joint alternative histories or scenario-dependent journal generation.

A directly relevant recent competitor is the auditable RAG architecture of Cosentino et al. (2026). Sections 3.1–3.2 describe persistent extraction, enrichment, retrieval and provenance records; Section 4.6 checks citation resolvability and failure recovery. Section 5 limits the interpretation of those results to the evaluated document-intelligence tasks. AccDSL should reuse the idea of persistent evidence and recoverable processing, while adding executable accounting and economic semantics. No performance comparison with that architecture has been run here.

## 4.6 Solvers and probabilistic execution

de Moura and Bjørner (2008), Sections 2–3, describes Z3's interfaces and theory-solving architecture. It supports choosing an SMT solver for logical choices and arithmetic constraints, but does not make arbitrary nonlinear simulation tractable. The first solver fragment is deliberately bounded: finite choices, exact linear arithmetic and selected calendar relations. Unsupported formulas remain in the representation with a solver diagnostic.

van de Meent et al. (2018) develops probabilistic programs through sampling, observations and inference; the inspected material includes the first-order language and weighted sampling discussion. Stan (Carpenter et al., 2017), its model language and sampler-output discussion, and Pyro (Bingham et al., 2019), Sections 2–3, show different ways to separate a probabilistic model from inference machinery. They are candidates for optional inference over an already-defined economic model. Exact accounting identities and discrete rule selection should not be turned into nearly satisfied likelihood penalties merely to fit an inference library. Nor should successful sampler execution be described as evidence that all interpretations were explored.

## 4.7 The remaining design problem

The combined literature provides algebra for journals, semantics for missing information, temporal storage, provenance, contract languages, rule languages, inference methods and extraction tools. The unresolved integration problem is to retain their distinctions while connecting them: a disclosure constrains economic histories; a history supplies contract and behavioural state; a scenario drives events; accounting rules produce journals; reporting rules render statements; and every step remains explainable.

The design is therefore an integration proposal with several local mathematical constructions. Its originality and empirical advantage have not been established by this survey. Its practical merit will depend on the acceptance cases, accounting review and comparative implementation work in Chapter 14.

# Chapter 5

# Architecture and reuse

## 5.1 One representation, several authoring methods

The architectural centre is a typed intermediate representation (IR), not a text parser. A document extractor, a human editor, a Python library and the AccDSL compiler all create the same records. This allows a modeller to begin with a reviewed structured file and later add convenient notation. It also allows source extraction to improve without changing accounting semantics.

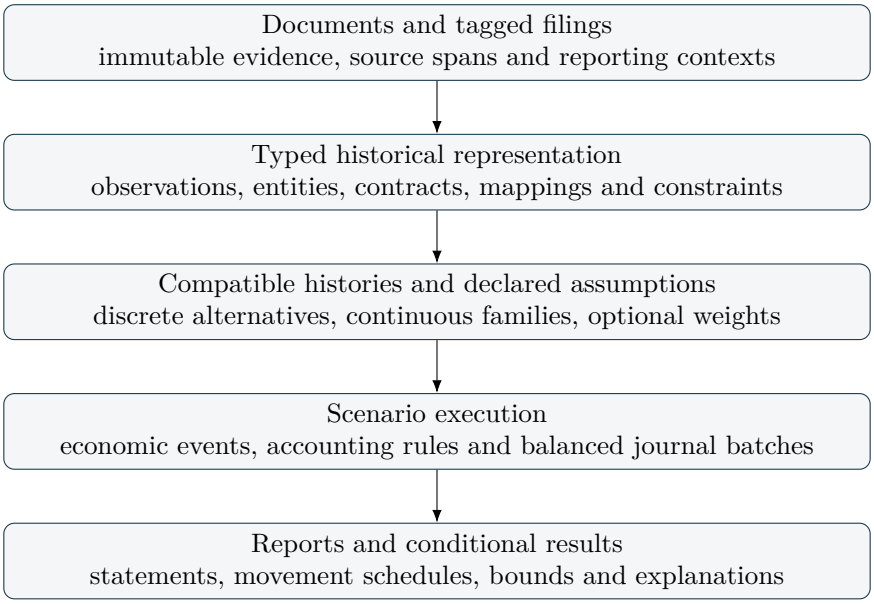



Figure 5.1: The simulation path. Every transformation records its inputs and assumptions; the source record remains available throughout.

Evidence storage and economic modelling must remain separate. If a company reports 100 of borrowing, a model can assume two loans of 60 and 40 without creating a new reported fact that the company disclosed those loans. The contract library consumes the assumed terms, the accounting module consumes the resulting events, and the explanation graph records the assumption at each affected result.

## 5.2 Why a DSL, and when another approach is better

The strongest baseline is a typed schema, SQL storage and ordinary Python code. That combination can implement the whole product. Its disadvantage is that repeated accounting requirements can become conventions scattered across functions: implicit currency conversion, missing period checks, hidden default values and duplicated rule selection. A bounded language makes these requirements part of typing and compilation. Its costs are language maintenance, diagnostics, migration and training.

| Approach | Useful role | Limitation for this product |
|---|---|---|
| Typed schema and Python | Reference executor, flexible economic models, initial authoring API | Domain restrictions require disciplined APIs or a validator; arbitrary code complicates static analysis and provenance. |
| XBRL and OIM | Tagged evidence, contexts, dimensions, taxonomies and exchange | Reported facts and reporting relationships do not recover hidden economic histories or specify scenario behaviour. |
| Ledger languages | Recording known transactions, balance checks, export and differential testing | Alternatives, source interpretation and contract exposure need additional semantics. |
| Knowledge graph / SQL | Relationships, lineage, temporal queries and evidence navigation | A graph or schema alone does not define event execution or the probability meaning of alternatives. |
| Rules / Catala | Dated policy logic, defaults, exceptions, traceable rule evaluation | Ledger state, event scheduling and ensembles require an enclosing runtime. |
| Probabilistic program | Priors, likelihoods and inference over model parameters | Needs declared probability measures; cannot turn unspecified ambiguity into probability or supply accounting rules. |
| Spreadsheets | Familiar review and small transparent financial models | Versioned evidence, shared unknowns and branching histories are difficult to govern as the model grows. |
| AccDSL over shared IR | Common domain checks, reusable templates, explicit alternatives and deterministic posting semantics | Requires implementation and evaluation; broad notation without stable semantics would add cost. |

The decision is to implement the IR, validators and Python reference API first, then the textual frontend. The same conformance cases must pass through both frontends. If the DSL fails to reduce ambiguity in representative authoring tasks, the IR and library remain useful. This is a reversible product decision, not a claim that a new language is inherently better than a library.

## 5.3 Open-source components

The following survey uses official repository descriptions, licence files and selected source interfaces archived on 2 October 2026. It is a source inspection, not an executed integration or performance evaluation. The machine-readable tool ledger records retrieval paths and available version metadata. Dependencies must be pinned when implementation begins; a branch name observed during this survey is not a release requirement.

| Component | Proposed use and checked boundary | Licence / decision |
|---|---|---|
| Arelle | Parse and validate XBRL/iXBRL. Inspected `ModelFact`, context and unit accessors. Preserve taxonomy extensions and full contexts when importing. | Apache-2.0 core; first-choice adapter. |
| Docling | PDF layout, text and table conversion. Inspected converter interface; retain page coordinates and extraction settings. | MIT code; assess model licences separately. Candidate. |
| Table Transformer | Table extraction baseline associated with PubTables-1M; check finance-domain performance before selection. | MIT; repository marked archived in inspected metadata. Baseline, not a maintenance default. |
| Beancount | Posting and transaction data types; export selected concrete branches and compare balances. | GPL-2.0 project metadata; integration and redistribution choice needed. Optional adapter. |
| Ledger | Known-journal reference and export. Inspected transaction balancing code and manual date/assertion behaviour. | BSD-style licence; optional reference tool. |
| hledger | Alternative ledger import/export and accounting checks. Repository documentation inspected. | GPL-3.0 project metadata; optional process adapter. |
| Catala | Potential backend for deterministic policy modules; defaults and exceptions have useful semantics. | Apache-2.0; prototype only after core execution semantics stabilise. |
| ACTUS | Contract schedules, payoffs and transitions; external risk-factor interface. Verify event schedules against ACTUS tests. | Service Apache-2.0; technical specification CC-BY-SA-4.0. Candidate contract adapter. |
| QuantLib | Calendars, cash flows and valuation. Floating coupon source separates fixing dates and pricing. Accounting treatment remains external. | Modified BSD-style licence; candidate valuation adapter. |
| OpenFisca | Versioned tax-benefit formulas as a rule-system reference; inspected variable/formula metadata. Corporate-tax coverage must be supplied. | AGPL-3.0; optional, not the core runtime. |

| Component | Proposed use and checked boundary | Licence / decision |
|---|---|---|
| Z3 | Finite alternatives, logical compatibility, linear constraints and conflict explanations. | MIT licence text; first solver candidate. |
| DuckDB | Local analytical queries over extracted facts and results. | MIT; candidate alongside SQLite transactional storage. |
| Stan | Continuous probabilistic models and diagnostics with an explicit target. Inspected model interface and paper. | BSD-3-Clause; optional inference backend. |
| Pyro / PyMC | Alternative probabilistic backends for model-specific inference. Pyro design paper and project descriptions inspected. | Apache-2.0 / Apache-2.0 licence texts; optional, no default selection. |

The licence entries identify the inspected projects, not a legal conclusion about every combination of code, models and datasets. The default product architecture can communicate with optional tools through explicit adapters. This does not, by itself, settle redistribution obligations. That question is resolved when a concrete dependency and distribution method are selected.

XBRL's Open Information Model supplies standardised facts with concepts, entities, periods, units and dimensions (XBRL International, 2021). Those facts are an excellent ingestion source. AccDSL retains their original contexts and decimals rather than flattening them immediately into a company-by-year table. Calculation relationships can help validate reporting totals, but are not a general description of the transactions that generated those totals.

## 5.4 Responsibilities that AccDSL must implement

AccDSL owns the evidence and temporal model, typed accounting concepts, shared unknowns and alternatives, rule-package selection, journal validation, consolidation provenance and report explanations. It also owns the distinction between feasible sets and distributions. These responsibilities cannot be delegated implicitly to whichever parser or solver is convenient.

External components receive narrow requests and return typed responses. A contract adapter returns scheduled events, cash-flow quantities, state changes and source/model references; it does not directly mutate the ledger. A valuation adapter returns a measure, currency, date, method and numerical status; the accounting package decides whether and where to recognise the change. A solver returns feasibility or bounds with scope and completeness status. A language model proposes source-linked mappings or interpretations; it cannot commit an unsupported assertion as fact.

The architecture consequently permits deterministic reference execution, alternative solvers and later acceleration without changing the meaning of a historical observation or a journal entry. Performance work begins only after the same semantic cases pass through these interfaces.

# Chapter 6

# Historical compatibility and future simulation

## 6.1 Define what is observed before inverting it

Let $D_k$ be the disclosures available under information cutoff $k$. Each numeric observation $d_i$ has a context: entity or group perimeter, concept, measurement basis, dimensions, currency, period and precision. These contexts are part of the observation operator, not additional columns that can be discarded after extraction.

A candidate history $h$ contains an opening economic and accounting state, events through the last historical date $T$, contract terms, accounting policies, consolidation relationships and latent quantities. Let $M$ denote the declared model scope: its event templates, admissible rule versions and allowed unknowns. It excludes arbitrary events that the chosen model has no way to represent. The reporting operator $O_i(h)$ produces the quantity to which observation $i$ refers, including its basis and perimeter.

Reported numbers are often rounded. If a number $v_i$ is rounded to the nearest unit $q_i$, an illustrative admissible interval is $[v_i - q_i/2, v_i + q_i/2)$ under the corresponding tie convention. This half-open endpoint choice is used only when that tie rule is declared; Section 3.11 gives the conservative closed enclosure when nearest rounding is assumed but the tie rule is unknown. The actual rounding convention, scale and XBRL accuracy metadata govern the imported constraint. A disclosure with unspecified precision does not acquire a convenient tolerance silently. Write $\mathcal{I}_i$ for the justified observation interval, and $C_M(h)$ for the joint accounting, contractual and semantic constraints. Then the historical compatibility set is

$$\mathcal{H}_M(D_k) = \{h \in \mathcal{H}_M : C_M(h) \ \wedge \ O_i(h) \in \mathcal{I}_i \text{ for every applicable } i\}. \tag{6.1}$$

Qualitative assertions enter as reviewed logical conditions, rather than numeric intervals. Incompatible source contexts are never forced into the same equation.

Equation (6.1) distinguishes three outcomes. An empty set means that the observations and declared model are inconsistent. A singleton means that this model identifies a history at its chosen resolution. A larger set preserves unresolved alternatives. An empty set does not prove that the company's accounts are false: a wrong extraction, missing event type or incorrect perimeter can cause the inconsistency. A singleton does not prove that the model contains the real history.

## 6.2 Why published statements cannot identify all transactions

The simplest derivation uses fixed account mappings and a fixed basis. Let $z \in \mathbb{R}^n$ contain unknown historical movement amounts. For $m$ disclosed aggregates, let $A \in \mathbb{R}^{m \times n}$ be the aggregation matrix and $y = Az \in \mathbb{R}^m$. If there is a nonzero vector $v$ with $Av = 0$, then

$$A(z + v) = Az + Av = y.$$

Both movement vectors give the same reports whenever both satisfy the remaining constraints. Thus the feasible family is the affine set $z + \ker A$ intersected with those constraints. Accounting requirements can shrink this family, but do not generally eliminate it. For example, cash collections in a receivables bridge are constrained by

$$AR_T = AR_0 + \text{credit sales} - \text{collections} - \text{write-offs} + \text{other classified movements}.$$

If the note does not distinguish acquisitions, currency effects and write-offs, several collections paths may fit the same closing balance.

There is a second ambiguity beyond aggregation. In Equation (1.1), identical historical movements arise from different contractual reset terms. Even recovering $z$ exactly would leave these terms unidentified. AccDSL therefore infers economically relevant states and contract terms alongside journal movements. A linear inverse solver over statement lines alone cannot solve this second problem.

## 6.3 The accounting state and its invariants

For one entity book with $n_a$ accounts in one functional currency, let $b_t \in \mathbb{Q}^{n_a}$ be signed account balances, with debit amounts positive and credit amounts negative. Exact decimal amounts can be represented as rationals with a power of ten denominator. A journal batch $j_t$ contains the signed functional-currency amounts posted by one atomic event or explicitly grouped event batch. Its balance condition is

$$\mathbf{1}^\mathsf{T} j_t = 0, \qquad b_t = b_{t^-} + j_t. \tag{6.2}$$

If $\mathbf{1}^\mathsf{T} b_0 = 0$, substitution gives $\mathbf{1}^\mathsf{T} b_t = \mathbf{1}^\mathsf{T} b_{t^-} + \mathbf{1}^\mathsf{T} j_t = 0$. Induction establishes conservation through every committed batch. Opening balances must therefore be supplied as a balanced, explained entry or a balanced imported state with unresolved composition explicitly identified.

This proof establishes a bookkeeping invariant. It does not establish that an amount belongs in the chosen account, that a contract exists, or that an accounting standard permits a treatment. Posting validation additionally checks account roles, ownership, currency conversions, applicability of rules and event preconditions.

The balance vector includes temporary revenue and expense accounts during the period. Period closing transfers their net balance to equity through balanced entries. A reporting function can include current profit in equity before closing, or use the closed book afterwards, but must not include it twice. Income and cash-flow reports use classified movements and exclude mechanical closing transfers from operating flows. Consequently statements are functions of balances, journals, contract state and reporting rules:

$$Y_{[u,v)} = R(b_{v^-}, J_{[u,v)}, X_{v^-}, \rho, [u, v)). \tag{6.3}$$

Here $\rho$ specifies the reporting basis and perimeter, and $X$ contains economic state beyond ledger balances. Cash-flow classification is part of $R$, not recoverable from the closing cash balance alone.

Each book has its own functional currency. Quantities such as shares, tonnes and foreign-currency units are typed separately and never added to this balance equation. A foreign-currency posting stores its original units and the functional-currency valuation with its rate source. Translation into a group's presentation currency and consolidation adjustments occur in a separate book under explicit rules.

## 6.4 Simulate events from economic state

For a chosen history $h$, the state at $T$ is $X_T(h)$. It contains balances, open contracts, lots or cohorts, unpaid accruals, commitments and other quantities needed for future evolution. Let $s$ identify a scenario definition, $U_t^s$ its supplied economic drivers, $a_t$ management's actions, and $\varepsilon_t$ future innovations. A behavioural policy $\pi$ chooses $a_t$ using the permitted information $\mathcal{F}_t$:

$$a_t = \pi(X_t, U_{\le t}^s, \mathcal{F}_t; \theta), \tag{6.4}$$

$$e_{t+1} = \mathcal{G}(X_t, a_t, U_{t:t+1}^s, \varepsilon_{t+1}; \theta), \tag{6.5}$$

$$(j_{t+1}, m_{t+1}) = \mathcal{A}(e_{t+1}, X_t, r_{t+1}), \tag{6.6}$$

$$X_{t+1} = \mathcal{T}(X_t, e_{t+1}, j_{t+1}, m_{t+1}). \tag{6.7}$$

The generator $\mathcal{G}$ emits events, such as a sale, rate reset, payment or default. The accounting function $\mathcal{A}$ applies rule package $r$ and returns journal postings plus measurement annotations $m$. The transition $\mathcal{T}$ updates economic and ledger state atomically. A default can therefore change the loan's legal status, expected recoveries, recognised allowance and cash collections through related, but distinct, records.

This ordering is a reference abstraction. Within a date, event dependencies and phases specify the actual order. Expectations used to measure an allowance or provision must be calculated from the information permitted at that date. They cannot use the realised future path of the simulation. An inner expectation model may use quadrature, analytic formulas or nested simulation; its approximation and seed belong in the measurement annotation.

Some scenarios specify a deterministic macroeconomic path while leaving idiosyncratic defaults random. Others supply a joint stochastic driver model. A risk-neutral measure used to value a derivative is distinct from the real-world measure used to simulate business outcomes. Every stochastic adapter declares its measure and conditioning information. The accounting function receives the resulting valuation with that declaration.

## 6.5 Preserve sets and distributions separately

Let $K_{h,s}$ be the probability law of future reports produced by the forward model conditional on history $h$, scenario $s$ and its declared behavioural assumptions. A deterministic simulation is the special case of a point mass. When no probabilities over historical explanations are justified, the natural answer is a family of laws,

$$\mathcal{K}_{M,s}(D_k) = \{K_{h,s} : h \in \mathcal{H}_M(D_k)\}. \tag{6.8}$$

The system can report each explanation, bounds on a deterministic outcome, or lower and upper event probabilities across this family. Those are different result types. A range of conditional means is not a confidence interval, and an envelope of conditional quantiles is not itself a single predictive distribution.

If a defensible probability measure $\mu(dh \mid D_k, M)$ supported on $\mathcal{H}_M(D_k)$ is supplied, histories and report outcomes must have specified measurable spaces. Assume that $K_{h,s}$ is a probability

kernel: it is a probability measure on report outcomes for each $h$, and $h \mapsto K_{h,s}(B)$ is measurable for each measurable event $B$. The predictive law is then

$$\mathbb{P}(Y^s \in B \mid D_k, M) = \int K_{h,s}(B)\, \mu(dh \mid D_k, M). \tag{6.9}$$

For discrete explanations this is the usual weighted sum. For continuous unknowns, integration uses the declared measure over those unknowns, not a count of solver solutions. Bayesian weights require a prior and a likelihood; expert weights remain labelled as elicited. A likelihood must not treat a statement total and all its components as independent repeated evidence. Deterministic equality constraints require a reduced parameterisation or proper conditional-measure treatment, rather than an undocumented Gaussian penalty with near-zero variance.

Suppose the fixed-loan explanation receives weight $w$ and the floating-loan explanation receives $1 - w$, solely as an illustration. Under the rate shock,

$$\mathbb{P}(I = 5) = w, \qquad \mathbb{P}(I = 7) = 1 - w, \qquad \mathbb{E}[I] = 7 - 2w.$$

The expected expense is useful but hides the two different outcomes. For a general scalar output with finite second moment, put $m(h) = \mathbb{E}[Y \mid h]$. Write $Y - \mathbb{E}[Y] = (Y - m(h)) + (m(h) - \mathbb{E}[Y])$. The cross term vanishes because $\mathbb{E}[Y - m(h) \mid h] = 0$. Taking expectations of the squared expression gives

$$\mathrm{Var}(Y) = \mathbb{E}_h[\mathrm{Var}(Y \mid h)] + \mathrm{Var}_h(\mathbb{E}[Y \mid h]). \tag{6.10}$$

The first term is uncertainty within a chosen explanation; the second captures differences between explanations. This decomposition requires the measure $\mu$. Without it, the system reports the family in Equation (6.8), rather than manufacturing a variance.

Weights themselves can be uncertain. For an admissible class $\mathcal{W}$ of measures, AccDSL can compute

$$\underline{p}(B) = \inf_{\mu \in \mathcal{W}} \int K_{h,s}(B)\, d\mu(h), \qquad \overline{p}(B) = \sup_{\mu \in \mathcal{W}} \int K_{h,s}(B)\, d\mu(h).$$

If every measure supported on the feasible histories is allowed and point masses are included, these bounds reduce to the infimum and supremum of $K_{h,s}(B)$ over histories: averages cannot lie outside the range of their integrands, and point masses attain, or approximate, the extremes. Narrower weight restrictions require an explicit justification and a separate optimisation problem.

## 6.6 Resolution and computational limits

For a requested horizon $H$ and scenario family $\mathcal{S}$, histories may be grouped only when they have the same conditional law for all requested outputs:

$$h_1 \sim_{H,\mathcal{S}} h_2 \quad \Longleftrightarrow \quad K_{h_1,s} = K_{h_2,s} \text{ on those outputs for every } s \in \mathcal{S}.$$

This definition explains why individual sales can sometimes be replaced by cohorts, while fixed and floating loans usually cannot be merged for an interest-rate stress. Exact equivalence can be difficult to prove. A grouping justified only by approximate numerical agreement carries an error criterion and is not labelled exact. Source provenance remains available even when execution shares a common state.

Bounds require comparable honesty. A linear program can provide certified extrema for a supported linear model. A set of feasible sampled histories provides observed extrema: its minimum is no smaller than the true infimum, and its maximum is no larger than the true supremum. It therefore supplies an inner exploration of possibilities, not a certified outer bound.

Interval methods may provide outer bounds that no single history attains. Every result records which of these meanings applies.

The initial implementation offers exact bounded choices and linear constraints, symbolic continuous families such as $\alpha \in [0, 1]$, and explicit scenario execution at selected feasible states. General nonlinear partial identification, automatic model discovery and exhaustive enumeration of all economic histories are research extensions, not promises made by the core language.

## 6.7 A solver formulation for ordinary rounded disclosures

The debt example used an exact principal. Real disclosures often do not. Suppose debt is shown as 100 and interest as 5, each rounded to the nearest unit with the declared half-open intervals $D \in [99.5, 100.5)$ and $I_0 \in [4.5, 5.5)$. Using an unknown fraction $\alpha$ and unknown total $D$ would introduce the product $\alpha D$. That expression is bilinear and is outside the promised linear fragment.

Instead use principal amounts $F$ and $V$. The compatible family satisfies

$$F, V \geq 0, \quad 99.5 \leq F + V < 100.5, \quad 4.5 \leq 0.05(F + V) < 5.5. \tag{6.11}$$

Future interest is $I_1 = 0.05F + 0.07V$. Its infimum is $0.05 \times 99.5 = 4.975$, attained with $F = 99.5, V = 0$. Its supremum is $0.07 \times 100.5 = 7.035$, approached as $F = 0$ and $V$ tends to 100.5 from below, but not attained. A closed interval is a valid enclosure, but its upper endpoint must not be reported as attainable. The exact solver check separately tests that no better value exists and whether each endpoint is attainable.

For $D > 0$, the amount representation is equivalent to the fraction representation through $D = F + V$ and $\alpha = F/D$. At zero principal the fraction is undefined and must be omitted or treated as irrelevant, not recovered. A known fraction restriction $\alpha \geq a$ becomes $F \geq aD$ when $D > 0$ and $a$ is fixed. If both $a$ and $D$ are unknown, that product remains nonlinear. Reformulation has a domain of validity; it is not permission to send arbitrary formulas to a linear solver.

The supported exact fragment contains rational linear equalities, weak and strict inequalities, and finite Boolean or enumeration choices. Guard partitions become finite disjunctions. Posting in integer cents introduces a different integer problem if performed before bounding. Consequently the request specifies whether bounds apply before quantisation, after a monotone declared rounding map, or to a bounded integer model. Approximate numerical valuations carry error metadata outside the exact rational kernel. Solver timeout, unsupported theory and unbounded objective have distinct responses; none is relabelled as infeasibility.

## 6.8 Building the family with explicit templates

The reconstruction input is a model-scope record, not an invitation to invent unlimited journals. It lists event templates, slots per entity and period, allowed contract classes, source mappings, opening-state domains, calendars, resolution and excluded operations. A template gives an incidence vector for postings and a transition for economic state. Northstar's annual bridges arise by substituting its collection, purchase and payment templates into opening-plus-movement identities.

For each allowed slot, create an activity flag and nonnegative amount; an inactive slot has zero amount. Instantiate the template's posting and state constraints, unify variables referring to the same observation or economic contract, and match report expressions to source domains

through validated context mappings. Add cross-note restrictions and choice-group dependencies to this same formula. No separate account-by-account reconstruction is allowed. Mark every constraint with its evidence, rule or assumption origin.

Enumerate the finite structural choices in a deterministic order. For each cell, ask for feasibility and retain the continuous formula, rather than enumerating real-valued solutions. An infeasible cell preserves a sufficient conflicting subset; an undecided cell preserves its formula and reason. Search is complete within the declared scope only when every finite cell has been decided, every applicable observation has been matched or explicitly qualified, and no resource truncation remains. A witness establishes existence in one cell. It does not establish complete search, and complete search does not establish that the scope includes the company's real history.

In Northstar the declared templates identify annual collections, purchases and supplier payments, but retain every $F \in [0, 100]$. Adding a verified note that $F = 60$ narrows this family by conjunction. Adding a proposed "all floating" reading to that revised view produces a rejected cell with the conflicting note attached. Adding a rule that expenses a capital contribution would instead fail treatment admission: inconsistent evidence and unsupported accounting have different diagnoses.

## 6.9 Which details may safely be combined

A resolution record names the outputs, horizon, scenario family, grouping key, required state, transformation and exact or approximate status. Safe exact transformations are deliberately few. Cash balances in the same book and currency can be added. Homogeneous inventory lots may be combined under a weighted-average policy if ownership, cost basis, impairment state and future costing treatment coincide. Debt principal can be added only within a group sharing currency, coupon/reset function, fixing and payment calendars, maturity, options, guarantees and measurement method. The transition and reporting maps must be additive on that group.

The last condition supplies a short proof obligation. If a transition obeys $T(x_1 + x_2, u) = T(x_1, u) + T(x_2, u)$ and reporting is additive for every admitted driver $u$, induction preserves equality after grouping. This does not apply to covenant thresholds, counterparty netting, nonlinear guarantees or lot-specific impairment. Two fixed loans with the same dates and rate can share an execution state under this test. The fixed/floating pair cannot; neither can two policies with different guarantees merely because their current reserves match.

A changed horizon, scenario, rule version or requested output invalidates the resolution approval and reruns the sufficiency check. Extending beyond a debt maturity requires schedule detail that an earlier short-horizon view could omit. Approximate cohorts retain the detailed source references and a stated error criterion tied to outputs. Unless that error is bounded or evaluated, the grouping remains an approximation awaiting evidence, not an exact quotient of the history family.

## 6.10 A finite probability model that can actually be evaluated

Optional probability weighting begins with disjoint economic hypotheses and a declared joint measure. As an illustration, assign fixed and floating hypotheses prior probabilities 0.6 and 0.4. The stress itself leaves these weights unchanged and gives mean interest $0.6(5) + 0.4(7) = 5.8$. Suppose genuinely new observed evidence has likelihood 0.2 under fixed and 0.8 under floating. The unnormalised posterior masses are 0.12 and 0.32; normalisation by 0.44 gives 3/11 and 8/11. These likelihoods are illustrative inputs, not extraction-confidence scores.

The finite reference algorithm enumerates joint hypothesis and innovation states, evaluates their declared joint mass and any observation likelihood once, checks the nonzero normalising constant, and propagates the entire statement vector on each state. It sums masses only across disjoint events. Two overlapping labels such as “fixed share at least one half” and “all fixed” are query views and cannot become two independent mixture components. Totals and their components share an observation block so their evidence is not counted twice. Continuous inference needs an additional parameterisation, density or conditional measure and an assessed numerical method; the finite algorithm supplies no guarantee for those extensions.

Under the prior, before the new observation is admitted, if the floating branch has an unsupported valuation operation, its mass 0.4 remains unresolved. The run may report the fixed result with successful mass 0.6, but cannot report 5 as the unconditional mean by renormalising away the failure. A separately requested mean conditional on successful execution must show that conditioning event and its mass. For unweighted families, an unresolved branch blocks a complete envelope over the family. Economic default is retained as a model outcome with remaining obligations, rather than being treated as a computational error to remove.

## 6.11 Numerical precision is part of a financial method

The affine family is useful for Northstar because the scenario fixes prices, rates and operating coefficients while the debt share remains unknown. Other contracts require roots or discount factors. Their numerical treatment must state what is exact, what is enclosed and when a posted amount is determined. An accounting name on a library call does not make its result exact.

Consider positive payments $C_i$ at positive times $t_i$ and a positive price $P$. The effective annual rate solves

$$f(r) = \sum_i \frac{C_i}{(1+r)^{t_i}} - P = 0, \qquad r > -1.$$

Each term discounts a payment to the valuation date. In this positive-payment case,

$$f'(r) = -\sum_i \frac{t_i C_i}{(1+r)^{t_i+1}} < 0.$$

Consequently a certified sign-changing bracket contains exactly one root. The method uses stated bracket bounds and an expansion cap, evaluates signs with outward error bounds, and bisects until the requested rate or value enclosure is reached. An uncertified sign, unavailable bracket or exhausted accuracy budget yields an unresolved numerical result. Mixed-sign cash flows can have several internal rates of return; the preceding monotonicity argument does not apply and cannot justify returning the first Newton iterate that converges.

Net proceeds 100 and a payment 121 in two years give exactly 10%. An immediate fee of 2 changes proceeds to 98 and changes that root. Annual-effective 21% for half a year has the exact accumulation factor $\sqrt{1.21} = 1.1$, so interest on 100 is 10. Simple proration would give 10.5 and computes a different contract. At annual-effective 10%, the half-year factor $\sqrt{1.1}$ is irrational. For rational time $p/q$ and positive $1 + r$, exact rational power comparisons can enclose the positive $q$th root of $(1 + r)^p$. A negative $p$ first requires the reciprocal. General exponential discounting needs a separately specified error-controlled method.

A posting quantum of 0.01 does not eliminate this uncertainty. The whole certified monetary enclosure must lie in one rounding cell under the admitted tie rule. An enclosure from 1.0049 to 1.0051 crosses a half-cent boundary and does not establish one posted amount. Refine within a declared budget or retain rounding unresolved. This rule separates evidence for a numerical answer from the accounting authority to use that answer.

## 6.12 Updating a law without counting evidence twice

Two financial histories can be compatible with the same reports without being equally probable. Once an explicitly justified law is supplied, genuinely new evidence may change it. For finite histories $h$, prior mass $p(h)$ and an admitted observation likelihood $L(D \mid h)$, Bayes' rule gives

$$p(h \mid D) = \frac{p(h)L(D \mid h)}{\sum_g p(g)L(D \mid g)}.$$

The denominator is the probability of that evidence under the declared model. If it is zero, the condition is impossible in that model; dropping incompatible histories and renormalising an empty collection cannot repair it.

For two histories with assumed prior masses 1/3 and 2/3 and likelihoods 3/4 and 1, the unnormalised masses are 1/4 and 2/3. Their sum is 11/12, so the posterior masses are 3/11 and 8/11. If the second history's cash calculation fails, the first still has mass 3/11 and unresolved cash retains mass 8/11. A report giving the successful history probability one would answer a different, success-conditioned question.

An observation has a semantic information identity as well as a file locator. Reimporting the same observation through another file does not create a second measurement. Repeating an identical admitted condition is idempotent: it keeps the law and information lineage unchanged. Submitting an already consumed observation as a fresh likelihood block is rejected as duplicate information. Equal selected member sets alone do not prove that two conditions have the same meaning. An admitted conditional kernel may divide a region's existing mass when it is refined; the number of new representatives never determines that division.

Continuous inference adds a measure choice. One must specify coordinates $z$, domain, base measure, prior, observation mechanism and any transformation Jacobian. Stuart (2010, Section 2.2, equations (2.5)–(2.8)) supplies the inverse-problem formulation, but does not establish identification or easy sampling for a mixed accounting model. In particular, an exact accounting equality can confine a model to a lower-dimensional set. Multiplying an ordinary Lebesgue density by its zero-volume indicator is not a posterior construction. One must parameterise the constrained model with its declared measure or supply an actual measurement-error model. Adding convenient small noise changes the target.

## 6.13 A continuous example whose posterior can be checked

A linear Gaussian model supplies a tractable comparison for later numerical inference. Assume the unknown economic quantity has a Gaussian prior $z \sim \mathcal{N}(m_0, V_0)$ with positive definite covariance $V_0$. Let the observation be $y = Hz + \epsilon$, where $\epsilon$ is independent Gaussian noise with positive definite covariance $R$. The measurement-error assumption is substantive: it describes this example, not rounding in every financial statement.

Multiplying prior and likelihood adds their quadratic forms. Terms that depend on $z$ are

$$\begin{aligned}&(z - m_0)^\top V_0^{-1}(z - m_0) + (y - Hz)^\top R^{-1}(y - Hz)\\ &\quad = z^\top (V_0^{-1} + H^\top R^{-1} H)z - 2z^\top (V_0^{-1} m_0 + H^\top R^{-1} y) + \text{constant}.\end{aligned}$$

Completing the square therefore gives posterior covariance $V$ and mean $m$ through

$$V^{-1} = V_0^{-1} + H^\top R^{-1} H, \qquad V^{-1} m = V_0^{-1} m_0 + H^\top R^{-1} y.$$

The prior and observation contribute precision and precision-weighted information. An implementation should solve factored linear systems rather than explicitly form these inverse matrices, and reject an invalid covariance or unsupported rank condition.

For $m_0 = 0$, $V_0 = 4$, $H = 1$, $R = 1$ and $y = 2$, posterior precision is $1/4 + 1 = 5/4$, variance is $4/5$ and mean is $(4/5)2 = 8/5$. A conditional cash quantity $100 + 10z$ has mean 116 and variance 80. A second *independent* measurement of 2 would give precision $9/4$, variance $4/9$ and mean $16/9$. Reusing the first measurement would produce those same numbers but would be wrong for the stated information. The analytic example can distinguish numerical error from information double counting.

For a bounded non-Gaussian scalar model, quadrature requires a stated domain or tail truncation and sufficient integrand error bounds to enclose the normalizer and requested moments. Without a positive lower normalizer bound, the ratio remains unresolved. Mixed models can enumerate finite choices and integrate each conditional continuous model, but combining branches requires their evidence factors, not draw counts. Optional MCMC needs target-specific parameterisation, tuning, independent chains, tractable correctness comparisons, convergence and Monte Carlo error checks. This proposal supplies no generally validated sampler or universal tolerance.

Nor can a more elaborate inference engine identify a quantity that the observation model never sees. If reports constrain only total debt 100, the likelihood may be constant in its floating share $x$. Its conditional prior then remains unchanged. A uniform prior is a chosen assumption, not a discovery from those reports. The inspected working version of Giacomini and Kitagawa (2021, Sections 2.1–2.3, equations (2.2)–(2.4)) motivates retaining posterior classes where a conditional prior cannot be revised. For Northstar there is a simple direct deduction: under all probability measures on $[0, 1]$, the expectation of $5 + 2x$ lies in $[5, 7]$, with endpoints attained by point masses. This deduction imports no asymptotic theorem from that paper's structural-vector-autoregression setting.

## 6.14 A stress is not an observed condition or an intervention

The interest-rate stress in Northstar fixes a future driver path. It supplies no new evidence about the historical debt share. An observational forecast instead conditions a joint law on information. A causal intervention needs structural equations and an exogenous disturbance law, because it replaces the mechanism determining a chosen variable while leaving other mechanisms in place (Pearl, 2009, Section 3.2, equations (5)–(9) and (15)–(17)).

An elementary example makes the difference numerical. Let $U$ equal zero or one with probability $1/2$ each, and assume

$$X = U, \qquad Y = 2X + U.$$

Observing $X = 1$ establishes $U = 1$ and hence $Y = 3$. Intervening to set $X = 1$ replaces the first equation. It leaves the original law of $U$, so $Y$ is 2 or 3 with equal mass and mean 2.5. Selection of a high-investment company and an imposed investment policy can therefore answer different questions even under one fully specified model. Financial filings alone do not establish these causal equations or identify their effects.

An intervention request must name its target, replacement equation, structural parents and disturbance law and satisfy acyclicity or a separately supported equilibrium rule. Correlated disturbances remain joint. The initial company profile supports imposed exogenous stress; the continuous and causal methods described here require separately admitted extension schemas and signatures. They do not silently expand the initial finite-law interface.

At every decision date, behavior and measurement receive the admitted historical view and

permitted simulated past. The scheduler's knowledge of a complete future stress path is not permission for a manager to foresee it. A perfect- foresight policy needs an explicit additional assumption. Common random numbers can improve a paired numerical comparison while establishing no causal effect in actual company data.

# Chapter 7

# The accounting language

## 7.1 Scope of the first language

The historical AccDSL Core 0 proposal describes entities and books, accounts, observations, shared unknowns, constraints, alternative interpretations, contract instances, accounting rules and reporting mappings. It has pure expressions and a restricted posting effect. Economic forecasting algorithms can live in typed plugins; unrestricted numerical code does not need to be reimplemented in the DSL. The historical grammar in Appendix B fixes the preserved teaching examples. The proposed portable records and first company profile below define the new authoring boundary; the first matched text frontend supports the declared debt fragment.

The historical Core 0 compiler has six ordered stages: parsing with source locations; name and module resolution; type and temporal-context checking; lowering to the IR; constraint and rule-applicability analysis; and execution-plan construction. The compiler never silently supplies an economic assumption to make a program compile. A program can be valid as a historical representation while requiring additional assumptions before a particular simulation can run. The proposed portable compilation sequence is specified separately in Section 7.3.

The language's meaning is defined by its IR and evaluator, rather than by the surface spelling alone. Formatting changes cannot change a model's semantic hash. A changed source span or reviewer decision changes evidence provenance even if the numerical model remains the same. The system therefore maintains separate evidence, semantic-model and execution identifiers.

For the preserved Core 0 examples, the complete grammar is supplied as `spec/core-0.lark`; the original design directory retains `spec/core-ir.schema.json` and `spec/core-contract.md`. They define reference resolution, expression arity, type/context checks, operations, failures and identity. Structural schema validation alone does not establish those semantic properties. ASCII identifiers and UTF-8 strings are used in this revision; currency suffixes, dates and decimal literals have explicit lexical rules. The grammar is checked by parsing the complete demonstration file, not by claiming that its parser also implements a compiler or solver.

## 7.2 Types that protect meaning

The following table uses the preserved Core 0 type vocabulary. Its distinctions motivate the more precise portable types introduced below.

| Type | Meaning and checks |
|---|---|
| `Money<C>` | Exact amount in currency C. Addition requires identical currency; conversion requires a dated rate and policy. |
| `Quantity<U>` | Units of shares, goods or other commodities. Quantities cannot be added to money or used as a functional-currency posting. |
| `Rate<B>` and `YearFraction` | Rate with basis B, plus a separately computed accrual fraction. Simple and compound accrual functions are distinct. |
| `Instant`, `Period`, `Calendar` | Instant observations and flows over intervals; interval operations preserve calendar conventions. |
| `Account<K,Book>` | Account role K and owning book. Cross-book effects require separate balanced batches linked by an economic event. |
| `Observation<T>` | Source assertion with contextual and precision metadata; accessing its value retains its constraint and provenance. |
| `Symbol<T,D>` | Shared unknown with domain D; repeated references denote the same value in a history. |
| `Choice<E>` | A finite structural choice over enumeration E. No probability is implied. |
| `Distribution<T>` | Explicit probability measure with parameter, conditioning and measure metadata. It is not a Symbol or a Choice. |
| `SourceRef`, `RuleRef`, `ModelRef` | Versioned evidence or implementation references; unresolved references prevent concrete execution. |
| `Measurement<T>` | Value plus valuation date, basis, method, assumptions and numerical status. Recognition remains a rule decision. |

The portable profile spells the rate type `Rate<convention>` and makes both calendar and accrual convention explicit in `YearFraction<calendar, convention>`. `InstantValue<T>` and `Flow<T, [start,end)>` distinguish dated stocks from interval flows; their context carries the calendar. An approximate numerical component of a historical `Measurement<T>` uses `Approximation` with its method, error description and environment references. An exact measurement remains exact; valuation context and recognition authority are separate in either case.

Core expressions use exact integers and rationals, Boolean logic, enumerations, dates, records, lists, comparisons and registered pure functions. A decimal literal is parsed exactly, not first converted through binary floating point. Numerical plugins can return approximate measurements, but must declare their error or numerical status and the rounding used when posting money.

Missingness is a tagged type with distinct values: `notDisclosed`, `notApplicable`, `unreadable`, `withheld`, and `unresolvedMeaning`. Zero is an ordinary known number. A missing datum can be replaced by a declared Symbol or assumption in a model branch; the source observation stays missing. An unknown is not necessarily random, and an estimate is not necessarily missing. On portable import, `notDisclosed`, `unreadable`, `withheld` and `unresolvedMeaning` become `Unknown` with the corresponding reason and obligation references. `notApplicable` becomes the explicit `NotApplicable` variant, never an unknown or zero.

Addition of monetary observations also requires compatible entity, perimeter, time, dimensions and basis. These checks operate on contextual expressions in addition to the currency type. An analyst can deliberately aggregate dimensions or transform a basis using an explicit function.

For example, `sumOver(segment, ...)` records the dimension being removed and requires the segment partition to be declared; it does not assume every disclosed segment total is additive.

## 7.3 A portable record has an exact meaning

A constructor and a textual frontend can describe the same financial model only if they exchange values whose identity is independent of a Python object or an incidental JSON writer. The proposed portable record therefore has six fields: `kind`, `schema`, `key`, `revision`, `body` and `dependencies`. Kind and schema identify a registered version; key is its stable, schema-scoped logical name; revision is a nonnegative integer. The body obeys that schema's required, optional and nullable fields. Unknown fields reject.

A typed reference has exactly kind, schema and digest. The digest covers the complete target record. Every body reference is also listed in dependencies; additional dependencies need a schema-declared role. The submitted closure is finite and acyclic. A receipt can refer to its request and old state, but the old state cannot refer forward to that receipt. Stable business names can relate counterparties without creating a cycle of content references.

Financial quantities use a reduced rational with signed numerator text $n$ and positive denominator text $d$, with $\gcd(|n|, d) = 1$ and zero represented by $0/1$. The decimal token 2.50 therefore denotes $5/2$ exactly. Revisions, ordinals and integer counts remain integers; a Boolean is never an integer. Approximate measurements instead retain decimal text, method, error description and environment. A rounded decimal cannot silently acquire exact status.

An absent required field is invalid. Unknown financial content is an explicit value with a reason and outstanding evidence references. It can be retained, but cannot satisfy a method that requires the missing fact. Null is permitted only where the particular schema gives it a meaning; not applicable, unknown and zero are distinct. A candidate also differs from a value admitted in a specific immutable view. Admission checks the actual decision and its ancestors; wrapping a candidate in an "admitted" type is insufficient.

Portable bytes use UTF-8 JSON with ASCII object keys sorted by their byte order, no insignificant whitespace, no floats, duplicate keys, nonfinite numbers or invalid Unicode scalar values. Decoded strings retain their code points. An escaped Unicode character and the same literal character encode equally; composed and decomposed spellings remain different unless an explicit named transformation relates them. Global NFC normalisation would silently change some source text and is therefore not part of this encoding.

The writer emits literal non-ASCII scalars, escapes quotation marks and reverse solidus, and escapes control characters U+0000–U+001F. It uses the five named JSON escapes where defined and lowercase `\u00xx` for the remaining controls; solidus is not escaped. Integers use minimal base-ten spelling without plus signs, leading zeroes or negative zero. Dependencies, module declarations by key, dimensions by axis and explicit all-of/any-of operand sets are sorted after rejecting duplicates. Other arrays, including journal lines, event schedules and diagnostic priority, retain order.

With these rules, the record digest is

$$\mathrm{SHA256}\big(\mathrm{UTF8}(\text{“accdsl.portable.v1/”} + \mathrm{schema}) \parallel \texttt{0x00} \parallel \mathrm{canonical}(\mathrm{record})\big).$$

This recognises representational identity, not economic equivalence or source truth. Original documents retain a separate hash of their raw bytes. A blob reference names that byte hash, algorithm and length; it is not a typed record reference. A submission record links those bytes to media type, transport and decoder, preserving a malformed request without pretending it was valid.

Historical Core 0 and replay examples keep their original encodings and IDs. A declared legacy import retains original bytes, decoder, old identity, portable output and field-by-field comparison. If one historical record maps to several portable records, an import mapping names every output and conversion. Import does not grant financial admission in a new view. This separation permits old examples to remain reproducible while the proposed portable representation becomes precise enough for two implementations to exchange.

## 7.4 The first company profile and its authoring boundary

The proposed `CorporateJointHistory/1` profile connects one legal entity and book per branch to annual/interim sources, revisions, working capital, payroll, tangible assets, simple financing, an explicitly declared tax model and complete statements. Its types preserve entity, perimeter and currency even where an operation is outside this initial scope. A real company cannot become eligible for the profile by dropping its subsidiaries, leases or contingencies.

Typed constructors must express every included declaration in the portable IR without opaque Python functions. The initially matched text frontend covers the bounded debt and constraint fragment. Whole-company textual authoring remains a subsequent frontend obligation; the whole-company typed model is part of the initial domain. Thus domain scope and authoring syntax are separate decisions. The following Core 0 listing teaches the shared debt family under its preserved historical grammar. It is not a promise that the first new parser supports every historical declaration. The matched portable debt example uses floating share $x$, while the following listing uses fixed share $\alpha$. The substitution $x = 1 - \alpha$ preserves the entire family, including its endpoint results.

Backend admission examines expanded expressions. Multiplication of an unknown quantity by an unknown price is bilinear even when hidden inside a function called revenue. The initial exact reconstruction fragment admits rational affine constraints, finite choices and bounded integers under explicit templates. Unsupported functions return a capability diagnostic. A type-correct historical representation may therefore be retained even when a requested forecast still requires an unavailable method or an additional admitted assumption.

The portable `CompileRequest` supplies source kind, source bytes or a typed module, schema and profile references, the full package lock, evidence/time view and requested capability target. Its stages are strict decoding and source locations; name/import resolution; dependency and cycle checks; static type and context checks; admission-layer checks; expanded-operation capability analysis; and canonical record production. The earlier six-stage workflow describes the historical Core 0 frontend; it does not add execution planning to this portable compiler or permit it to call a solver.

The result contains candidate IR, source map, unresolved inputs, capability analysis and a validation receipt bound to every input. Invalid structure or types yield no executable IR. Valid structure with an unresolved financial input can yield a retained candidate whose affected target has execution admission false. Source maps retain decoded Unicode code-point half-open ranges for text or constructor paths for typed input. Formatting can change source-map identity without changing financial-model identity.

For a matched example, both authoring routes must describe debt 100, floating share $x \in [0, 1]$ and conditional interest $100((1 - x)/20 + 7x/100) = 5 + 2x$. Endpoints are 5 and 7; $x = 1/2$ gives 6. Changing the floating assumption from 7% to 9% changes the upper endpoint to 9. Independent arithmetic is needed because two frontends could share the same wrong constant. Agreement between them alone does not establish correctness.

## 7.5 A small model with a continuous interpretation family

The following source uses a synthetic observation whose full context is stored in the evidence registry. The example imports the demonstration debt package, which uses one-year simple accrual, no fees and immediate rate resets. It is not an IFRS rule package.

All dependencies are instantiated in `fixtures/demo-debt-package.json`: the source reading, balanced opening state, currency precision, contract state and dates, reset and interest operations, closing and report mappings, request and expected outputs. `fixtures/borrowing.acc` is the complete source, including the retained earnings account and an `opening` declaration. The shorter listings below explain its declarations in stages. They do not replace that full source.

```
module BorrowingExample version "1";
use DemoDebt version "1";
source note7 = external("synthetic-note-7");
entity Acme functional USD;
book Local entity Acme basis DemoDebt version "1";
account Cash : Asset book Local;
account Debt : Liability book Local;
account InterestExpense : Expense book Local;

observation debt : Money<USD> instant @2025-12-31
  = 100 USD from note7;
symbol alpha : Real in [0, 1]
  because "The fixed-rate proportion is not disclosed";
```

The remaining declarations use this same unknown fraction in both loan components. A change to the fraction therefore changes the whole debt mix consistently.

```
let fixedPrincipal = alpha * value(debt);
let floatingPrincipal = (1 - alpha) * value(debt);
constraint split:
  fixedPrincipal + floatingPrincipal == value(debt);

contract loans : DebtMix book Local {
  principal = value(debt);
  fixed_fraction = alpha;
  fixed_rate = 0.05 per year;
  initial_floating_rate = 0.05 per year;
}
alternative allFixed { assume alpha == 1; }
alternative allFloating { assume alpha == 0; }
scenario rateShock {
  set market_rate = constant(0.07 per year);
}
```

The parent model retains every $\alpha \in [0, 1]$. The two named alternatives are useful conditional views of that family, not a declaration that the endpoints exhaust it. Requesting the parent family's exact bounds produces $[5, 7]$ for the next year's expense under `rateShock`. Requesting a conditional run of `allFixed` supplies $\alpha = 1$. Requesting a weighted mixture requires a separate weight model; the two labels do not create equal weights.

An observation declaration is a reference to evidence plus a proposed typed reading. The compiler checks that the registry context agrees with the declared period and type. If the evidence is rounded, `value(debt)` denotes a shared constrained value rather than a silently exact 100. This synthetic registry explicitly declares an exact value, which keeps the illustration deterministic.

## 7.6 Posting rules and explicit exceptions

An event type declares the information needed by a rule. The example below assumes an event representing interest accrued and paid on the same date. Separate accrual and settlement events are needed when those dates differ.

```
event InterestPaid(principal: Money<USD>,
                   rate: Rate<annual_simple>,
                   fraction: YearFraction) book Local;

rule recogniseInterest for InterestPaid as e
when e.principal >= 0 USD {
  let charge = roundMoney(simpleInterest(e.principal, e.rate, e.fraction),
                          currencyScale(USD), "half_even");
  require charge >= 0 USD;
  post {
    debit InterestExpense charge;
    credit Cash charge;
  }
}
```

The rule cannot perform file I/O, mutate another entity or sample a random number. It returns a proposed batch. The runtime verifies balance and all preconditions before committing state. Negative interest requires an explicit supported rule; it is not mishandled by this intentionally restricted example. Rounding is a declared demonstration convention, not a universal accounting default.

Rules can declare `overrides ruleName`. The resulting override graph must be acyclic. For a concrete event, the evaluator determines all applicable rules and removes any that are overridden by another applicable rule. Exactly one maximal rule must remain for a required recognition operation. An empty set of applicable rules produces `UnsupportedTreatment`; more than one maximal rule produces `RuleConflict`. File order and a hidden numeric priority never break ties.

If a guard depends on a symbolic value, the solver partitions the current feasible family into the guard and its negation, subject to the existing constraints. This is a logical partition, not a random draw. An incomplete solver returns `UnknownApplicability`, not a guessed rule. Statically provable disjointness or total coverage can be cached; otherwise the runtime performs the same checks on every concrete event.

## 7.7 Alternative readings, assumptions and updates

An alternative adds constraints and possibly a supported semantic mapping to its parent model. Its record contains a rationale, supporting spans, opposing evidence, author, creation time and scope. Child alternatives inherit all parent constraints; they cannot quietly discard an inconvenient disclosure. A deliberate source rejection is a separate review decision with an explicit reason and a new evidence view.

Two independent alternatives may coexist, while two mutually exclusive readings of the same text require an explicit choice group. The representation stores these relationships in a constraint graph. It does not take a Cartesian product blindly. For example, interpreting an instrument as a liability affects both its balance-sheet account and the treatment of its periodic payments; those choices must be linked through one shared treatment variable.

New evidence narrows an existing family by conjunction when it concerns the same definitions and reporting basis. If it changes the basis or corrects an earlier assertion, the system creates a

revised evidence view and records the relationship. Posterior reweighting, when enabled, is a separate operation with likelihood provenance. A branch that becomes inconsistent is retained as historically considered and rejected, with the evidence that rejected it.

## 7.8 Modules and stable intermediate representation

A package identifies its namespace, semantic version, dependencies, effective dates, jurisdiction, supported entity classes, recognition operations and measurement policies. Standard packages define allowable treatments; industry packages add economic objects and specialised rules; company packages map source concepts and declare evidenced elections. A company extension cannot override a mandatory standard rule merely because its package is loaded last. An override is valid only at a declared extension point and within the selected standard's policy domain.

The IR has separate node classes for evidence, contextual observations, expressions, domains, constraints, alternatives, contracts, economic events, measurement results, rules, journal batches, report mappings and runs. Each node has a stable identifier and explicit input references. No expression stores executable host-language text. Plugins are referenced by versioned identifiers with typed signatures. This permits a second implementation to execute the same supported program without evaluating arbitrary embedded code.

The IR serialisation version is independent of the DSL version and of any accounting-standard edition. Migrations preserve the original payload and record changed meanings. A syntax migration can be automatic when semantics are unchanged. A changed recognition policy requires a new model version, even if the source file still parses.

## 7.9 Errors are part of the interface

The compiler and runtime return typed diagnostics with a source span, affected entity and period, dependency chain and suggested local action. The principal diagnostic categories have distinct recovery paths:

| Diagnostic | Meaning and next action |
|---|---|
| `TypeMismatch`, `ContextMismatch` | Resolve incompatible units, periods, entities or measurement bases; do not coerce implicitly. |
| `MissingEvidence` | Supply a source or label a model input as an assumption. |
| `InfeasibleHistory` | Inspect the conflicting evidence and model constraints; preserve the rejected candidate. |
| `RuleConflict`, `UnknownApplicability` | Resolve rule scope or preserve a justified conditional partition. |
| `UnsupportedTreatment` | Add a reviewed capability or leave the affected output incomplete. |
| `UnbalancedBatch` | Reject the atomic update and diagnose the posting rule. |
| `SolverIncomplete`, `NumericalFailure` | Preserve the target and inputs; repair or change the computational method. |
| `InsufficientSimulationState` | Add declared conditional assumptions or request a family of results. |

Warnings can describe low extraction confidence or an approximate economic model, but they do not replace these errors. A required unknown input prevents a concrete unconditional result. The analyst can add a declared conditional assumption, request bounds, or leave that output incomplete. Every path has a different, explicit result type; there is no generic fallback to zero.

# Chapter 8

# Reconstruction and execution procedures

## 8.1 Ingest, reconcile and construct histories

The historical procedure starts with a filing manifest. It records documents, reporting dates, publication dates, revisions and expected statements and notes. Tagged filings are parsed through an XBRL adapter; PDF and HTML sources are segmented into text, tables, captions and references. The original bytes and extraction output remain immutable. Re-running an improved extractor creates a new extraction version with links to the same source.

For each table, the importer first resolves headers, units, signs, periods and dimensions. It then proposes concept mappings. A deterministic validator checks subtotal arithmetic, duplicate facts, comparative consistency and cross-note references. Every proposed repair creates a new reading. The system may suggest that an OCR value is 10 rather than 70, but only a reviewed correction or an explicit alternative allows that reading into a validated model.

Reconciliation proceeds from local totals to movement schedules. A debt bridge may contain opening debt, issuance, repayment, acquired debt, interest accretion, currency effects, fair-value changes and closing debt. Every component receives a movement type. An unexplained residual is stored as a residual with a source and period; it is not automatically classified as repayment or issuance. A model can introduce an unknown decomposition of that residual, subject to evidence and economic constraints.

The reconstruction engine then performs the following finite procedure within a declared model scope:

1. Build shared variables for observations, opening states, contract terms and supported movement amounts. Attach original rounding domains.
2. Add identities, roll-forwards, account restrictions, contractual relationships and cross-period links. Add each semantic reading as a condition on the same joint model.
3. Propagate types, units, intervals and simple equalities. Reject context mismatches before invoking a numerical solver.
4. Ask the solver for feasibility. On infeasibility, return a sufficient conflicting subset of constraints, its source spans and model assumptions. A solver's conflict core need not be the smallest possible explanation.
5. Retain finite alternatives and continuous solution families symbolically. Materialise

example histories only when needed for review or execution.

6. Check whether each requested simulation has sufficient economic state. Missing maturities, reset dates or cohort composition become declared assumptions or explicit unresolved inputs.

Unrestricted journal enumeration would be unbounded: arbitrary offsetting transactions can be inserted without changing reports. The model therefore declares allowed event types, time resolution, relevant accounts, lot/cohort resolution and limits on latent contract instances. These limits describe the represented family; they are not evidence that all real possibilities were searched. The compiler canonicalises syntactic duplicates and equivalences established by its supported rewrite rules. It does not decide arbitrary economic equivalence. Probability measures attach to declared economic hypotheses or parameters, never to a count of encodings; adding a duplicate representation therefore cannot create extra probability mass or coverage.

The first implementation uses symbolic linear bridges and finite structural choices. Nonlinear economic models can be evaluated on feasible states through plugins. A solver timeout preserves the unresolved family and returns `SolverIncomplete`; it does not prune the family as impossible. Search budgets belong to run configuration and are recorded with the results.

## 8.2 Condition a run without rewriting history

A run selects an evidence view and cutoff, a historical family, a forecast starting date, scenario definitions, accounting packages, management policies, output queries and numerical settings. A run may use a concrete history, an exact symbolic family, a finite set of conditional histories, or weighted draws from a declared distribution. The output records which mode was used.

Suppose no debt maturity is disclosed. An analyst may run a conditional case that assumes all debt matures after the forecast horizon. The run is valid under that assumption even though the historical representation remains incomplete. The interface must display the assumption next to the result and provide sensitivity cases when the assumption is material. It must not label the assumed maturity as a discovered historical fact.

Management behaviour normally uses only information available at the decision date. A perfect-foresight stress case is a separate, explicitly labelled policy. Scenarios should share the same underlying innovations when estimating paired differences where appropriate, with stable stream identifiers so branch order does not change the draws. Common random numbers improve the comparability of paired simulations; they do not imply that every branch has the same economic event counts or shock distribution.

## 8.3 Schedule and commit events

Each event carries an economic occurrence time, accrual interval when relevant, settlement time, entity, contract or cohort identifier, currency, inputs and provenance. Its ordering key consists of date, declared phase, dependency edges and a deterministic tie identifier. The reference phases are: opening and scheduled contractual changes; market observations and rate resets; business events and decisions; accrual and measurement; settlement; period closing and reporting. A contract can require a more specific dependency. These phases are an engineering starting point, not a universal description of market timing.

Two events may share a tie order only if their order is either specified or irrelevant to the result. If payment before remeasurement and payment after remeasurement differ, an unresolved

ordering is a model ambiguity or an error. The engine does not conceal it behind alphabetical event IDs.

```
execute_event(operation_request):
    validate exact input closure, bindings and historical admission
    validate separate posting authorization, journal and delta
    lookup stable entity/book/branch/business-event key
    if exact committed fingerprint: return original receipt
    if different committed fingerprint: return event conflict
    compare expected opening state and head generation
    atomically recheck key/head and commit journal, delta, state, receipt
    return receipt, emitted dependent events and diagnostics
```

The atomic unit includes the economic state and its accounting effects. If posting fails, neither state changes. A repeated event ID with identical inputs returns the existing result; the same ID with different inputs is a conflict. Here the inputs include the request identifier, expected head and execution settings, not only the financial payload. Section 8.5 defines the stable key and full fingerprint. Consolidation runs after entity books reach the required date and stores its adjustments separately. A failure in one scenario branch cannot mutate another branch's state.

## 8.4 From an economic event to permission to post

A delivery can change revenue, inventory and receivables at once. Before any book changes, the method needs to know which contract was performed, which opening state it consumes, which evidence supports that reading and which accounting profile applies. Balance alone cannot supply those facts. AccDSL therefore separates the economic basis, permission to use a treatment, proposed entries and the transaction that makes them durable.

The event basis binds an immutable event to opening state, context, branch, time view, chart, authority, method, evidence, admissions, assumptions and view. Its business identity is the four-tuple of entity, book, branch and business event. The branch declares whether it is synthetic or real. A synthetic teaching profile can justify a teaching posting; it cannot authorize a real book merely because the resulting arithmetic is identical.

The method first evaluates reporting framework, jurisdiction and dates; then entity and transaction scope; contract and recognition facts; elections; numerical capability; and account/context bindings. Every material finding survives. A false required fact is ineligible, an unknown required fact is unresolved, and an event outside the method is outside scope. A known false finding must not disappear behind a later unknown one. Missing operative authority never becomes source-supported permission.

The native operation `accounting.assess@1` remains read-only assessment. Its `assess_only` request and false posting permission cannot be promoted by editing a flag. The separate `accounting.authorize@1` operation produces a `PostingAuthorization` from the complete basis and a supported posting profile. It permits posting only when the treatment is eligible, the method is fully specified for the event and the branch mode matches the profile's evidence scope. An unresolved or unsupported assessment is still a useful completed assessment, but it contains no permission to post.

An authorized proposal contains both a journal and an independent economic delta, each bound to the same basis and authorization. Journal lines name account role, debit or credit, positive exact quantity, currency/context, entity, book, event and method. Zero lines are omitted; a negative effect reverses sides. The journal balances within each currency and book. The delta separately checks quantities and contractual memory: inventory movements consume available

lots, cash follows settlement, share counts follow ownership events and debt face amount remains distinct from carrying value.

The commit arguments have exactly four references:

```
AccountingCommitArguments/v1 {
  basis_ref, authorization_ref, journal_ref, delta_ref
}
```

The expected opening state and branch generation appear once, in the outer operation request. A reference to a native assessment in the authorization field fails its reference type. False posting permission fails authorization. An unknown posting schema returns unsupported operation with no journal commit. All bindings are checked before looking up a prior receipt.

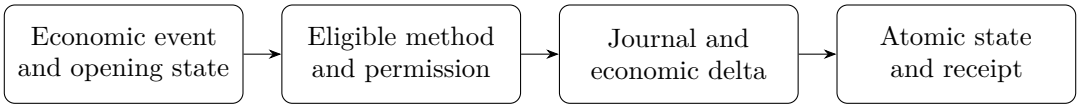



Figure 8.1: Each step answers a different question. Accounting assessment may inform the permission decision, but only the final transaction changes the book.

For a delivery, the basis identifies the exact performance obligation, quantity, allocation and control finding. The opening state contains cumulative revenue, billing, collection and inventory. The selected method derives the recognition and lot movements; the report later consumes that same committed path. Combining a journal from one method version with a delta or opening state from another is invalid even if the totals happen to balance.

## 8.5 A lost acknowledgement must not create a second sale

Suppose a sale commits but the response is lost. The client knows the request, but cannot tell whether the server committed it. Retrying under a new event ID could duplicate revenue. Retrying the original request must instead return the original immutable receipt.

The lookup key is

$$(\text{entity},\ \text{book},\ \text{branch},\ \text{business event}).$$

Payload, event version, opening state and request settings belong to the complete fingerprint, not this key. A renamed run or bumped event version cannot evade a conflict. A separately commissioned hypothetical projection uses a new branch namespace; reopening the same projection uses its existing one.

A correction of an already committed event is a new business event linked to its predecessor and an admitted correction method, such as reversal and replacement or a dated adjustment. It preserves the original receipt. The accounting treatment must apply to the actual correction; absent treatment authority leaves it unresolved rather than licensing a generic reversal.

Commit first validates input shape, closure, bindings and authorization in the requested historical view. It then looks up the stable event key. An equal fingerprint returns the old receipt with no additional commit; a different fingerprint returns an event-identity conflict. Only an absent key proceeds to the expected-state and generation comparison. This order is essential: after a successful sale the current head has changed, so checking freshness before replay would reject the very retry needed to resolve the lost response.

A new candidate is assessed and constructed without mutation. One transaction then rechecks key absence and the expected head, inserts journal, delta, closing state, dependencies and

receipt, and advances the head. It commits all or none. If another client has committed the identical event, return that receipt. A different event that advanced the head makes this request stale. A changed payload under the same key conflicts. Diagnostic or scratch records never count as economic transitions.

A timeout consequently has three possible observations: committed, not committed or unknown. Unknown retains the original key, fingerprint and expected head for receipt lookup or exact retry. Refreshing the head changes the request and cannot be presented as the same retry. Missing committed dependencies are an integrity failure, not permission to reconstruct an approximately similar receipt.

## 8.6 Funding, tax and other simultaneous relations

Some financial models have circular dependencies. Interest affects taxable profit; tax and investment affect funding needs; funding affects interest. The reference executor first tries to resolve these through explicit timing. For example, beginning-of-period debt can determine current interest, with new borrowing affecting the next accrual period. That is acceptable only when the model's timing assumption is stated and appropriate.

Where the economic model genuinely requires simultaneous decisions, the dependency graph identifies the strongly connected component and delegates it to a declared equation or optimisation solver. The solver must return a solution, residuals, admissibility checks and uniqueness or multiplicity status. A contraction iteration requires a justified convergence condition; an arbitrary fixed iteration count is not a mathematical solution. Multiple admissible equilibria become alternatives when material. No solution produces a model outcome such as financing failure, or an explicit numerical failure, depending on the cause.

Funding policy must include available facilities, limits, covenants, collateral and allowed capital actions. The program cannot issue unlimited debt to keep cash positive. An inability to meet payments may be a legitimate default outcome. Default ends or changes contractual and accounting processes under the relevant event rules; it is not quietly discarded as a bad simulation draw.

## 8.7 Worked example: identical history, different future journals

Return to the company in Chapter 1. Immediately after the historical interest payment and closing, cash is 95, debt is 100 and equity is −5. There are no other assets, liabilities or transactions. Next year's expense is 5 in the all-fixed branch and 7 in the all-floating branch under the rate shock. The generated entries and closing statements are:

| Output | All fixed | All floating |
|---|---:|---:|
| Debit interest expense / credit cash | 5 | 7 |
| Closing cash | 90 | 88 |
| Closing debt | 100 | 100 |
| Current-year profit | −5 | −7 |
| Closing equity after profit transfer | −10 | −12 |
| Cash change | −5 | −7 |

The balance-sheet identity holds in each branch: $90 = 100 - 10$ and $88 = 100 - 12$. The demonstration reporting profile classifies interest paid as operating cash flow. A different

permitted reporting profile changes that classification, while preserving the cash movement and source event. This example proves the intended distinction and its arithmetic; it does not estimate the probability that either debt structure describes a real company.

## 8.8 Worked example: a bond has several relevant amounts

A two-period zero-coupon bond raises cash of 100 and promises 110.25 at maturity. Assume a 5% effective rate per period, no transaction costs, no default, and amortised-cost treatment under a demonstration rule. The effective-interest schedule follows directly from $100(1.05)^2 = 110.25$:

| Date | Opening liability | Interest accretion | Closing liability |
|---|---:|---:|---:|
| Issue | 0 | 0 | 100 |
| End of period 1 | 100 | 5 | 105 |
| End of period 2, before payment | 105 | 5.25 | 110.25 |
| Redemption | 110.25 | 0 | 0 |

Issuance debits cash and credits the liability by 100. Each accretion debits interest expense and credits the liability. Redemption debits the liability and credits cash by 110.25. Legal face amount, carrying amount and market fair value remain separate fields. A market-rate shock may change fair value without changing this contractual payment schedule or this demonstration amortised-cost path.

Cash-flow reporting uses the actual payment event and its economic components; the annual accretion is non-cash. The applicable reporting package decides how to classify the maturity payment, rather than inferring it solely from the liability debit. Fees, modifications, early redemption, embedded options, hedges and foreign currency require additional event and measurement rules.

Ordinary equity issuance similarly separates cash, share quantity and legal capital. Ten new shares at 12, with par value 1, produce cash of 120, share capital of 10 and premium of 110 under a simple par-value profile. Shares outstanding increase by ten units. Direct issuance costs of 3 reduce the specified equity account and cash under the chosen illustrative treatment. The accounting rule and the cost's eligibility must be explicit; this example does not prescribe every jurisdiction's treatment.

## 8.9 Reporting and explanations

The reporting engine produces the balance sheet, income statement, comprehensive income, cash-flow statement, changes in equity and note schedules when the necessary modules are present. Each output carries its basis, perimeter, period, scenario and interpretation. A comparative table cannot combine an original prior-year figure and a restated current-year comparative without displaying that choice.

An explanation is a dependency graph with typed edges: extracted from, interpreted as, assumed by, constrained by, generated by, measured by, posted by, aggregated into and supersedes. A cash-flow line must lead back to payment events; a fair-value movement must lead back to valuation inputs; a consolidated amount must show eliminations. A provenance path establishes how a result was obtained. It does not by itself establish that the underlying economic model is correct.

## 8.10 The corporate reference package as operations

The Northstar example defines a closed, reproducible corporate package rather than a universal standard. Its events separate performance, invoicing and settlement even where the example makes the first two simultaneous. Every payload includes entity/book, currency, economic date, period, source or assumption IDs and an information-view ID. Amounts are exact before the declared posting quantisation. Quantities and money are checked separately.

| Event | Preconditions and state change | Posting |
|---|---|---|
| Deliver/invoice | Goods owned and available; control transferred; quantity removed at selected cost; unconditional amount due added | Debit receivable, credit revenue; debit cost of sales, credit inventory |
| Collect | Receipt no greater than selected unpaid receivable; reduce that balance; excess must be separately classified | Debit cash, credit receivable |
| Receive goods | Accepted owned quantity at supported unit cost; create supplier obligation and inventory lot | Debit inventory, credit payable |
| Pay supplier | Identified obligation and sufficient available cash; reduce both | Debit payable, credit cash |
| Payroll | Service period and cash obligation evidenced or assumed | Debit payroll expense, credit cash; accrual variant uses a payable |
| Acquire plant | Control obtained, cost known, in-service date/life/residual recorded; add asset cohort | Debit plant cost, credit cash or payable |
| Depreciate | In-service asset with remaining depreciable amount; apply dated schedule, capped by that amount | Debit depreciation, credit accumulated depreciation |
| Interest | Contract, reset state and accrual fraction known; create interest obligation, then settle if cash available | Debit interest, credit interest payable; debit payable, credit cash |
| Current tax | Reference tax base positive; 20% teaching rule; apply prepaid tax before further payment | Debit tax expense; credit prepaid tax or payable; settlement separately affects cash |
| Dividend | Explicit authorised distribution, sufficient funds and permitted capital action | Debit retained earnings, credit dividend payable; then settle |
| Close | Required period operations completed; preserve original movements | Transfer temporary income/expense balances to retained earnings |

The current example combines accrual and settlement when they share a date, but the separate obligation remains the general operation. Customer advances are debit cash/credit contract liability and become revenue only on the supported performance event. Inventory costing requires an admissible policy; negative inventory and collection beyond an identified claim are

rejected. Non-cash acquisitions, write-offs, impairment and currency movements require their own classified templates. An unknown residual never enters one of these operations merely to force a reconciliation.

Reporting maps asset and contra-asset balances into the balance sheet and liability/equity balances with their opposite sign. Before closing, current profit is included once in equity; after closing it is already in retained earnings. Income uses period movements in income/expense accounts, and OCI uses its own designated accounts. Cash flow sums cash legs by economic operation: Northstar's collections, suppliers, payroll, interest and taxes are operating, plant payments investing, and debt/equity proceeds or repayments and dividends financing. Non-cash depreciation is excluded from direct cash flow and added back only in the indirect reconciliation. Note schedules use the same contracts, lots and classified movements as the statements.

## 8.11 Scenario meaning and information access

An `ExogenousStress` supplies a path or joint driver law while holding the historical evidence model fixed. `ObservationalConditioning` additionally supplies actual observations and a joint probability model; the finite method in Chapter 6 evaluates the conditional distribution when its normalising mass is positive. `StructuralIntervention` requires a named structural model and replaces designated equations, retaining its exogenous noise. An absent structural model gives `UnsupportedScenario`, not an intervention inferred from an accounting identity (Pearl, 2009, Section 3.2.1). The textual Core 0 `scenario` declaration is an exogenous stress only; the other modes require the explicit typed API records.

At each decision date the runtime constructs an information view containing eligible disclosures, committed state through that date, currently observed drivers and the declared forecasting model. Future realised innovations and future outcomes are inaccessible. A stress path may be announced in advance, but this is an explicit visibility policy. Otherwise management sees only its current and past values. A bank's allowance at the opening date may use a forecast distribution of defaults; it must not inspect the outer simulation's future realised default. Inner valuation randomness has a separate stream domain from outer realised outcomes.

Funding follows a declared waterfall: available cash, committed eligible facilities up to their limits, and separately authorised capital actions. Each action checks collateral, covenants and settlement timing. If an amount remains due, record an economic funding failure and the unpaid obligation. If required valuation or accounting is unsupported, stop the affected dependencies as incomplete. The branch manifest preserves the last committed state, completed outputs, unavailable outputs and reason. These distinctions let an analyst see a meaningful bad outcome without confusing it with broken computation.

# Chapter 9

# Accounting methods for an operating company

## 9.1 A common core with distinct accounting books

The common core governs evidence, time, money, events, journals and provenance. Reporting standards govern recognition, measurement and presentation within that core. The design preserves three levels of meaning: economic terms such as a promise to pay cash; an accounting treatment under a particular standard; and a company's reported label or extension. These levels are connected by explicit rules and mappings, rather than collapsed into one universal chart of accounts.

A rule package includes a standards register with jurisdiction, issuer class, effective dates, transition provisions, permitted elections, paragraph references, implementation version and tests. The package selector uses the entity, reporting basis, period and adopted elections. It must distinguish a standard's mandatory effective date from permitted early adoption. For example, the IFRS Foundation's IFRS 18 overview states an effective date of 1 January 2027 with earlier application permitted (IFRS Foundation, 2026c). A 2026 historical report cannot be assigned that presentation regime solely because the latest installed package includes it.

IFRS, US GAAP and ASBE packages share economic contracts and events where appropriate, but produce separate accounting books. A reconciliation between books consists of explained adjustments with origins in differing treatments, not a renaming of accounts. The first implementation can preserve all three source bases even while only a selected subset has executable rules. Such a source is represented as observed information with unsupported simulation treatments, rather than being silently converted to the supported basis.

The official IFRS overviews for IFRS 9, IAS 32 and IFRS 17 establish the relevant scope of financial-instrument measurement, presentation and insurance contracts (IFRS Foundation, 2026d,a,b). They are orientation sources, not substitutes for reviewed paragraph-level implementations. The retained official Ministry of Finance ASBE 21 page now supports selected lease provisions. The official 2017 ASBE 22 text also supports the selected credit-loss, interest and write-off distinctions worked through in Chapter 11 (Ministry of Finance of the People's Republic of China, 2017). Entity-specific issuance and amendment chains and broader ASBE authority remain incomplete. Chapter 12 compares the dated ASBE 25 premium-allocation and incurred-claim provisions with the retained AASB 17 text (Ministry of Finance of the People's Republic of China, 2020). The Ministry's 2025 explanation supports the stated entity-class timetable, while Ping An's reports supply an actual adoption and comparative-restatement example (Section 12.7). Direct verification of the operative issuance and implementation notices

remains open, including the detailed conditions of the later simplifications. Current ASC access is also unresolved. Historical amendments and issuer policies inform proposed comparisons without substituting for operative authority. These source limits are addressed explicitly in the following dated-method discussion.

## 9.2 Non-financial companies

The initial industry package covers sales, collections, purchases, payments, inventory movements, payroll, capital expenditure, depreciation and basic financing. Revenue recognition is a separate rule from invoicing and cash collection. A service contract may generate a performance event before an invoice, creating a different asset from an unconditional receivable. Customer advances create obligations before revenue. Returns, rebates and variable consideration need separate assumptions and subsequent adjustments.

Inventory state includes quantity, cost, location, ownership and lot or cohort. A costing method is selected within the reporting package's permissible domain. An aggregate stock can support a simple model only if its approximation is declared. A commodity-price shock can expose differences between old and new inventory that an average value suppresses. Fixed-asset state likewise requires acquisition dates, components, useful lives, residual values, depreciation method, tax basis and impairment history when those affect the requested forecast.

Industry extensions add economic mechanisms without bypassing the accounting core. Construction requires performance and completion states; extractive industries require reserves, depletion and restoration obligations; utilities may require rate-regulation treatment; software businesses may require contract liabilities and development-cost decisions. The module manifest states what is represented, what is executable, and what remains aggregated.

## 9.3 A matched inventory case across reporting profiles

Standards packages need operation-level decisions. For interchangeable goods, the inspected AASB 102 text incorporating IAS 2 permits FIFO or weighted average in paragraph 25 and explains both in paragraph 27 (Australian Accounting Standards Board, 2015). The reproduced ASBE 1 text specifies FIFO, weighted average or specific identification in Article 14; its official Ministry version could not be retrieved in this review, so that reproduction has provisional source status (Ministry of Finance of the People's Republic of China, 2006a). Caterpillar's company-authored 2024 report states that it principally uses LIFO and supplies a FIFO comparison (Caterpillar Inc., 2024, Note 1.C). That is evidence of a disclosed US-GAAP company policy; full ASC 330 paragraph authority remains an acquisition task. It is not a rule that every US company must use LIFO.

Take the same purchases in three separate books: ten units at 5, followed by ten at 7, then delivery of twelve units. FIFO charges $10(5) + 2(7) = 64$ to cost of sales and leaves inventory 56. LIFO charges $10(7) + 2(5) = 80$ and leaves inventory 40. Periodic weighted average uses cost $120/20 = 6$, charges 72 and leaves 48. The physical deliveries and supplier cash flows agree; profit and carrying inventory differ by the selected costing rule. A concept mapping cannot remove that difference. The FIFO/average selector in an IAS 2 profile rejects LIFO as an unsupported election under the inspected paragraph. The US company profile admits its evidenced LIFO policy while retaining its narrower authority. An ASBE package remains source-provisional until the attributed text and applicable amendments are officially verified.

Inventory impairment illustrates another required memory. Under inspected AASB 102 para-

graphs 28–34, an item costing 56 written down to net realisable value 50 records expense 6. If the conditions reverse and net realisable value becomes 54, reversal is 4, capped by original cost; the previous write-down must therefore remain in state. The reproduced ASBE 1 Article 19 also addresses reversal within the earlier provision. This paragraph is not used to assert the US treatment, whose primary authority was unavailable. Source limitations constrain the treatment claim, while the separate-book method and exact costing calculations remain fully specified.

## 9.4 Operating events

The operating methods use the selected revenue, inventory and asset provisions of Australian Accounting Standards Board (2022c, 2015, 2022a). The lease and gross-tax case uses the dated lease, income-tax and cash-flow compilations (Australian Accounting Standards Board, 2023c,b,a). Each rule still needs the entity, period, transaction facts and elections described below.

Every event uses the event-basis, authorization and commit contracts described in Section 8.4. The event inputs described here are required in addition to identity and date. Each object identifier resolves to its contract, context and admitted source or assumption. A purported cash collection does not establish a sale; a purchase order does not establish receipt.

Revenue, billing and collection are separate transitions. A delivery event names the contract, performance obligation, quantity, allocated price and control evidence. Recognition first releases a related contract liability; earned consideration not yet unconditional creates a contract asset. Both credit revenue. A missing control finding or unresolved price constraint leaves the recognition unresolved.

An invoice or right-to-payment event names the contract, bill and unconditional amount. It debits the receivable and credits an existing contract asset, with any excess recorded as a contract liability. A collection names that receivable, settlement and amount and debits cash while crediting the receivable. An excess payment needs a separate customer-advance method; it cannot be credited to revenue merely because cash arrived. These transitions retain cumulative performance, billing and collection independently.

Purchasing starts when the company receives owned goods, not when it places an order. The event records lot, quantity, unit cost and ownership evidence, debits inventory and credits the payable. Consigned or unreceived goods cannot pass that ownership condition. Consumption or sale identifies an available lot allocation, quantity and admitted cost formula; it debits cost of goods sold or an eligible production asset and credits inventory. Overselling does not create stock. Supplier payment separately debits the payable and credits actual cash. Insufficient cash blocks that settlement and can trigger a separately modelled default; it never creates a hidden loan.

A tangible-asset event records component cost, residual value, useful life, depreciation method and the date it becomes available for use. Initial debit to PPE has a corresponding cash or payable credit; depreciation later debits expense or an eligible asset and credits accumulated depreciation. Employee service records earned pay, deductions and employer obligations for the service period, separately from settlement. Unknown labour or tax law cannot become a zero payable.

Closing derives revenue, expenses, OCI and owner movements from those events using admitted statement mappings. It checks cash, inventory, financing and tax roll-forwards rather than introducing a balancing entry. These event fields form the method's required inputs; each identifier resolves to the same contract, context and evidence used by its authorization.

AASB 15 paragraphs 9, 22, 31, 35–43, 47–59, 73–86 and 105–108 provide the selected revenue conditions; the retained compilation dated 31 December 2022 applies to annual periods beginning on or after 1 January 2023 and before 1 July 2026, covering the stated for-profit FY 2024 example. Variable consideration is estimated under a selected method and admitted only subject to the significant-reversal constraint, not recognized solely because its unconstrained expectation is positive. Principal versus agent, returns, warranties, contract modifications and distinct bundles require their own scoped decision records. If their facts are unresolved, retain alternative eligible methods; do not borrow the simple delivery rule by label.

The following schedules make the distinction between performance and billing explicit. For one performance obligation with price 100, cumulative revenue 60, unconditional bills 50 and collection 30, AR=20, CA=10, CL=0 and cash=30. Revenue 60 is represented by those three assets. Billing before performance changes intermediate balances but not the final result. For bills 80 instead, AR=50, CA=0, CL=20; net assets remain 60. Cumulative performed-service cost 40 then 70 posts 40 then 30, not 110. A downward revision follows the correction method and keeps the old result tied to its original view.

For this simple contract, let $R$ be cumulative recognised revenue, $B$ unconditional billings and $C$ collections, with $0 \le C \le B$. The unpaid bills determine accounts receivable (AR); performance in excess of billing is a contract asset (CA), while billing in excess of performance is a contract liability (CL). After each event,

$$\text{AR} = B - C, \quad \text{CA} = \max(R - B, 0), \quad \text{CL} = \max(B - R, 0), \quad \text{cash received} = C.$$

This recurrence applies per contract, with no impairment, variable consideration or unbilled prepayment; it is not permission to offset unrelated contracts. For billing 80 before delivery, debit AR 80/credit CL 80. Recognizing service 60 later debits CL 60 and credits revenue 60. Collecting 30 debits cash 30/credits AR 30, leaving AR 50 and CL 20, without a second receivable or an artificial contract asset. If service comes first, recognize CA 60/revenue 60, then bill AR 80 against CA 60 and CL 20. Both orders end at the same state. Cash received before an unconditional bill needs an explicit advance event that debits cash and credits CL and records the advance applied to later billing; it cannot pass through collect(AR).

Inventory uses identified cost for non-interchangeable items or the admitted cost formula for interchangeable items. AASB 102.9, 23, 25 and 28–34 support lower of cost/net realizable value, relevant cost formulas and write-down/reversal conditions. A method never chooses LIFO merely to fit a reported balance. In a second lot example, purchases are 10 units at 2 and 10 at 3, followed by sale of 12 units. FIFO gives cost 26 and stock 8 units/cost 24. Weighted average gives cost 30 and stock 20. Under the same given selling price 48, margin differs 22 versus 18. Both alternatives require their own permitted policy, consistent application and evidence; matching total cash does not remove the difference. NRV 16 for the remaining FIFO lot records expense 8, inventory allowance 8; recovery to NRV 22 reverses 6, leaving allowance 2. Recovery to 30 is capped at original cost 24. US LIFO remains a separately sourced question.

The retained AASB 116.7/15, 29–30 and 48–62 support cost and depreciation. Asset 100, residual 10, three-year even use gives 30 per full year. A year-two estimate change after first-year depreciation to residual 10 and three remaining years gives (70−10)/3=20 prospectively; an error in original cost is a correction, not an estimate change. Component replacements derecognize the replaced component's remaining amount before recognizing its replacement; no perpetual double asset. Borrowing-cost capitalization, impairment, provisions, employee benefits and share-based payment are specified as additional scoped methods below, not silently assigned zero in real-company cases.

## 9.5 Apple: asset balances do not identify asset movements

The preceding methods need receipts, lots, useful lives and settlement dates. Apple's disclosures supply some of the balances those events must explain, but they do not supply the events themselves. Its FIFO and straight-line policies narrow the permitted cost calculations without identifying the actual inventory lots or asset cohorts (Apple Inc., 2024, Note 1, printed p. 34). Comparing the notes with the cash-flow statement shows why filling those missing inputs by subtracting balances can produce a wrong history.

### 9.5.1 An inventory adjustment is not purchases or payments

Between 28 September and 28 December 2024, components fall from 3,627 to 2,792 million dollars while finished goods rise from 3,659 to 4,119 million. Total inventory falls from 7,286 to 6,911: the component changes, −835 and +460, sum to −375 million (Apple Inc., 2025, Note 5, printed p. 9). Those two classes do not reveal quantities, unit costs or the transfers and sales that produced the balances.

The quarter's cash-flow statement reports +215 million for inventories within its reconciliation of net income to operating cash (Apple Inc., 2025, printed p. 5). That is an indirect reconciliation adjustment, not total cash paid to suppliers. Even replacing it with minus the stock change would give +375, a different quantity. The difference $215 - 375 = -160$ million remains unexplained by these selected disclosures. In fiscal 2024, inventory rose from 6,331 to 7,286, while the cash-flow adjustment was −1,046; relative to the −955 stock-change shortcut the difference is −91 million (Apple Inc., 2024, printed pp. 31 and 33). Neither residual is an identified currency effect, write-down or other journal entry. Production, consumption, ownership, valuation and payment histories must be established before they can explain the differences.

### 9.5.2 Testing a proposed fixed-asset history

The fixed-asset notes distinguish three measurements. The table shows gross property, plant and equipment (PP&E), its accumulated depreciation and the remaining carrying amount, all in millions of dollars (Apple Inc., 2024, Note 5, printed p. 39) (Apple Inc., 2025, Note 5, printed p. 9).

| Amount | 30 Sep 2023 | 28 Sep 2024 | 28 Dec 2024 |
|---|---|---|---|
| Gross PP&E | 114,599 | 119,128 | 120,615 |
| Accumulated depreciation | 70,884 | 73,448 | 74,546 |
| Net PP&E | 43,715 | 45,680 | 46,069 |

Accumulated depreciation is shown here as a positive deduction: gross less that deduction equals net. During fiscal 2024, gross cost increases 4,529, accumulated depreciation increases 2,564 and net PP&E increases 1,965. Yet the note reports PP&E depreciation expense of 8.2 billion, and the cash-flow statement reports capital payments of 9,447 million. Its broader depreciation-and-amortisation adjustment is 11,445 million (Apple Inc., 2024, printed pp. 33 and 39). These amounts differ in meaning; the 8.2 billion also has coarser displayed precision. The change in accumulated depreciation cannot be substituted for the year's depreciation charge.

Consider a deliberately limited asset book to test a tempting reconstruction. Let $G_0, G_1$ be opening and closing gross cost and $B_0, B_1$ accumulated depreciation, recorded as positive amounts. Let $A$ be gross additions, $R$ the original cost of retired assets, $d$ the depreciation

charge and $S$ the accumulated depreciation removed with those assets. Assume there are no other movements and all depreciation is expensed. For this illustration, use the displayed amounts as exact, including $d = 8{,}200$ million. The two balances must then satisfy

$$G_1 = G_0 + A - R, \qquad B_1 = B_0 + d - S, \qquad 0 \le S \le R. \tag{9.1}$$

The inequality requires a nonnegative carrying amount for the retired assets in this cost model. If additions are equated with the 9,447 of cash payments, the gross balance requires $R = 9{,}447 - 4{,}529 = 4{,}918$. The depreciation balance requires $S = 8{,}200 - 2{,}564 = 5{,}636$. The proposed book removes more accumulated depreciation than original cost, by 718 million. It is infeasible under these combined assumptions. That finding rejects the proposed history; it does not establish an error in Apple's filing or identify which assumption must change.

The result is not sensitive to small rounding differences under a specified rounding model. If the million-dollar balances and payment are rounded to the nearest million, and 8.2 billion to the nearest 0.1 billion, the excess $d - 9{,}447 + (45{,}680 - 43{,}715)$ lies in the closed outer envelope $[666.5, 769.5]$ million. Its lower endpoint is $8{,}150 - 9{,}447.5 + 45{,}679.5 - 43{,}715.5 = 666.5$. This is a conditional precision check, not a verified issuer rounding rule; the exact tie convention remains unknown.

Allowing additions to be partly unpaid provides two simple repairs. Retain the same selected opening and closing balances, 8,200 depreciation and 9,447 cash payments. Assume retirement at the start of the year with zero proceeds, depreciation of surviving assets, and new acquisitions at year end. There are no opening capital payables, taxes or other asset movements. The figures below are constructed histories, not estimates of actual Apple transactions.

| Annual movement (USD million) | History 1 | History 2 |
|---|---:|---:|
| Gross additions | 10,529 | 11,529 |
| Gross cost retired | 6,000 | 7,000 |
| Accumulated depreciation removed | 5,636 | 5,636 |
| Retired carrying amount, expensed | 364 | 1,364 |
| Cash paid for additions | 9,447 | 9,447 |
| Additions remaining payable | 1,082 | 2,082 |

Both histories satisfy equation (9.1); their gross additions can also be allocated to match the three reported asset classes. For History 1, acquisition debits PP&E cost 10,529 and credits cash 9,447 and a capital payable 1,082. Depreciation debits expense and credits accumulated depreciation by 8,200. Retirement debits accumulated depreciation 5,636 and loss 364, while crediting gross PP&E cost 6,000. Net PP&E rises 1,965 and cash falls 9,447, so total assets fall 7,482. Liabilities rise 1,082 and equity falls 8,564 through the two expenses: $1{,}082 - 8{,}564 = -7{,}482$. History 2 has the same asset change but liabilities rise 2,082 and equity falls 9,564. These are coherent subsystem consequences, not a fit to all of Apple's statements or an exhaustive family of real asset histories. Neither history receives a probability or preference from this arithmetic.

The following quarter supplies a new snapshot, not the missing annual movements. Its gross, accumulated and net changes are 1,487, 1,098 and 389 million, whereas capital payments are 2,940 and broad depreciation/amortisation is 3,080 (Apple Inc., 2025, printed pp. 5 and 9). The inspected interim note does not separately identify PP&E depreciation expense. The lease note adds a related distinction: finance-lease right-of-use assets are included in PP&E, while operating-lease right-of-use assets are in other noncurrent assets. The 1.0 billion of noncash right-of-use acquisitions covers the disclosed lease population (Apple Inc., 2024, Note 8, printed p. 42); adding it all to PP&E or to cash capital expenditure would erase that distinction.

The method therefore preserves reported balances, cash payments, expense and unresolved movement components separately. A future cost or funding query still needs lot and asset ages, useful lives, residual values, impairment and settlement facts. The real filing has supplied constraints and ruled out one simple history, while leaving several different explanations open.

## 9.6 Lease and tax integration

A lease connects asset use, a financing obligation and income tax. The following schedule extends the operating-company example by separating these three measurements. The retained lease and tax dossiers supply the dated provisions, modification case and assumptions. The official lease compilation is dated December 2023 despite the older filename. The right-of-use (ROU) asset represents the admitted use of the leased asset. Its cost, the lease liability, accumulated depreciation, remaining payments, rate version and option judgments are separate state fields. Tax retains carrying amounts, legal tax bases, reversal dates, enacted/substantively-enacted rate evidence, recovery support, and original profit/OCI/equity allocation. The deferred tax asset (DTA) and deferred tax liability (DTL) are retained gross before any permitted offset presentation. A liability schedule is not a tax-base schedule.

For the two annual 121 payments at 10%, initial liability and ROU are 210. Year-one interest is 21, liability after payment 110, depreciation 105 and ROU 105. Year two interest is 11, and both balances finish at zero. A same-asset modification just after year one creates two future payments 144 at revised 20%, PV 220, so liability increases 110 and ROU increases 110 to 215. A payment index reset has different rate rules; it cannot reuse that modification blindly.

To demonstrate connection to company statements, assume a synthetic entity with opening cash/equity 500, no other opening balances, and annual cash service revenue 400 less cash operating costs 100, all recognized in their years. There are no dividends, other assets, debt, losses or permanent differences. Add the base lease above. Assumed tax law deducts cash lease payments against taxable income, attributes those deductions to the lease liability, gives zero tax bases to ROU/liability, and taxes profits at enacted 25%. Sufficient usable profits and all recognition conditions for DTA are assumed. Tax is paid the following year. No current-period tax payment is invented from tax expense.

At commencement debit DTA 52.5 and credit DTL 52.5 in addition to the lease recognition. At year-one close DTA is 27.5 and DTL 26.25. The DTA reduction 25 is deferred expense; DTL reduction 26.25 is deferred benefit, net benefit 1.25. Taxable profit is 300−121=179; current tax/payable 44.75. Pretax book profit is 300−105−21=174, total tax 43.5 and profit 130.5. At year two, deferred expense 1.25 combines with current tax 44.75 to give total tax 46 on pretax profit 184.

| Closing balance | 2024 | 2025 |
|---|---|---|
| Cash | 679 | 813.25 |
| ROU asset | 105 | 0 |
| Deferred tax asset | 27.5 | 0 |
| Lease liability | 110 | 0 |
| Deferred tax liability | 26.25 | 0 |
| Current tax payable | 44.75 | 44.75 |
| Equity | 630.5 | 768.5 |

2024 assets 811.5 equal liabilities 181 plus equity 630.5; 2025 assets 813.25 equal liabilities 44.75 plus equity 768.5. For this case, elect financing classification for lease interest under the

retained AASB 107 regime. The operating and financing cash totals below depend on that presentation election. Direct operating cash is 300 then 255.25 after the first tax payment; financing lease cash is −121 each year; no investing cash arises. Indirect operating cash in year one is profit 130.5 plus depreciation 105 plus interest 21 minus deferred benefit 1.25 plus unpaid current tax 44.75 = 300. Year two gives profit 138 +105 +11 +1.25 =255.25. The second tax payable settles next year. Zero OCI and no owner transactions mean the equity change equals cumulative profit. This case tests gross tax, cash timing and financial-statement coherence, not a real entity's tax law.

## 9.7 Apple: recognising tax and settling the obligation

The lease example separates a tax charge from the payment that follows it. Apple's fiscal 2024 disclosures show why that separation matters when several tax jurisdictions, deferred balances and a large disputed obligation enter the same accounts. Its annual income-tax expense was 29,749 million dollars; net cash paid for income taxes was 26,102 million (Apple Inc., 2024, printed pp. 29 and 33). The 3,647 difference is not, by itself, an increase in tax payable. Deferred tax and other balance movements must first be distinguished from current tax and settlement.

### 9.7.1 The expense and balance reconciliations answer different questions

Note 7 divides the 2024 expense between current and deferred amounts for federal, state and foreign taxes. In millions of dollars, the reported components are as follows (Apple Inc., 2024, Note 7, printed p. 40). Parentheses in the source represent negative expense, or a tax benefit.

| Tax category | Current | Deferred | Total |
|---|---|---|---|
| Federal | 5,571 | −3,080 | 2,491 |
| State | 1,726 | −298 | 1,428 |
| Foreign | 25,483 | 347 | 25,830 |
| Sum | 32,780 | −3,031 | 29,749 |

The current charge is therefore 32,780 and the deferred benefit 3,031. The tax payable has its own scope. Note 6 reports current income taxes payable of 26,601 and noncurrent income taxes payable of 9,254 at 28 September 2024, against 8,819 and 15,457 a year earlier. Combining both maturity classes gives 35,855 against 24,276, an increase of 11,579 (Apple Inc., 2024, Note 6, printed p. 39). Current expense less net cash taxes is only 32,780 − 26,102 = 6,678. Reclassifying an amount between current and noncurrent cannot explain the difference in their combined total.

To expose the missing movements, let $T$ denote that combined tax payable and $N$ the net deferred tax asset reported in Note 7. Let $C$ be current tax expense, $D$ deferred tax expense (negative for a benefit), and $P$ net cash taxes paid, all for the same annual period and in millions of dollars. Define $U$ and $V$ as the residuals needed to connect the reported amounts:

$$\Delta T = C - P + U, \qquad \Delta N = -D + V. \tag{9.2}$$

These are diagnostic definitions, not assertions that every tax balance has the same recognition or settlement scope. The first relation gives $U = 11{,}579 - 6{,}678 = 4{,}901$. Setting that residual to zero would produce the wrong closing payable. Assigning it to currency translation, uncertain tax positions or another journal would require the missing transaction and classification evidence.

The deferred balance illustrates the second relation. At September 2024, gross deferred tax assets of 34,873 less a valuation allowance of 8,866 leave 26,007. Deducting deferred tax liabilities of 6,805 gives net deferred tax assets of 19,202, compared with 17,251 a year earlier (Apple Inc., 2024, Note 7, printed p. 41). The increase is 1,951, so $V = 1{,}951 - 3{,}031 = -1{,}080$. The deferred benefit in profit does not explain the whole net-balance movement. Its allocation among currency, equity, other comprehensive income and other causes remains unresolved here.

Note 6's 19,499 deferred-tax-asset presentation is also different from Note 7's 19,202 net position. Their difference of 297 is an arithmetic consequence, not a separately disclosed liability line or authority to offset jurisdictions. The reported temporary differences—including research expenditure, deferred revenue, leases and depreciation—need their own tax bases and recovery conditions. The note reports indefinitely available Irish foreign-tax and California R&D credit carryforwards, but also a valuation allowance for those credits and some other differences. An indefinite term does not establish that an asset is recoverable (Apple Inc., 2024, Notes 6–7, printed pp. 39–41).

### 9.7.2 A recognised charge, restricted assets and a later payment

Apple attributes its large 2024 State Aid charge to the September 2024 court decision confirming the European Commission's recovery decision. This is the issuer's account of the dispute and its accounting consequence. Note 7 describes a 15.8 billion dollar payable to Ireland, partly offset by a 4.8 billion US foreign tax credit and an 823 million decrease in unrecognised tax benefits (Apple Inc., 2024, Note 7, printed pp. 39–40).

The narrative calls the net charge 10.2 billion; the statutory-rate reconciliation gives the more precise amount 10,246 million. Subtracting the displayed narrative centres gives $15{,}800 - 4{,}800 - 823 = 10{,}177$, which is 69 million lower. That is compatible with the coarser inputs: if the two billion-dollar amounts are rounded to the nearest 0.1 billion and 823 to the nearest million, the component calculation lies in the closed outer interval $[10{,}076.5, 10{,}277.5]$ million, containing 10,246. The 10.2 billion charge is compatible with it as well. This is a conditional rounding check, with the issuer's precise rule and tie convention unverified; the centre difference does not establish a conflicting source assertion.

The tax charge does not make the settlement funds disappear at recognition. At 28 September 2024 the escrow comprised 2.6 billion dollars of cash and cash equivalents and 13.2 billion of current marketable securities, restricted from general use (Apple Inc., 2024, Notes 4 and 7, printed pp. 37 and 39). The annual liquidity discussion separately identifies the obligation and the funds held to settle it. The following quarter reports both a payment and the remaining obligation (Apple Inc., 2024, printed p. 26) (Apple Inc., 2025, printed p. 16).

| State Aid obligation and settlement | EUR billion | USD billion |
|---|---:|---:|
| Obligation at 28 Sep 2024 | 14.2 | 15.8 |
| Payment during the following quarter | 11.0 | 11.9 |
| Obligation at 28 Dec 2024 | 3.3 | 3.4 |

The three rows are different dated quantities. In dollars, a movement equation with opening obligation 15.8, payment 11.9 and closing obligation 3.4 requires an additional net movement of $3.4 + 11.9 - 15.8 = -0.5$ billion. Under nearest rounding to 0.1 billion, its outer interval is $[-650, -350]$ million, excluding zero. The same calculation in euros gives $3.3 + 11.0 - 14.2 = 0.1$ billion, with outer interval $[-50, 250]$ million, which includes zero. Neither calculation supplies a common exchange rate or an exact exchange-gain entry. The interim tax discussion mentions

foreign-currency revaluation effects related to the decision, but does not allocate this 500 million dollar residual (Apple Inc., 2025, printed p. 16).

The remaining funds also retain their asset classifications. At December 2024, restricted cash and cash equivalents were 126 million dollars and restricted current marketable securities 3.3 billion (Apple Inc., 2025, Note 4, printed p. 8). Their displayed sum, 3,426 million, is compatible with the rounded 3.4 billion obligation. It is not all cash available for ordinary operations. The opening 2.6 billion of restricted cash alone could not have funded the reported 11.9 billion payment; securities sales, maturities, transfers and valuations need their own histories.

For an isolated settlement illustration, suppose 11,900 million of restricted cash is available and the amount paid equals the liability's carrying value at settlement. Debit tax payable and credit restricted cash by 11,900. Assets and liabilities each fall by 11,900; equity and current-period tax expense do not change in this posting. This assumes the required cash conversion has already occurred and omits any associated gain, loss or fee. It explains settlement of an existing obligation without pretending to reconstruct Apple's funding and currency entries. The quarter separately reports tax expense of 6,254 and net cash taxes of 18,651 million (Apple Inc., 2025, printed pp. 1 and 5); neither should be replaced with the Ireland payment or treated as another copy of the preceding year's charge.

### 9.7.3 Uncertain tax positions do not supply a payment distribution

The last part of Note 7 preserves another population: gross unrecognised tax benefits. Its 2024 movement is $19{,}454 + 1{,}727 - 386 + 2{,}542 - 1{,}070 - 229 = 22{,}038$ million. The increases concern prior- and current-year positions; the decreases concern changed prior-year positions, settlements and expired limitation periods. Interest and penalties are excluded from this rollforward (Apple Inc., 2024, Note 7, printed p. 41). The 1,070 settlement-related decrease is a movement in that gross balance, not an independently identified total of cash paid to tax authorities.

Of the rounded 22.0 billion gross amount, the issuer says that 10.8 billion, if recognised, would affect its effective tax rate. It also describes a reasonably possible decrease of approximately 5–13 billion over the next twelve months, primarily concerning transfer pricing and deemed repatriation tax, while noting uncertainty about examination timing. These are different conditional statements. The range is neither a probability law nor a promised cash payment or expense release. A forecast still needs jurisdictional outcomes, recognition, interest, penalties and settlement dates. Retaining those unknowns keeps the tax note connected to the company's funding needs without turning management's dated assessment into an invented future transaction.

## 9.8 Returning to the company forecast

The Northstar case remains the broader ordinary-company teaching baseline: collections, purchases/payments, inventory, payroll, PPE, financing, tax, statements, annual/interim and revision views are already specified. Its 2024 collection/purchase/payment solution is 245/155/145. The 2025 forecast in the Northstar reference fixture opens with cash 74.6 and equity 90.6, spends 20 on capital expenditure at period end, pays dividend 8 and uses a base floating rate 7%. Its simplified tax is settled in the year earned. The running case uses fixed-rate share $\alpha$; here floating-rate share $x = 1 - \alpha$ describes the same uncertainty. For 2025 the same unknown

floating fraction $x \in [0,1]$ determines every statement amount:

$$\begin{aligned} I(x) &= 5 + 2x, & \text{pretax profit}(x) &= 82 - 2x, \\ \text{tax}(x) &= 16.4 - 0.4x, & \text{profit}(x) &= 65.6 - 1.6x, \\ \text{operating cash}(x) &= 48.6 - 1.6x, & \text{closing cash}(x) &= 95.2 - 1.6x, \\ \text{equity}(x) &= 148.2 - 1.6x, & \text{assets}(x) &= 288.2 - 1.6x. \end{aligned}$$

Receivables 55, inventory 40, net PPE 98, payables 40 and debt 100 complete the balance sheet. Liabilities are 140 for every $x$, so the common cash/equity slope preserves the accounting identity. There is no default distribution on $x$. At a 9% floating rate the all-floating history has interest 9, profit 62.4, cash 92, equity 145 and assets 285. The annual cash bounds still cannot establish daily payment solvency without event-date cash flows.

The lease/tax example extends the interaction evidence without changing the original Northstar fixture or pretending that its teaching 20% tax formula is an IAS 12 method. In a real-company package, each required method is admitted separately and unsupported operations remain visible. The extended obligations for pensions, impairment, provisions, valuation and share-based payment are in the additional corporate methods that follow; omission from a finite company case never proves the actual company has none.

## 9.9 Employee benefits

The additional methods below use the selected IAS 19, IAS 36, IAS 37, IFRS 2, IFRS 13 and IAS 23 provisions in the original 2023 EU incorporating regulation (European Commission, 2023). This is historical mechanism support; later amendments and actual entity facts remain separate applicability inputs.

DefinedContributionState retains service periods, promised contribution formula, contributions due/paid and applicable legal terms. For earned service requiring contribution 10, debit employee expense 10/credit payable 10; payment settles it. Calling a plan defined contribution needs the limited-obligation facts; a guaranteed benefit or additional funding obligation cannot disappear by name.

The defined-benefit obligation (DBO) measures accrued employee benefits under the stated actuarial assumptions. DefinedBenefitState retains employee/cohort service, benefit formula, projected salary, retirement/turnover/mortality assumptions, discount-curve currency/term, obligation, plan-asset fair value, asset ceiling, contributions, benefits and remeasurements. IAS 19.67–74 requires projected unit credit and the stated service attribution; .83 supplies discount-rate requirements. Assumptions are separately admitted measurements, not constants provided by the accounting rule. A one-employee toy plan promises 100 for each completed service year, paid after year two, with certain completion, zero salary growth and supported 10% discount. At year-one end accrued benefit 100 due in one year has PV 1000/11. In year two that opening obligation accretes 100/11 and the second earned benefit adds 100; obligation 200 then settles 200. Merely using current salary times years without timing would give the wrong year-one obligation.

A separate funded-plan roll-forward supplies an independent statement example: opening DBO 100, plan assets 80 and net liability 20; current service 12; interest 5 on DBO, interest 4 on assets at 5%; contributions 10 at year end; benefits 8 paid from plan at year end; actuarial obligation loss 7; actual asset return 6. DBO closes 100+12+5−8+7=116, assets 80+6+10−8=88, net liability 28. Expense is 12+1=13, OCI loss is 7−(6−4)=5. Debit service expense 12/credit net pension liability 12; debit interest expense 1/credit liability 1; debit OCI 5/credit liability 5; debit liability 10/credit cash 10. End liability 20+13+5−10=28. Plan-paid benefits reduce

both DBO and plan assets and do not cause a second employer cash payment. IAS 19.113, .120–127 supports these components; remeasurement OCI is not recycled to profit under 122. A surplus requires the asset ceiling and applicable minimum funding analysis, not recognition of any calculated negative liability.

Unknown actuarial inputs give conditional obligations or measurement_unresolved. Curtailment, settlement, amendment and multi-employer/insured arrangements need their own fact conditions and measurement-date remeasurement. A buyout is not settlement if obligations remain. Sensitivities vary one assumption only when that conditional meaning is explicit; correlated risks require joint scenarios.

## 9.10 Impairment and provisions

ImpairmentState retains asset/CGU boundaries, carrying amounts, allocated goodwill, corporate assets, recoverable-amount inputs and previous impairments/depreciation. IAS 36.9–10 determines indicators and mandatory annual tests. Recoverable amount is the higher of value in use and fair value less disposal costs (.18); it is not the lower of two pessimistic scenarios. VIU needs the supported cash flows, forecast horizon/growth, discount and exclusion of inappropriate financing/tax or uncommitted restructuring effects under 33–57. Compute conditional values if those estimates are unknown; do not infer them from historical profitability.

For a cash-generating unit (CGU), assets excluding goodwill 100 plus goodwill 20, recoverable amount 90: loss 30 first removes goodwill 20 and then reduces other assets 10 subject to their floors under 104–105. Debit impairment 30/credit goodwill 20/credit asset allowance 10. If the asset floor is 95 the requested reduction to 90 is impermissible; apply the remaining allocation/recognition conditions, not a forced plug. For a standalone cost-model asset with carrying 80, recoverable 60, recognize 20 loss. At a later measurement before further depreciation, supported recovery to 90 permits reversal only 20 to the no-impairment 80 ceiling (.117–119). Goodwill impairment is not reversed (.124). Different dates require the hypothetical unimpaired depreciation path, not the old nominal cost as cap.

ProvisionState retains obligating past event, present legal/constructive duty, outflow assessment, estimate distribution, settlement dates, risk/discount basis, reimbursements and evidence. IAS 37.14 requires present obligation, probable outflow and reliable estimate; .36–47 governs best estimate/risk/discount. An intended future operating expenditure without that duty is not a provision. Assume an established obligation to pay 121 in two years, deterministic amount, appropriate 10% rate and no double-counted risk: recognize expense/provision 100, accrete interest 10 then 11, settle 121. The passage-of-time increase is borrowing cost under 60. If the obligation is an eligible asset restoration cost, initial debit follows that asset's method rather than always expense; later changes need the applicable restoration interpretation. Contingent liabilities/disclosures remain visible when recognition conditions are not met; no generic 0.5 threshold may substitute for the relevant standard's required analysis.

## 9.11 Share-based compensation

AwardState distinguishes equity versus cash settlement, grant/vesting/service dates, granted units, grant-date fair value, service/nonmarket/market conditions, expected units vesting, cumulative expense and changes/cancellations. IFRS 2.7–11, 15–23 supports the employee equity-settled method. With 100 units, grant value 3, two-year service condition and all expected to vest, year-one cumulative expense 150 and equity 150. At vesting only 80 satisfy the service condition: cumulative 240, second-year expense 90. Debit employee expense and

credit equity. Failure of a market condition is different under 21; if the employee satisfies other vesting conditions, it does not automatically reverse the grant-date measured service expense. Unknown conditions cannot be represented by a single forfeiture percentage without their type and evidence.

Cash-settled awards remeasure the liability under 30–33. Same 100 expected units, half service and current unit fair value 4 gives liability/expense 200 at year one; 80 actual units and value 5 at settlement gives total 400, second expense 200, then debit liability 400/credit cash 400. Reusing the equity grant value 3 would be wrong. Actual award valuation requires the specified model/market inputs, option rights, volatility, dividends and calibration; the fair values 3/4/5 are case inputs. Modifications and settlement-choice arrangements require their own conditions, with the original minimum expense and increment analyzed under the applicable provisions rather than overwriting the original grant.

## 9.12 Fair value and capitalization boundaries

A ValuationRequest names the unit of account, market/date, asset/liability characteristics, principal/accessible market, technique, input hierarchy, counterparty/own non-performance risk where relevant, calibration and sensitivity. IFRS 13.24, .42–43 and .67–80 guide the selected measurement. Quoted identical asset price 7 for 10 units gives 70 in a qualifying Level 1 case; a blockage discount based only on holding size cannot reduce it to 63 under 69/80. A present-value illustration 111 due next year at an independently supported 11% rate gives 100; that equality does not establish the rate, market comparability or accounting classification. Historical EIR, an impairment discount, risk-neutral pricing and a physical expected return answer different questions.

Borrowing-cost capitalization requires a qualifying asset, eligible costs, commencement/suspension/cessation dates and the specific/general-borrowing calculation, before moving interest into cost. IAS 23.8–25 supplies these rules in the retained historical EU text: for specific borrowing, eligible actual costs less temporary investment income; for general borrowing, the applicable weighted rate and cost ceiling. Commencement requires expenditures, borrowing costs and necessary preparation activities; extended inactive periods can suspend it, and substantial completion ends it. Assume specific borrowing 100 outstanding during a full eligible construction year at 5%, with an admitted expenditure schedule and temporary investment income 1 on temporarily unused funds during that period. Net capitalized cost is 4. Real admission still needs the dated standard, actual schedule and eligibility evidence. It is not a default 5% add-on to all assets. Internally developed intangibles likewise need research/development separation and all applicable recognition criteria; a forecast of future sales does not establish an asset.

The extension interface returns the stated measured components, proposed journal, economic delta, disclosures and unresolved inputs under the event-basis, authorization and commit contracts. Calculation success does not permit another operation to fill a missing authority/fact with a default. Real-company acquisition/tax/employee/valuation evidence and the full exceptions remain explicit internal preparation, with proposed answers and competing treatments retained for any later authorized domain review.

## 9.13 Rule selection

The proposed selection procedure distinguishes operative authority from historical source support. The retained sources include official AASB compilations, the original EU incorporating regulation and official ASBE lease and income-tax texts (Australian Accounting Standards

Board, 2022e,c, 2023c,b; European Commission, 2023; Ministry of Finance of the People's Republic of China, 2019, 2018, 2006b, 2022). Historical US lease additions are separately identified in the retained amendment source (Financial Accounting Standards Board, 2016).

AuthorityRequirement contains issuer/framework, jurisdiction, entity class, annual-period/application interval, paragraph and local source Ref, incorporating instrument, amendment chain, transition, elections and unresolved dependencies. RuleSelection binds those requirements to the accounting event and evidence view. The algorithm evaluates chronological applicability first, then scope and facts, then available elections/transition, then measurement/presentation. If two supported rules can apply, retain the explicit alternative and required election; if one is contradicted by facts, mark ineligible. Missing current authority gives unresolved, never eligibility through a familiar rule name.

An unchanged historical report remains an assertion under its own basis and vintage. A harmonized comparison creates an adjustment book and reconciliation, with changed recognition/measurement explained. Label mapping alone cannot convert US GAAP to IFRS or ASBE. The four clocks and source version remain part of each rule dependency; later knowledge cannot leak into an original as-of view. Transition entries reconcile opening assets, liabilities, tax and equity, rather than changing old source bytes or plugging an unexplained amount.

## 9.14 IFRS and incorporating jurisdictions

The retained official AASB texts provide explicit IAS/IFRS incorporation for their stated entity/version scope. AASB 9/15 compilation dates are 31 December 2022 with the source covers' application window; AASB 16 compilation 8 is 31 December 2023, AASB 112 compilation 7 is 29 September 2023. Their filenames are not date evidence. AASB 102's older retained edition is selected historical support for inventory, not current 2024 applicability certification. The retained EU 2023/1803 original provides official EU-adopted IFRS 3/10/IAS 19/21/36/37/IFRS 2/13 passages inspected in this method investigation; its historical adoption scope and later amendments remain separate. The retained amendment investigation supplies additional source evidence, not a blanket acceptance of all rules in these standards.

Initial real-method admission is therefore claim-specific: the selected revenue, lease/tax, financing, credit, group/FX and insurance cases must resolve each named applicability prerequisite. For synthetic demonstrations, declare the chosen dated provisions and assumptions and use the synthetic book mode. No assertion of whole-IFRS compliance follows from those entries.

The base lease in the saved dossiers (two annual 121 payments, 10% rate, no qualifying exemptions or extra costs) has liability 210 initially, 110 after year one, 0 after year two. IFRS-equivalent cost-model ROU is 210/105/0; expenses 126 then 116. Income tax needs its own legal deductions and gross temporary differences, as the connected corporate case demonstrates. Revenue timing, inventory cost formula, financial instrument classification, control and contract boundaries are required rule predicates, not accounting options chosen after seeing which outcome fits best.

## 9.15 US GAAP proposed methods and authority limitation

The saved official FASB ASU 2016-02 Section A and later 2020-05/2021-09 amendments contain inspected historical ASC 842 additions. Their covers explicitly say an ASU is not authoritative. Current Codification access returned 403 again in this source investigation; the proposed methods cannot be marked current-authority-admitted. ASU amendments, historical FAS 109,

issuer accounting policy and a model's memory are all insufficient substitutes for the operative ASC text and full applicable amendment chain.

The same lease facts include ten-year remaining equipment life, fair value 1000, alternative use to lessor, no ownership transfer, no purchase option, two-year term and PV 210. Under the inspected historical 842-10-25-2/25-3 provisions, the stated facts support the proposed operating classification. Historical 842-20-25-6 and 35-3 yield single lease expense 121 per year, liability 210/110/0 and ROU 210/ 110/0. First year debit lease expense 121/credit liability 21/credit ROU 100; payment debits liability 121/credits cash 121. Next expense 121 decomposes into liability interest 11 and ROU reduction 110. The finance comparison uses the separate depreciation 105 plus interest 21/11 path. Changing classification facts is not a free election. The 75%/90% heuristics in the historical guidance cannot become universal requirements. This complete numerical contrast is useful despite the remaining current-authority limitation.

Proposed ASC 740 temporary-difference treatment retains tax base, reversal dates, enacted rates, valuation allowance and source-specific allocation/presentation. The saved historical FAS 109 and ASU 2019-12 provide inspected enactment support, not a completed current ASC 740 implementation. Under the matched PPE case, book 800/tax 600 at assumed 25% yields DTL 50 and current tax 75 on taxable 300. A future 30% rate that is merely substantively enacted but not legally enacted cannot be imported from the AASB 112 rule into this proposed US treatment; the declared 2025 closing temporary difference is 300, giving DTL 75 at the enacted 25% rate versus DTL 90 at 30%. This deferred-tax comparison is separate from current tax 75 on taxable profit 300. Actual legal enactment remains a separate jurisdictional fact.

The proposed CECL method for in-scope amortised-cost assets estimates lifetime expected cash shortfalls from origination using past/current/forecast evidence, with a documented reversion method beyond a reasonable/supportable forecast horizon and applicable instrument exceptions. It retains allowance roll-forwards, write-offs, recoveries, purchased-credit-deteriorated handling and off-balance commitments separately. It must not copy IFRS 9's 12-month/SICR staging to the same record. A candidate contrast: lifetime discounted loss 6 with 2 attributable to default events in the next year gives IFRS general Stage 1 candidate 2 versus CECL lifetime candidate 6, conditional on scope and method support. No current ASC 326 text was retrieved, and the attempted official amendment path returned 403; this contrast remains an explicitly unadmitted proposed reading, not a source-verified US requirement or posting example. Full current ASC 326, insurance 944, consolidation 810, FX 830 and relevant later amendments are specific internal source tasks before corresponding real admission.

## 9.16 ASBE lease adoption and deferred tax

For an ASBE lease, selecting the lease standard and selecting the related income-tax treatment require different application dates. Article 68 of the revised ASBE 21 states a general date of 1 January 2019, but the Ministry of Finance's 2018 issuance notice specifies the enterprise groups. Paragraph I requires application from that date for enterprises listed both domestically and overseas, and for overseas-listed enterprises preparing financial statements under IFRS or ASBE. Other enterprises applying ASBE must apply the revision from 1 January 2021 (Ministry of Finance of the People's Republic of China, 2018, issuance notice, para. I).

Paragraph II permits earlier application for enterprises whose parent or subsidiary is listed overseas and prepares its overseas financial statements under IFRS or ASBE. Their early date cannot precede simultaneous application of the specified 2017 financial-instruments and revenue standards. Paragraph III discontinues the 2006 lease standard and its application guidance when the enterprise applies the revision. Thus a 2020 reporting date alone does not establish

use of the revised lease standard for an enterprise in the second group: its qualifying early adoption would need evidence. The retained official Chinese texts include the 2018 notice and attachment and the Ministry's later ASBE 21 page, published on 28 October 2019 (Ministry of Finance of the People's Republic of China, 2018, 2019). They replace the earlier secondary reproduction as support for the inspected wording.

Articles 4–8 require control of identified use, economic benefits and direction; article 6 includes both conditions for substantive substitution and the express presumption when difficult to determine. Articles 14–18 specify commencement, ROU cost and PV, using implicit rate or incremental borrowing rate if the former cannot be determined. Articles 20–23 separate cost-model depreciation and periodic interest; articles 25–29 distinguish reassessment/index changes/modifications and their discount-rate rules. Articles 30–33 define the exemptions; a two-year term is not short-term. Articles 60–67 specify transition choices and disclosures; the chosen transition cannot be inferred from a present closing liability (Ministry of Finance of the People's Republic of China, 2018, attached standard, arts. 4–8, 14–33, 60–67).

For the declared base lease, these inspected recognition/measurement provisions support proposed ROU 210/105/0 and liability 210/110/0 with expense 126/116, under the same stated facts, depreciation assumption and fixed rule version. A same-asset modification is tested against 28–29, rather than labeled a separate lease without increased scope and commensurate consideration. A 2019 first-adopter using article 61(2) records cumulative opening changes without adjusting comparatives, while 61(1) is retrospective; comparison datasets must retain that distinction and the entity's actual election.

The tax question is why equal opening lease balances can still require separate deferred-tax assets and liabilities. The Ministry's original ASBE 18 defines tax bases and temporary differences in articles 4–9 and contains initial-recognition exceptions in articles 11 and 13. Article 13 also limits a deferred-tax asset to the probable taxable profit available to use the deduction. Articles 16–23 govern measurement, recovery or settlement assumptions, subsequent review, allocation and noncurrent classification (Ministry of Finance of the People's Republic of China, 2006b). The original text must be read with later interpretations.

Interpretation 16 removes the initial-recognition exemption for a single transaction that is not a business combination, initially affects neither accounting profit nor taxable income or loss, and creates equal taxable and deductible temporary differences. Section I(a) expressly includes qualifying initial lease liabilities and right-of-use assets; the corresponding deferred-tax liabilities and assets are recognized separately under ASBE 18. Equal carrying amounts alone do not establish equal temporary differences, and the asset's recoverability condition still applies. Section IV makes this treatment effective from 1 January 2023; Section I(b) permits early application from the year of issuance, 2022, with disclosure (Ministry of Finance of the People's Republic of China, 2022, secs. I(a)–(b), IV). This tax date is distinct from the enterprise's lease-standard adoption date. The operative text now supports the treatment previously illustrated by the retained official 2023 tax application case.

For the synthetic two-year lease, retain the assumed payments of 121, 10% discount rate and straight-line depreciation. All amounts use the same illustrative monetary unit. Assume that tax deductions are allocated to the liability, both tax bases are zero, the applicable tax rate is 25%, and sufficient taxable profit satisfies the deferred-tax asset recognition condition. Also assume that the transaction meets Section I(a)'s conditions and falls within its application period. At commencement the ROU asset and liability are each 210. The taxable ROU difference gives a deferred-tax liability (DTL) of 52.5; the deductible liability difference gives a deferred-tax asset (DTA) of 52.5. The entry debits DTA and credits DTL by those amounts. At the first annual close, liability 110 and ROU 105 give DTA 27.5 and DTL 26.25. Reducing the DTA by 25 creates deferred-tax expense; reducing the DTL by 26.25 creates a benefit, leaving net

benefit 1.25. Both balances reverse by expiry. Applying the old exemption to omit both opening deferred balances is wrong for these qualifying facts and dates: it conceals the required gross balances and the later tax effect.

Transition also needs more than an opening journal entry. Section I(b) requires adjustments for qualifying transactions between the earliest comparative opening and application. For qualifying existing balances at that opening, cumulative effects adjust opening retained earnings and the relevant financial-statement items. Consider a separate transition example with cash 500, ROU 105, lease liability 110 and equity 495, with no deferred tax yet recognized. Under the same assumed tax bases, rate and recoverability, recognition debits DTA 27.5, credits DTL 26.25 and credits opening retained earnings 1.25. The resulting balance sheet is

$$\underbrace{500 + 105 + 27.5}_{\text{assets } 632.5} = \underbrace{110 + 26.25}_{\text{liabilities } 136.25} + \underbrace{495 + 1.25}_{\text{equity } 496.25}.$$

The equity adjustment records the cumulative effect at the comparative opening. If the remaining lease year falls within the comparative year, the DTA of 27.5 and DTL of 26.25 reverse during that year, giving comparative deferred-tax expense 1.25. The opening adjustment alone therefore does not complete the transition. Neither ASBE 21's article 61(2) lease-adoption election nor Interpretation 16 Section III's no-comparative-adjustment rule for its separate share-payment topic may be substituted for Section I(b)'s tax transition.

These are gross recognition and measurement amounts. Separate recognition and noncurrent classification do not determine whether an actual balance sheet may offset DTA and DTL. That decision needs the applicable presentation provisions and entity/tax-authority facts. Actual tax law, tax bases and rates, recoverable profit, the enterprise's adoption and elections, and later amendments remain to be established before applying the example to an issuer. Full ASBE 22 credit, consolidated statements, group combinations and insurance revisions also retain specific source dependencies; similar amounts do not establish full IFRS equivalence.

## 9.17 Matched-case outcomes and decision

| Input changed or withheld | Required consequence |
|---|---|
| Framework changes with same lease facts | Preserve different expense/ROU paths and source strength, not just rename accounts |
| Method source present but target entity's adoption unknown | Unresolved applicability, no real posting authorization |
| ROU/liability initially equal | Still evaluate gross tax differences and later divergent balances |
| Historical rule supports method but a later reporting year is requested | Recheck operative amendments/transition; no automatic extension |
| Issuer policy discloses LIFO, regulator text unavailable | Retain issuer assertion; do not upgrade it into universal US authority |
| Foreign basis label replaced by local label | Reject as a standards conversion without recognition/measurement bridge |

These decisions establish a common procedure and specific proposed treatments. The recovered ASBE lease notice and tax interpretation support the stated application schedules and conditional example. Broader authority coverage remains incomplete: operative US sources, ASBE sector methods, actual tax law and presentation conditions, and the applicable amendment and entity-election evidence still require internal investigation.

# Chapter 10

# Financing, ownership and the group boundary

A company can preserve the same operating assets while changing who supplies its funding and who shares its results. The following cases move from debt cash flows to ownership, then to the reporting boundary of a group. Each numerical example states its own contract; the coupon-bearing convertible below and the later zero-coupon compound example are different instruments.

## 10.1 Debt and hybrid financing

The contract layer for a debt instrument stores issuer and holder perspective, currency, issue and settlement dates, issue proceeds and legal face amount. Its payment terms include coupon/index/spread, reset and fixing dates, day-count convention, payment calendar, amortisation schedule and maturity. Its rights and protections include seniority, collateral, covenants, options, conversion terms, guarantees and linked hedges. Missing terms are explicit unknowns. A single carrying amount cannot stand in for this record.

The accounting layer stores the classification decision and its evidence, initial measurement, carrying amount, accrued interest, effective-rate method, unamortised fees or discounts, fair-value disclosures, impairment or credit adjustments where applicable, and presentation mappings. Contractual interest, effective-interest expense, realised cash payments and fair-value changes are separate quantities. The zero-coupon example in Chapter 8 provides a minimal conformance case for this separation.

The lifecycle must support issuance, recognition, accrual, reset, coupon, principal repayment, modification, derecognition, default, recovery and conversion. A modification event first determines which accounting operation the selected standard requires; it does not always mean issuing a new loan. An embedded option can affect valuation and classification without being exercised. A derivative linked as a hedge requires both the economic contract and any applicable accounting designation; economic risk reduction alone does not establish hedge-accounting eligibility.

Hybrid instruments make interpretation particularly consequential. IAS 32's overview discusses liability/equity classification and the separation of a compound instrument (IFRS Foundation, 2026a). The executable classification must consider the full terms and the selected standard version. US GAAP or ASBE treatment must be implemented under its own rules rather than inferred from an IFRS classification. Where the terms are not disclosed sufficiently, alternative

classifications can coexist only if each is actually compatible with the known terms and applicable rules.

## 10.2 Equity and ownership

Equity has both monetary accounts and security quantities. The security register records share class, issued and outstanding quantities, treasury holdings, par or stated value where applicable, voting rights, dividend rights, redemption features, options and conversion relationships. Monetary equity accounts include contributed capital, reserves, retained earnings and other components required by the selected basis. These records evolve together through issuance, repurchase, cancellation, split, dividend and conversion events.

A stock split changes quantities and per-share measures without necessarily changing aggregate monetary equity. A cash dividend affects distributions and cash rather than current operating expense under the corresponding ordinary equity treatment. Repurchases and employee share awards require their own rules. Earnings per share uses a dated quantity schedule and rules for potential ordinary shares; simply dividing profit by the closing share count is insufficient when shares change during the period.

Funding simulations must connect the quantity register to market assumptions. An equity issue at a scenario-dependent price changes cash raised, shares and dilution. The model must state whether it targets a fixed cash amount, a fixed share count or a capital ratio. These are different management decisions even when the historical statement would report the same aggregate proceeds.

## 10.3 A compound bond exposes classification dependencies

Consider an issuer receiving 100 for a two-year bond paying annual coupons of 5 and principal 100 at maturity, with a holder option to convert that principal into ten ordinary shares. Assume its currency is the issuer's functional currency, no other options, fees or contingent settlements, and that the conversion feature qualifies as equity under the selected rule. A comparable liability without conversion yields 10%. The inspected IAS 32-equivalent text requires separate liability and equity components and measures equity as the residual (Australian Accounting Standards Board, 2022b, paragraphs 28–32, AG30–AG32).

The liability's present value is

$$L_0 = \frac{5}{1.1} + \frac{105}{1.1^2} = \frac{11050}{121} \approx 91.3223, \qquad E_0 = 100 - L_0 \approx 8.6777.$$

Issue debits cash 100 and credits these two components. Under the stated effective-interest schedule, the first coupon leaves liability $1.1L_0 - 5 = 1050/11 \approx 95.4545$; after the second coupon it is 100. Redemption debits liability and credits cash 100. Alternatively, conversion at maturity debits liability and credits equity 100 while issuing ten shares; the original equity component remains in equity. AG32 supports the absence of a gain or loss on that maturity conversion. Exact rational valuations are kept before the declared monetary rounding.

A variable-number share settlement, cash alternative or other clause can change classification. The contract record must therefore retain these terms; it cannot select liability/equity accounts independently of the coupon, conversion and settlement rules. Early repurchase and induced conversion invoke separate operations: the inspected AG33–AG35 text describes allocation and the induced-conversion consideration difference. They are not implemented by changing the

maturity date of the simple example. US-GAAP and ASBE classification require their own dated authority; the IFRS-oriented split is never silently reused under another basis.

## 10.4 Schedules and initial classification

The following liability and ownership methods use selected dated AASB 9 and AASB 132 provisions (Australian Accounting Standards Board, 2022e,b). Issuer, holder, contractual cash flow and accounting measurement remain separate throughout the schedule.

DebtContract contains parties, currency, principal, issue/settlement dates, payment calendar, fixed/index/spread/reset terms, compounding/day-count, business-day adjustment, principal changes, fees, options, security, priority and legal default/remedy terms. Each term has evidence/admission provenance. The economic schedule emits due cash independently of the recognition method. A market-value adapter may value that schedule but cannot choose amortised cost, FVOCI, FVTPL or liability/equity classification.

An investor's classification evaluates business model and contractual cash-flow features; a debt label alone cannot establish the AASB 9.4.1.2/4.1.2A conditions. At initial recognition the selected 5.1.1 rule measures fair value and qualifying costs as appropriate. An unknown day-one observable-value condition yields unresolved under 5.1.1A/B5.1.2A. The retained acquisition cases distinguish FVTPL expensed fees, amortised-cost fees, original EIR and POCI credit-adjusted EIR. No rate method inferred from market value can overwrite the contract rate.

For an issuer with a plain amortised-cost debt, net proceeds 100 and one legally due payment 121 at the end of two annual periods, the exact EIR solves 100=121/(1+r)^2, hence r=10%. Issue debit cash 100/credit debt 100; year-one debit interest 10/credit debt 10; year-two debit interest 11/credit debt 11; redemption debit debt 121/credit cash 121. Outstanding contractual payment stays 121 until settled even while carrying value is 100 or 110. Total interest 21 equals cash outflow 121 minus proceeds 100. An immediate transaction fee 2 changes net proceeds to 98 and the EIR root; retaining 10% is wrong. The numerical root method is in the numerical-method discussion; an irrational rate is not silently an exact rational.

A fractional period uses the declared convention. Annual-effective 21% over half a year produces factor 1.1 and interest 10 on 100; simple proration gives 10.5 and is wrong for that contract. A floating coupon resets from the admitted fixing available at the reset date, not a later observed rate. Actual day-count and holiday-calendar versions are immutable inputs. Unknown fixings preserve conditional paths, rather than defaulting to the latest value.

## 10.5 Modification, extinguishment and default

At a modification, retain original and replacement terms, counterparty, fees, effective date, legal-release evidence and original EIR. Determine derecognition before measurement. AASB 9.3.3.1–3.3.3 and B3.3.1–B3.3.6 govern the selected issuer-liability cases. The quantitative ten-percent test compares the PV of new terms, including specified borrower/lender fees, at original EIR against remaining original PV. It is not a universal asset test, nor proof that every change below ten percent is immaterial; changed currency, rights and other qualitative terms need the applicable analysis. Insufficient evidence remains unresolved. Third-party payment without legal release does not extinguish debt.

On the first anniversary above, old remaining 121 has PV 110. Replace with 133.1 due at the original maturity date, one year after the modification, with no fees and an explicit supported

substantial-terms conclusion: PV 121 differs by exactly 10%, satisfying the inclusive test. Assume separately supported new fair value 118; debit old debt 110, debit extinguishment loss 8, credit new debt 118. The new EIR is 133.1/118−1, not automatically the old rate. If the replacement is instead a stated non-substantial same-currency reduction to 115.5, no fees or other changes, PV 105 leads to debit debt 5/credit modification gain 5 under the B5.4.6 recalculation mechanism, then interest 10.5 and settlement 115.5. The actual change below ten percent still needs the qualitative finding. Repurchase of carrying amount 110 for 108 extinguishes the repurchased obligation: debit debt 110/credit cash 108/credit gain 2. Unamortised fees belong in the carrying amount before that comparison, not a second gain adjustment.

If cash available is 80 against debt 121 due, record the due obligation and an unpaid/default economic state; payment 80 leaves contractual unpaid 41 only when that partial payment actually occurs. Do not cancel the remaining liability, recognize a waiver or issue a loan unless a separately admitted event authorizes it. Covenant acceleration, cross-default, cure periods, collateral realization and restructuring require the actual legal terms and priority waterfall. A scenario may explore specified alternatives with explicit assumptions; it may not silently select the cheapest settlement as an accounting fact.

## 10.6 Equity and compound instruments

EquityInstrument retains class, voting/distribution/liquidation rights, issued, treasury and outstanding counts, par/legal-capital requirements and conversion terms. Share counts are integer quantities separate from currency balances. Issue 100 ordinary shares at 2 with no fees: debit cash 200/credit equity 200, issued and outstanding 100. Repurchase 10 at 3 for treasury: debit treasury-equity 30/ credit cash 30; outstanding 90, issued 100, treasury 10. Reissue 5 at 4: debit cash 20/ credit treasury-equity 15/credit other equity 5; outstanding 95, treasury 5. There is no trading profit under the selected AASB 132.33 rule; a statutory par split requires that jurisdiction's facts. Dividend declaration creates a liability only at the admitted point at which the obligation exists; proposal of a dividend is not itself payment or automatically an expense.

For a qualifying fixed-for-fixed issuer conversion option in functional currency, AASB 132.28–32 separates liability fair value and residual equity. Assume total issue proceeds 100, standalone debt fair value 90, zero coupon, maturity two years and cash redemption 100 or fixed 10 ordinary shares; no extra derivative, tax or fee. Issue debit cash 100/credit debt 90/credit conversion-equity 10. Effective interest accretes debt 90 to 100. At maturity conversion under original terms debits debt 100 and credits issued-share equity 100; the original equity 10 stays in equity (a permitted internal transfer is not income). Shares increase 10 and no redemption cash leaves. Redemption instead debits debt 100/credit cash 100; unexercised equity is not remeasured into profit. AG 32 supports no conversion gain/loss in that stated case. Variable share count, cash alternatives, foreign currency and changed conversion terms require classification/modification analysis; they cannot be forced into this example.

Financial statement mappings separate interest, modification/extinguishment results, owner transactions and cash settlement. A financing-liability note reconciles opening carrying amount, cash proceeds/repayments, effective interest, fees, conversions, modifications and FX. Total shares and earnings-per-share denominators need dates and dilution conditions; outstanding shares at year end are not a substitute for a weighted period denominator. Full option valuation, hedge accounting, EPS and legal-capital regimes remain explicitly additional methods; the proposed records preserve their required inputs.

## 10.7 Control and an elimination book

The consolidation procedure selects the control graph at each date, aligns supported accounting policies and presentation currencies, combines controlled entity books, and posts acquisition, elimination and non-controlling-interest adjustments in a separate book. A control conclusion needs evidence and a rule version. Ownership percentage alone is insufficient. Unknown purchase allocation, currency translation or unmatched intragroup differences remain named unresolved adjustments rather than unexplained balancing entries.

For a simplified acquisition of a different subsidiary from the goodwill case that follows, assume control is established, all net assets are already at the relevant acquisition measurements, no tax or goodwill arises, and non-controlling interest is measured proportionately. A parent has cash 60, investment 40 and equity 100; an 80%-owned subsidiary has cash 30, plant 20 and equity 50. Consolidation debits subsidiary equity 50, credits investment 40 and credits non-controlling interest 10. Group assets are cash 90 plus plant 20, and group equity is 100 attributable to the parent plus 10 to other owners.

A subsequent intragroup loan of 10 shifts cash from parent to subsidiary, creates a receivable and payable, and is eliminated by debiting payable and crediting receivable 10. Group totals remain 110. An unmatched loan currency or settlement date would block that exact elimination. If the parent instead sells its entire interest for 44 immediately after the initial state and loses control, the group removes subsidiary cash 30, plant 20 and non-controlling interest 10, recognises cash consideration 44 and gain 4; remaining cash/equity is 104. More complex acquisitions and retained interests require additional measurement operations; this example fixes the structure and arithmetic without certifying a complete consolidation standard.

## 10.8 Perimeter and consolidation state

The acquisition, control and currency methods below use the inspected IFRS 3, IFRS 10 and IAS 21 provisions in the retained 2023 EU text (European Commission, 2023). The acquisition with goodwill is a new example; it does not change the preceding zero-goodwill case.

ControlAssessment records investor/investee, relevant activities, rights and their substantive/protective character, variable returns, ability to use power, principal/agent analysis, effective interval and evidence. IFRS 10.5–9 requires power, returns and their connection, with reassessment on changed facts. An 80% share count is evidence to analyze, not an automatic substitute for control. Unknown control yields separate conditional perimeters and unresolved group results; no entity disappears merely to make reconciliation easy.

PerimeterVersion binds control intervals, ownership classes, report cutoffs, functional/presentation currency, policy-alignment adjustments and the exact source entity books. ConsolidationAdjustment is a distinct group book with matched counterparty/event references; legal-entity journals remain intact. The procedure is: select time-compatible entity books; apply supported policy alignment; translate them; add like balances/movements; eliminate investment against acquisition equity; eliminate intragroup balances, income/cost and unrealised gains; allocate profit/OCI to owner/NCI; reconcile all statements and movement notes. IFRS 10.19–23/B86–B96 support these steps. Reporting-date differences need B92–B93's conditions and intervening adjustments; a generic nearest-date join is invalid.

## 10.9 Acquisition and ownership case

Assume a business combination, no common-control transaction, parent obtaining control and 80% of ordinary residual interests for cash 96. Acquiree identifiable assets fair value 160 and liabilities 60 give net 100. There is no prior interest, contingent consideration, tax difference, separately unrecognized intangible, transaction cost or impairment; those absences are synthetic facts. Select proportionate-share NCI under IFRS 3.19: NCI 20 and goodwill 96+20−100=16 under .32. In parent books debit investment 96/credit cash 96. In the consolidation overlay, debit acquiree acquisition equity 100 and goodwill 16, credit investment 96 and NCI 20. Only post-acquisition income enters group performance. Asset measurement exceptions, deferred tax and a bargain gain reassessment under .36 would change the calculation; goodwill cannot absorb an unexplained missing balance.

If post-acquisition net profit is 20, all accrued in retained earnings with no dividend/OCI, NCI increases 4 to 24 and parent's attributable profit is 16. Parent buys another 10% for 15 while retaining control: transfer 12 from NCI (10% of current net 120), credit cash 15 and debit parent equity 3. Remaining NCI 12; goodwill 16 unchanged; no new profit. IFRS 10.23/B96 makes this an equity transaction. This example has equal ordinary rights; preference rights need their actual allocation and cannot use a simple ownership fraction.

In a separate alternative starting before that additional purchase, sell all 80% for 130 when net assets 120, goodwill 16, NCI 24 and OCI zero. On loss of control, group gain=130+24−(120+16)=18. Derecognize individual assets/liabilities and goodwill, remove NCI, recognize consideration and gain. The separate parent book's investment gain is not added again. If 20% is retained and 60% sold for 100 with retained fair value 35, group gain=100+35+24−136=23; recognize the retained investment 35. Its subsequent associate/financial-asset method needs separately established significant-influence/classification facts. IFRS 10.B98–B99 also requires the appropriate prior-OCI treatment; this case assumes none.

## 10.10 Intercompany inventory and tax

Assume parent's external inventory cost 60 is sold to its controlled subsidiary for 100 on credit, all still held at report date, tax ignored in the base case. Parent records AR 100/revenue 100 and COGS 60/inventory 60; subsidiary inventory 100/ AP 100. Consolidation debits AP 100/credits AR 100 and debits revenue 100/credits COGS 60 and inventory 40. Group inventory 60, revenue and COGS zero, and no group profit arises. When subsidiary later sells externally for 120, group COGS is 60 and profit 60, not 20; release the previously eliminated 40 in the continuing adjustment roll-forward. Reversing a closing overlay and recreating a new one must not recognize the profit twice.

If the purchaser's independently supported tax base is 100, group carrying amount 60 creates deductible difference 40. At assumed applicable 25% and with all IAS 12 DTA conditions satisfied, recognize group DTA 10/deferred-tax benefit 10; reverse on outside sale. The original tax expense and cash obligations in the seller's tax jurisdiction remain. Different rates, recoverability or tax-law transfer rules change the answer; tax is not simply eliminated with group profit. For an upstream subsidiary sale, attribute elimination and related tax effects to the appropriate subsidiary/NCI performance; downstream parent's profit elimination does not mechanically reduce NCI by the same fraction.

## 10.11 Functional currency versus group presentation

EntityCurrencyState names the functional currency and its evidenced rationale, foreign-currency nominal balances, monetary/nonmonetary classification, historical or fair-value measurement dates, rates and sources. IAS 21.21–23 and .28 require transaction-date translation and closing-rate remeasurement of monetary items; historical-cost nonmonetary assets retain transaction-date rates. A functional currency change is prospective under .35–37. Hyperinflation, lack of exchangeability and hedge accounting require separate applicable methods; do not import the historical .26 text into a later amended period without checking it.

Example: buy goods for foreign 100 at functional 1.2 per foreign unit; debit inventory 120/credit payable 120. Closing rate 1.3 gives payable 130 and FX expense 10, while inventory remains 120 before any NRV test. Pay at 1.25: debit payable 130/ credit cash 125/credit FX gain 5. Cumulative expense 5 matches 125 cash less 120 asset cost. Translating the inventory again to 130 would be wrong for this declared basis.

Group presentation is a different operation under IAS 21.39–41. A subsidiary has opening net assets 100, equity 100, no debt or distributions, annual profit 20 entirely earned at rate 1.25, and closing assets 120. Opening historical rate 1.2 and closing 1.3 give translated assets 156, contributed/opening equity 120 and profit 25; translation OCI 11 reconciles the difference. It is explained as $100\times(1.3-1.2)+20\times(1.3-1.25)=11$, not a free balancing plug. If ownership is 80%, parent/NCI shares of profit are 20/5 and OCI 8.8/2.2 under the stated equal rights. Use transaction rates; an average is acceptable only as a justified approximation, not when significant fluctuation invalidates it. Goodwill and fair-value adjustments follow IAS 21.47's foreign-operation treatment, rather than freezing them automatically in the parent's currency.

Group reports retain the translation reserve by entity, ownership and event. Loss of control requires the appropriate reserve reclassification under IAS 21.48 and IFRS 10.B99; a no-control-loss change transfers the relevant attribution in equity under applicable conditions. Cash translation effects reconcile cash separately from operating/investing/financing movements. An intercompany monetary balance's exchange result cannot simply be eliminated as if no currency exposure existed; IAS 21.45 and net-investment .32 require the substantive classification.

# Chapter 11

# Credit losses and the bank balance sheet

The funding and default distinctions become central in a bank. A loan creates an asset, customer payments affect settlement liquidity, and a forecast of credit loss affects an allowance before cash is lost. The first case connects funding to a simple allowance; the later dated loan schedule separates credit stages, interest recognition and eventual recovery.

## 11.1 Banks: accounting capital is not regulatory capital

A bank package needs instrument and cohort states for loans, securities, deposits, wholesale funding, repos, derivatives, collateral and commitments. For interest-rate analysis, currency and repricing buckets matter alongside remaining maturity. For credit analysis, exposure, default state, loss severity, recoveries, guarantees and their dependencies matter. For liquidity analysis, deposit behaviour, collateral eligibility, encumbrance and payment timing matter. These states can be represented at an explicitly chosen aggregation level; the public record rarely identifies every contract.

Loan origination can debit loans and credit a customer deposit in the bank's book. Subsequent transfer to another bank produces settlement flows. Treating all loan growth as an immediate cash outflow would therefore encode a specific funding assumption, not a universal banking identity. A system-wide model must also link counterparties and settlement assets; single-company simulation records external counterparties without claiming to close the whole economy.

An allowance calculation is a dated measurement based on forecast information. For a single horizon, let EAD be a fixed exposure at default, PD the probability of default, and LGD the mean loss fraction conditional on default. Ignoring discounting, the illustrative expected loss in currency units is

$$L = EAD \times PD \times LGD.$$

With exposure 100, default probability 0.02 and loss given default 0.4, it produces 0.8. Changing those inputs to 0.04 and 0.5 produces 2, so the allowance increase is 1.2 if there are no other movements. This calculation is a demonstration model. A standard-specific expected-credit-loss implementation must determine horizon, staging or other classification, discounting, forward-looking information, write-offs and purchased-credit-impaired or other special cases. The simple formula is not an IFRS 9 or US GAAP compliance implementation.

Regulatory capital is computed in a separate prudential layer with its own perimeter, eligibility, deductions, risk weights and transition provisions. Accounting equity feeds a reconciliation

to capital; it is not automatically Common Equity Tier 1. Liquidity and capital constraints influence the management policy, which in turn generates asset sales, funding or capital events. A breached ratio can be an economically meaningful scenario outcome. No hidden capital injection is permitted to repair it.

## 11.2 A bank's cash, allowance and capital in one path

The bank reference state contains settlement cash 30, gross loans 100, allowance 0.8, deposits 110 and equity 19.2. Originating a loan of 20 first debits loans and credits the borrower's deposit. A subsequent transfer of 15 to another bank debits deposits and credits settlement cash. Loan origination and external cash use are consequently distinct events. Receive loan interest 6 in cash and credit deposit interest 2 to customer balances. Remeasure the allowance to 2.4, posting an additional loss of 1.6. The resulting state is cash 21, gross loans 120, allowance 2.4, deposits 117 and equity 21.6. Net assets are $21 + 120 - 2.4 = 138.6 = 117 + 21.6$.

Now 25 of deposits runs off. The declared funding policy uses cash first, then a committed facility with limit 10. It draws 4 and settles 25, leaving cash zero, deposits 92 and wholesale debt 4. A runoff of 40 instead leaves 9 unpaid even after the maximum facility draw. The simulator records default and retains the unpaid deposit obligation. It does not assume unlimited central-bank or shareholder funding.

The prudential calculation is a separate, illustrative policy. Deduct 2 from accounting equity to obtain capital 19.6; apply a 75% risk weight to gross loans for risk-weighted assets 90. The resulting ratio is 19.6/90. Neither the deduction nor the weight is a Basel default. Actual prudential packages require jurisdiction, exposure class, eligibility, buffers and dated rules, with a reconciliation from the accounting book. The structure follows the need to model capital and liquidity jointly in Behn et al. (2019, Section 2), without transferring that paper's estimated preferences or shock parameters to this bank.

For a richer credit adapter, partition borrower cohorts by currency, vintage, credit state, remaining maturity, collateral and measurement category. At date $t$, supply scenario probabilities, conditional default-time probabilities, exposure paths, conditional loss severity and discount factors. Its expected loss calculation has the declared form

$$L_t = \sum_s p_s \sum_{k \in H} d_{t,k} \Pr(\tau = k \mid s, \mathcal{F}_t) \mathbb{E}[\mathrm{EAD}_k \mathrm{LGD}_k \mid \tau = k, s, \mathcal{F}_t].$$

The conditional product expectation retains dependence between exposure and severity. The horizon $H$ concerns eligible default times; associated cash shortfalls may continue later. The rule package decides twelve-month versus lifetime default scope, credit-state transfers, interest basis and write-off recognition. Changes in allowance must be attributed to origination, derecognition, transfers, revised expectations, currency and write-offs. Contract and prudential support can be specified now, while full IFRS 9/CECL/ ASBE classification and paragraph-level validation remain later obligations.

## 11.3 State and event order

The dated credit methods use the selected amortised-cost, interest and impairment provisions of AASB 9 (Australian Accounting Standards Board, 2022e). The following loan has different inputs from the earlier liquidity example, so its allowance and final equity are calculated separately.

BankState contains cash/reserves by settlement venue and currency; deposits by customer, access rights and maturity; assets by contract, gross carrying amount, allowance, original EIR, accrued cash flows, collateral and credit state; undrawn commitments; funding lines and covenants; accounting equity; regulatory mappings; and evidence/information views. LoanState separately retains origination credit risk and current risk, SICR finding, credit-impaired/POCI status, forbearance, modification, write-off and recovery history. Aggregating by today's loan balance alone loses both staging evidence and later response to shocks.

Each time step first admits information known then, forms permitted credit and liquidity forecasts, chooses behavior under that information, and schedules due events. Accrual, actual settlement, impairment measurement and regulatory queries have separate business identities. A deposit received from outside debits cash/reserves and credits deposit liability. A loan credited to the borrower's deposit debits loans and credits deposits without initially transferring reserves; a transfer to another bank then debits deposits and credits reserves. A loan paid directly outside instead debits loan/credits reserves. These two funding paths cannot be treated as identical cash events. Deposit interest debits interest expense/credits deposit or accrued payable; withdrawal requires settlement funds.

## 11.4 Expected credit loss

For each admitted scenario j and date t, calculate contractual due flows minus the cash expected to be received, including supported collateral/enforcement cash and timing; discount the shortfall under the applicable original EIR or specified exception. With mutually exclusive exhaustive scenario weights w_j, ECL is the sum of w_j times those present-value shortfalls. Default-time hazards must be converted to disjoint default-time probabilities; summing cumulative PDs over dates counts the same default repeatedly. Collateral recoveries cannot be credited both in LGD and as a second recovery cash flow. A shorthand PD×LGD×EAD is admissible only if its definitions and discounting reproduce the specified cash-shortfall calculation; it is not a universal replacement.

Expected credit loss (ECL) measures the discounted expected cash shortfall. A significant increase in credit risk (SICR) changes the general method's default horizon. Purchased or originated credit-impaired (POCI) assets follow a separate method. For the general approach, twelve-month ECL considers lifetime shortfalls arising from possible defaults during the next twelve months; it is not just unpaid cash in those twelve months. SICR uses the change in lifetime default risk since initial recognition, reasonable forward-looking information and the applicable rebuttable presumptions. Do not install a two-times-PD or 30-day rule as an unconditional universal trigger. Lifetime ECL follows a supported SICR finding; cure needs a supported reversal of that finding. Subsequently credit-impaired assets use original EIR on amortised cost for interest in subsequent periods; POCI has a separate credit-adjusted-rate and cumulative-change method.

The 2017 ASBE 22 text makes the two horizons explicit. Article 47 defines credit loss as the present value of all contractual cash flows less expected receipts; a late receipt can therefore create a loss even when its nominal amount is paid in full. Article 48 limits twelve-month ECL by the date of the possible default event, while retaining the resulting cash shortfalls over the asset's life. Article 52 assesses changes in default risk, rather than changes in the allowance, and Article 58 requires probability weighting, time value and reasonable, supportable information (Ministry of Finance of the People's Republic of China, 2017, Articles 47–48, 52, 58). These provisions support the following dated comparison. Article 84 states 1 January 2018, but applying the standard to a particular company also requires its issuance schedule, subsequent amendments and adoption facts.

The general measurement interface requires scenario_cashflows, law_ref, information_view, horizon_admission, credit_assessment and discount_rule. An unweighted history set can produce conditional losses or loss bounds, but cannot produce an unbiased probability-weighted ECL by silently using equal weights. Return measurement_unresolved for the ECL claim. Trade receivables and certain other assets use the separately applicable simplified approach; the general loan staging rule must not overwrite AASB 9.5.5.15–16.

## 11.5 Full loan and allowance roll-forward

Assume a fixed-rate non-POCI amortised-cost loan held to collect its contractual cash flows, originated for 100, with one terminal payment 121 two years later, original EIR 10%, no fees, collateral, modifications, FX or taxes. Since no payment falls due before maturity, an earlier default in this example is a separately established qualitative credit event, such as financial distress under the admitted default definition; it is not a missed instalment. Article 52 of ASBE 22 requires a definition consistent with credit-risk management and consideration of qualitative indicators (Ministry of Finance of the People's Republic of China, 2017). At origination the assumed lifetime scenarios are: default in the first twelve months with probability 0.1, default during the following year with probability 0.1, and no default with probability 0.8. Either default gives terminal collection 60.5. The first twelve-month ECL includes lifetime loss from the first scenario only: $0.1\times(121-60.5)/1.1^2=5$; initial lifetime ECL would be 10. Debit loan 100/ credit cash 100; debit impairment 5/credit allowance 5. Gross 100, allowance 5, net 95.

At year-one accrual the loan has survived the first window and is not yet credit-impaired: debit gross loan 10/credit interest 10. Survival alone would give remaining default probability $0.1/0.9=1/9$. A subsequent, separately admitted credit forecast revises that conditional probability to 0.2; it is an authored forecast update, not a Bayes calculation from survival alone. Its lifetime weights are 0.8/0.2, with a separately established SICR finding; $\text{ECL}=0.2\times60.5/1.1=11$. Debit impairment 6/ credit allowance 6. Gross 110, allowance 11, net 99. The ratio change alone is not the defined SICR test. Only twelve months remain, so twelve-month and lifetime ECL both equal 11 under this revised forecast. This amount cannot distinguish the two classifications.

Immediately afterward, a distinct, ordered credit-impairment event changes expected maturity collections to 60.5. This is a forecast mean, not a guaranteed or maximum recovery. Required allowance is $110-60.5/1.1=55$; debit impairment 44/credit allowance 44. Net 55. ASBE 22 Article 59(5) measures this non-POCI credit-impaired asset by gross balance less discounted expected receipts. Articles 38 and 39(2) then make amortised cost, after the allowance, the basis for subsequent-period interest (Ministry of Finance of the People's Republic of China, 2017).

The bank is assumed still to have reasonable prospects of fuller recovery before maturity. The separate finding that some contractual cash will not reasonably be recovered first arises at settlement on this branch. If that finding arose earlier, ASBE 22 Article 43 would require the corresponding earlier write-off. Expected collections alone do not justify postponing an already-required write-off.

In the next full year interest revenue is $55\times10\%=5.5$. The gross contractual accretion is 11. One consistent decomposed posting debits gross loan 11, credits interest revenue 5.5 and allowance 5.5: gross 121, allowance 60.5, net 60.5. The allowance increase of 5.5 is attributed to discount unwind in its roll-forward, with no additional impairment expense in this posting. This reconciles gross and net-interest bases without counting the same expected loss again. Collect 60.5: debit cash/credit gross 60.5. When no reasonable expectation of remaining recovery is established,

debit allowance 60.5/credit gross 60.5. Both balances reach zero. A mere delinquency is not that write-off finding. Subsequent unexpected cash recovery is recorded with its own recovery-income method and evidence; it does not restore an old live receivable by default.

| Moment | Cash, opening 200 | Gross loan | Allowance | Deposit liability | Equity, opening 20 |
|---|---|---|---|---|---|
| Initial loan and allowance | 100 | 100 | 5 | 180 | 15 |
| Year-one accrual and SICR | 100 | 110 | 11 | 180 | 19 |
| Credit impairment | 100 | 110 | 55 | 180 | −25 |
| Maturity accrual | 100 | 121 | 60.5 | 180 | −19.5 |
| Collection and write-off | 160.5 | 0 | 0 | 180 | −19.5 |

Total interest 15.5 less impairment 55 is loss 39.5, equal to 100 advanced minus 60.5 received. The negative equity state survives; no missing branch or capital plug restores solvency. These five dates jointly test allowance, recognition and bank balance-sheet arithmetic. The same loss cannot also be deducted directly from gross before the stated write-off.

For a non-derecognized loan modification, AASB 9.5.4.3 requires gross PV at original EIR and a modification result, then allowance and SICR reassessment under 5.5.12. For example gross 110 changed to a single future 110 at original 10% gives revised gross 100 and modification loss 10 before any allowance change. ASBE 22 Articles 42 and 56 similarly separate remeasurement of a non-derecognised asset from its credit-risk reassessment (Ministry of Finance of the People's Republic of China, 2017). This is separate from the financing method's issuer-liability ten-percent test. A derecognition decision requires its own asset conditions and cannot reuse that threshold.

## 11.6 Funding, capital and decision policies

Liquidity simulation settles the dated net payment queue by venue/currency, respecting encumbrance, available cash, committed line terms and admissible actions. Haircut/eligibility and transferability come from a versioned policy; market value alone is not available liquidity. With cash 20, due withdrawal 30 and an admitted usable facility 8, even a full draw leaves shortfall 2. Record the default/unresolved settlement according to the event's legal model. Raising exactly 2 of new equity without an investor event is an invented financing source. Interbank contagion requires explicit counterparties and settlement priorities; it cannot be inferred from aggregate deposits.

A RegulatoryAssessment reads an accounting view and an independent jurisdiction, date, perimeter, eligibility, deductions and exposure-weight policy. It outputs capital components, RWA/exposure measures, liquidity measures and reconciliations with their distinct denominators. Illustratively, equity 20 minus an assumed regulatory deduction 2 yields capital 18; assumed RWA 125 gives 14.4%. Those inputs are synthetic, not a Basel rule, national capital approval

or default threshold. No binding capital-policy numbers have been inspected for a target bank here. Use the Basel/national source and actual bank facts before real admission.

Behavior under capital/liquidity constraints is a separate economic model. Specify feasible actions, objectives, information set and ties before solving; failure to find an action returns policy_failure or default under the stated contract. Halaj/Behn sources can motivate candidate behavior but cannot calibrate it for an individual bank. A forecast experiment must compare simple fixed-book, contractual runoff and explicitly estimated policy baselines with time-safe data, before claiming a predictive improvement. CECL and other bases retain separate authority/method branches in the dated-authority discussion.

# Chapter 12

# Insurance service, finance and claims

An insurer receives cash before much of its service is delivered. Cash premium, service revenue and profit therefore follow different schedules. A short coverage example first connects gross insurance to reinsurance. The subsequent two-year cases introduce unearned profit, loss allocation and finance one change at a time.

## 12.1 Insurance: cohorts and future service

An insurer requires policy cohorts or groups with issue dates, coverage, premiums, benefits, expenses, lapses, claims development, guarantees and reinsurance relationships. Asset portfolios need their own contracts and valuations. The liability state must distinguish expected future cash flows, discounting, risk adjustments or margins, unearned profit measures where applicable, incurred claims and movements between opening and closing states. The IFRS 17 overview's distinction between measuring future cash flows and recognising profit as service is provided motivates this separation (IFRS Foundation, 2026b).

The module API accepts policy/cohort state, economic assumptions, non-economic assumptions and information date. It returns a measurement vector, movement attribution and proposed recognition events. Experience differences, changes in assumptions, finance effects and service effects remain separately identifiable. Reinsurance held has its own record and measurement; it is not an automatic percentage offset to every insurance liability.

Scenarios can alter discount curves, mortality, morbidity, lapse rates, inflation, catastrophe losses and policyholder behaviour. Their joint dependence and cohort exposure must be declared. A portfolio total can match history while hiding very different guarantee and duration profiles. This makes alternative economic histories especially valuable, but also limits what public-data reconstruction can identify. A full insurance package needs actuarial and accounting review beyond the core release.

## 12.2 Insurance service and reinsurance in a complete cohort

The example uses one-year policies and a one-year proportionate treaty starting when those policies are recognised, with uniform service for each. Both groups qualify for premium allocation through the one-year conditions in paragraphs 53(b) and 69(b); paragraphs 62–62A determine treaty recognition. Assume no acquisition cash flows, investment components, significant financing or onerous remaining coverage. Claims and recoveries are expected within a year of the claims being incurred. The inspected AASB 17 text incorporating IFRS 17 specifies the movements in paragraphs 53–59, allocation in B126 and separate reinsurance requirements in 63–70A (Australian Accounting Standards Board, 2022d). Gross and ceded groups retain distinct identities.

Begin with capital and cash of 20. Receive premium 120 and recognise a remaining-coverage liability of 120. Pay reinsurance premium 24 and recognise a coverage asset of 24. After half the uniform service, recognise revenue 60 by releasing gross coverage and reinsurance expense 12 by releasing its coverage asset. Estimated incurred claims are 40, of which 30 is paid; the remaining-claim risk adjustment is assumed to be 2. Gross claims expense is therefore 42 and incurred-claims liability is 12. Assume the treaty reimburses half the gross expected claims, recovery cash of 15 has arrived, the risk transferred is valued at 1, and reinsurer non-performance has zero expected effect in this example. The reinsurance claims asset is $20 + 1 - 15 = 6$, and recovery income is 21.

Cash is $20 + 120 - 24 - 30 + 15 = 101$. Assets are cash 101, remaining reinsurance coverage 12 and claims recovery 6, totalling 119. Liabilities are remaining gross coverage 60 plus incurred claims 12, totalling 72. Profit is $60 - 42 - 12 + 21 = 27$ and equity is $20 + 27 = 47$. These balances reconcile without netting reinsurance against the insurance liability. If counterparty credit risk changes, the recovery measurement changes through its own inputs; it does not mechanically halve every gross liability or risk adjustment.

Longer-duration business adds a fulfilment-cash-flow vector, discount curve, risk adjustment, contractual service margin, loss component and coverage-unit schedule. Select portfolio and annual cohort, assess onerous grouping, project cash flows within contract boundaries, discount with the declared financial assumptions, measure non-financial risk separately, and attribute subsequent movements to new business, experience, assumptions, finance and service. Future-service changes adjust the supported service-margin operation subject to the loss rules; current/past-service effects enter their specified result components. Unsupported variable-fee, transition or loss-recovery operations must stop those dependencies explicitly. A PAA example does not implement these longer-duration methods by implication.

An actuarial reserve adapter also needs a concrete estimation contract. For cumulative claims $C_{i,k}$ by origin year $i$ and development year $k$, the chain-ladder point method estimates each factor from matched observed cells, $\widehat{f}_k = \sum_i C_{i,k+1} / \sum_i C_{i,k}$, then projects the latest cumulative claim along the remaining factors. A triangular example with rows $(100, 150, 180)$, $(120, 180)$ and $(140)$ gives factors 1.5 and 1.2, outstanding amounts 36 and 112 for the unfinished rows, and total 148. Mack (1993, Sections 2–3, equations (1)–(3)) makes explicit the conditional-mean and variance assumptions and independence across origin years; its aggregate-error discussion retains dependence from shared estimated factors. A single final-development observation cannot supply an empirical tail variance without an additional assumption. The adapter therefore returns the point projection and marks uncertainty estimation unavailable in this tiny example. It supplies neither a full predictive distribution nor an IFRS risk adjustment. Calendar effects, inflation, changing claims handling and tail selection require separate sensitivity and validation before real use.

## 12.3 Group state and general measurement

The following cases apply selected general-measurement and service-allocation provisions from the retained AASB 17 compilation (Australian Accounting Standards Board, 2022d). They extend the premium-allocation example without presuming actuarial calibration or a complete insurer model.

The general method distinguishes fulfilment cash flows (FCF), including an explicit risk adjustment (RA), from the contractual service margin (CSM), which holds unearned profit. The liability for remaining coverage (LRC) concerns future service; the liability for incurred claims (LIC) concerns service and insured events already experienced. Keeping these components separate makes a change in future estimates different from an unpaid claim.

Contract records retain insured risks, premiums/benefits, service and settlement dates, options/guarantees, cancellation/repricing rights and acquisition costs. Group admission evaluates portfolio/similar-risk management, onerous status and required issuance cohorts under AASB 17.14–24, recognition under 25–28 and boundary under 34. Cash outside the boundary belongs to future contracts, not this group's profit. GroupState retains expected dated cash flows, current discount curves, inception rates, explicit risk adjustment, CSM, loss component, LRC/LIC, coverage units, service experience and assumption vintages. Same premium totals are not sufficient state for merging different guarantees or claim tails.

General measurement follows 32–38: fulfilment cash flows are probability-weighted future net outflows, adjusted for timing/financial risk without double counting, plus explicit non-financial risk adjustment. CSM removes initial unearned gain under its conditions; an initial net loss is recognized under 47, not negative CSM. Paragraphs 40–44 separate remaining coverage and incurred claims. For a nonparticipating group, accrete CSM at the specified locked-in rate, apply future service changes at their required rate to CSM up to its available amount, record any excess loss and allocate current service over current plus future coverage units under B119. B96–B97 distinguish future service from incurred-claim changes, financial changes and current experience; not every revision is a CSM adjustment.

A MeasurementInput must resolve cash-flow/law, discount, risk-adjustment method, service grouping and coverage-unit evidence. Missing probabilities or RA support permit conditional sensitivity cases but block a source-supported measurement. Risk adjustment is the compensation the entity requires under 37; a standard deviation, confidence quantile or arbitrary percentage is only a candidate method until justified for that entity. Disclosures and the confidence-level equivalent, where applicable, are separate obligations. Market-consistent financial inputs and entity risk compensation must not be conflated.

## 12.4 Complete two-year general-measurement case

Take a two-year nonparticipating group with one premium 120 received at inception, claims expected/actually incurred and paid 40 each year, zero discount rate, RA 10 initially released 5 with each equal service year, no acquisition costs, investment component, lapses, taxes, finance changes or reinsurance. Coverage units are 1 in each year because equal benefits and coverage are assumed. Before premium payment, expected net cash outflow is $80-120=-40$, RA 10, FCF $-30$ and CSM 30, with net initial carrying amount zero. After the premium debit cash 120/ credit LRC 120. Its components are remaining expected claims 80, RA 10 and CSM 30.

Each year releases expected service claims 40, RA 5 and CSM 15: debit LRC 60/ credit insurance revenue 60. Incur claims 40: debit service expense 40/credit LIC 40; payment debits LIC 40/credits cash 40. End of year one: cash 80, LRC 60 (claims 40, RA 5, CSM 15), LIC 0

and equity/profit 20. End year two: cash 40, liabilities 0, cumulative equity/profit 40. Revenue 120 and claims 80 reconcile to premium 120 less claims cash 80. No investment component has been mistaken for revenue.

At year-one close, a separate pre-next-service change increases the remaining claim estimate from 40 to 60, RA unchanged. FCF rises 20; consume remaining CSM 15 and recognize service loss 5. LRC rises from 60 to 65, comprising FCF 65, CSM 0 and loss component 5 as a tracked portion, not an added liability on top. Debit service loss 5/credit LRC 5 with component transfer CSM→FCF 15. A decrease of 8 in the revised future claim estimate, before any additional service/finance movement, first reverses loss 5 and restores CSM 3. FCF 57 plus CSM 3 gives LRC 60. Credit loss reversal 5/debit LRC 5. This tests 48–52 ordering. Later systematic allocation of loss component to releases/finance must reduce it to zero by coverage end; it cannot be amortized as revenue a second time.

In a separate case, an already-incurred unpaid claim estimate rises 40 to 48. Debit service expense 8/credit LIC 8; do not consume future-service CSM. A discount-rate change is financial under B97; its profit/OCI route follows the selected permitted finance policy. The zero-rate base case does not remove the need for separate current and locked-in rate calculations in a real group.

For the unreversed onerous alternative, carry the loss component through the remaining service year. Opening LRC 65 comprises future claims 60 and RA 5, with CSM 0 and loss component 5 within that LRC. A declared proportional allocation assigns 5/65 of each expected claim/RA release to the loss component, consistent with paragraphs 49–52 and B124's revenue exclusions. The expected claim release allocates 60/13 to loss reversal and the RA release 5/13; the total is 5, leaving loss component 0. Debit LRC 65, credit revenue 60 and credit loss reversal 5. Separately incur and pay claims 60 through LIC. Period profit is 60−60+5=5; combined with first-year profit 20 and the earlier onerous loss 5, lifetime profit is 20. Closing cash 20 equals equity 20 and all liabilities are zero. Treating all 65 as revenue while also reversing loss 5 overstates profit by 5.

## 12.5 Nonzero discount and service allocation

A second complete case keeps premium 120, two equal coverage units and RA 10, but expected and realized claims are 44 after one year and 48.4 after two years. Both the inception and current discount curves remain a flat 10% throughout; there are no new estimates, options, fees, taxes, investments or OCI election. This RA schedule is an authored measurement input, not an actuarial calibration: it declines by 5 per year, with its entire change in the service result under 81. All other insurance finance effects go to profit or loss. The retained source supports the mechanics in 44, 80–88, B119 and B123–B124.

At inception the present value of the two expected claims is

$$\frac{44}{1.1} + \frac{48.4}{1.1^2} = 40 + 40 = 80.$$

FCF before premium is 80+10−120=−30, CSM 30, and after premium LRC 120. In year one the claim PV accretes 8; CSM accretes 3 at the locked-in rate. Allocate the resulting CSM 33 across the two current/future units, releasing 16.5. Revenue is expected claims 44 plus RA 5 plus CSM 16.5=65.5. Service expense 44 and finance expense 11 leave profit 10.5. After paying 44, cash 76 and LRC 65.5 (claim PV 44, RA 5, CSM 16.5) reconcile to equity 10.5.

In year two claim PV accretes 4.4 and CSM accretes 1.65. Release all remaining CSM 18.15, RA 5 and expected claims 48.4, giving revenue 71.55. Service expense 48.4 and finance expense

6.05 leave profit 17.1. Final cash/equity 27.6 equals premium 120 minus claims 92.4; all group liabilities are zero. Total revenue 137.05 less finance expense 17.05 equals premium 120. Setting revenue equal to cash premium each year or releasing 15 of CSM each year would ignore the specified financing. These are separate component calculations followed by debit finance expense/ credit LRC for accretion, debit LRC/credit revenue for service and the stated LIC recognition/settlement entries. Cash is deliberately assumed uninvested.

## 12.6 PAA, reinsurance and valuation methods

The preceding premium-allocation case states separate eligibility and recognition timing for gross insurance and reinsurance. PAA under 53–59 requires the one-year coverage or nonmaterial-difference condition; a short payment lag is not the eligibility test. LRC starts with received premium and applicable acquisition adjustments, then releases insurance revenue under the prescribed service pattern. LIC still measures incurred obligations using the applicable fulfilment-cash-flow rules; the allowed short-settlement discount simplification needs its own conditions. Emerging onerous facts require a loss assessment; a premium balance cannot be presumed sufficient.

A one-year policy can leave claims unpaid for several years. The distinction between the period of insurance service and the period until settlement is therefore essential even when PAA is available. The official ASBE 25 text separates three timing tests. Article 50 permits PAA when each contract's coverage is no longer than one year, or when the stated no-material-difference condition holds. Article 54 concerns a different interval: the time between each portion of service and the related premium due date. If that interval does not exceed a year, the remaining-coverage liability need not be adjusted for financing. Article 56 instead measures from claim incurrence to expected payment or receipt. Its option to omit the time-value and financial-risk adjustment requires settlement within a year of incurrence (Ministry of Finance of the People's Republic of China, 2020, Articles 50, 54, 56). The retained AASB 17 compilation makes the corresponding distinctions in paragraphs 53, 56 and 59(b).

Consider a separate group of issued policies with exactly one year of coverage. All amounts are illustrative currency units. Premium 120 is due and received when coverage starts, with no earlier onerous trigger; group recognition is therefore at that date under Article 16. Assume valid insurance grouping, uniform risk release, no acquisition costs, investment component, reinsurance, tax, other expenses or other movements, and no facts indicating onerous remaining coverage. Elect the permitted remaining-coverage financing exception. After half a year, recognise revenue 60 and retain remaining coverage of 60. Articles 53 and 57 support this roll-forward and time allocation; Article 57 requires a different allocation when risk release is not predominantly time-based (Ministry of Finance of the People's Republic of China, 2020). These assumptions isolate the claim measurement from other changes in the group's result.

At that reporting instant a claim is first incurred, with an expected payment of 60.5 exactly two years later. This is a mean estimate, not a certain settlement. Assume an appropriate flat annual effective discount rate of 10% and a separately estimated non-financial risk adjustment of 5 in current measurement units. The inputs are authored for the example; applying them to an insurer would require market and actuarial evidence. Articles 21, 23, 25 and 26 require current probability-weighted cash flows, appropriate discounting and a separate non-financial risk adjustment (Ministry of Finance of the People's Republic of China, 2020). The claim's cash-flow present value is $60.5/1.1^2 = 50$, so its liability is $50 + 5 = 55$. The risk adjustment is not discounted a second time. The claim has only just been incurred, so no intervening finance accrual enters this measurement.

Record debit cash 120/credit remaining coverage 120 on premium receipt, then debit remaining coverage 60/credit insurance revenue 60 for the service provided. Recognising the claim gives debit insurance service expense 55/credit incurred claims 55, consistent with Article 31. The group then has cash 120, remaining coverage 60, incurred claims 55 and a cumulative result of 5: $120 - 60 - 55 = 60 - 55 = 5$. These are balances attributable to the group at mid-year, not a complete insurer forecast. Using the undiscounted $60.5 + 5 = 65.5$ for this two-year claim would overstate the required liability by 10.5 and understate the result by the same amount. The shortcut is wrong under the stated rules and assumptions. A zero rate would remove that discounting difference while leaving the separate risk adjustment intact. Articles 86(3) and 97 also require disclosure of the distinct claim components and the PAA/financing choices (Ministry of Finance of the People's Republic of China, 2020).

This comparison uses identified editions. The Ministry page reproduces the 2020 ASBE 25 text; Article 120 states 1 January 2023, but that sentence does not settle every entity group's adoption schedule. Section 12.7 distinguishes the Ministry's later explanation of that timetable from a company's reported adoption. The retained December 2022 AASB 17 compilation applies to annual periods beginning on or after 1 January 2023 but before 1 July 2026. The shared result in this example establishes a scoped measurement comparison. Applying it to a particular entity still requires its applicable rules and contract facts; later amendments and broader equivalence require separate evidence.

Reinsurance has its own group, boundary, recognition time, counterparty risk, cash-flow, CSM/cost and possibly loss-recovery component under AASB 17 paragraphs 60–70A. Do not multiply every gross insurance liability by a quota share: excluded claims, limits, deductibles, timing, reinstatements and counterparty non-performance matter. Claim 100 with an eligible 50% recovery but expected 90% collectability gives conditional expected recovery 45, not 50; altering transferred insurance risk and reinsurer non-performance are different inputs. This simple calculation does not establish treaty recognition or RA by itself. Gross liability 100 remains; reinsurance recovery is a separate asset/measurement with its own authority.

For direct participation, test B101's conditions using the actual underlying items, substantial-share and variation judgments. Retain underlying fair values, entity fee share and guarantee/risk-mitigation choices; paragraphs 45 and B101–B118 specify the variable-fee adjustments to CSM, with onerous and mitigation exceptions. A local component illustration of a qualifying share: underlying-value increase 20 and entity's fixed 10% share gives candidate variable-fee change 2; eligibility, guarantees and other FCF changes must be evaluated before posting. The full option/guarantee valuation and transition calculations are further supported profiles, with source and numerical obligations, not an assumed linear fee model.

The claims-development example above supplies an estimation input, not a recognition rule. Its point projection is 148, but the single observed final factor leaves its variance unresolved. In the general method the factors are

$$\widehat{f}_k = \frac{\sum_{i \in O_k} C_{i,k+1}}{\sum_{i \in O_k} C_{i,k}},$$

where $O_k$ contains origin years with both required observed cells. Each incomplete row is projected through the remaining factors. The same triangle therefore has ultimate claims 180, 216 and 252, with unpaid amounts 0, 36 and 112. Tail development, calendar inflation, zero or negative cells and dependence need explicit extensions. The conditional means, variance scaling and independent-origin-year assumptions behind the inspected Mack error formulas (Mack, 1993, equations (1)–(3), Theorems 1–3) cannot turn this tiny triangle into a predictive law or a justified insurance risk adjustment.

## 12.7 Adoption dates and the restatement of 2022

A standard's effective date and the economic date of a reported balance need not coincide. An insurer adopting new rules in 2023 may report a revised balance for December 2022 while retaining the earlier report as a record of what it originally disclosed. The system must preserve both versions and explain which one a calculation uses.

The Ministry of Finance's explanation dated 12 June 2025 describes the original ASBE 25 timetable. It specifies application from 1 January 2023 for enterprises listed both domestically and overseas, and for overseas-listed enterprises preparing financial statements under IFRS or ASBE. Other enterprises applying ASBE were scheduled to apply the revised standard from 1 January 2026. Its account of the 2025 notice says existing adopters, including enterprises that had adopted early under the 2020 notice, must continue applying it. Insurers facing genuine difficulties and seeking deferral must submit written reasons to the Ministry's Accounting Department and the relevant institutional supervision departments of the National Financial Regulatory Administration. That statement does not establish an automatic deferral merely because preparation is difficult (Ministry of Finance of the People's Republic of China, Accounting Department, 2025, answers on background and implementation arrangements).

The same explanation limits the new simplification choices to unlisted enterprises; matters outside those choices remain subject to ASBE 25. It names grouping, measurement, presentation and transition as affected areas, including comparative information and the transition date. These are source-supported descriptions of the Ministry's explanation. The operative 2020 and 2025 notices have not yet been directly verified in this collection, so the explanation does not supply the detailed conditions needed to implement those choices (Ministry of Finance of the People's Republic of China, Accounting Department, 2025, answer on the relationship between the notice and the standard).

Ping An provides a concrete adoption history. Its 2022 ASBE annual report states that the group would adopt the revised insurance standard for accounting years beginning on 1 January 2023. Its 2023 report confirms adoption on that date and restatement of comparative information (Ping An Insurance (Group) Company of China, Ltd., 2023, Note XIV.2, pp. 315–316) (Ping An Insurance (Group) Company of China, Ltd., 2024, Note IV.34, p. 192). The following balances all refer to the consolidated group at 31 December 2022, in RMB millions. The first column comes from the 2022 report; the second is the restated comparative in the 2023 report. Changes are calculated from those columns; the asset, liability and parent-equity changes also match the three rows of Note IV.34's reported bridge (Ping An Insurance (Group) Company of China, Ltd., 2023, pp. 146–148) (Ping An Insurance (Group) Company of China, Ltd., 2024, pp. 141–143 and 192).

| Balance | Original | Restated | Change |
|---|---|---|---|
| Assets | 11,137,168 | 11,009,940 | $-127,228$ |
| Liabilities | 9,961,870 | 9,823,944 | $-137,926$ |
| Total equity | 1,175,298 | 1,185,996 | $10,698$ |
| Parent shareholders' equity | 858,675 | 869,191 | $10,516$ |
| Non-controlling interests | 316,623 | 316,805 | 182 |

The distinction between total equity and equity attributable to the parent matters. Assets less liabilities rise by $-127,228 - (-137,926) = 10,698$. Of that increase, 10,516 belongs to parent shareholders and 182 to non-controlling interests. Treating the parent-equity row as total equity would leave an unexplained difference of 182. These are changes between reported accounting bases for the same date, not cash received in a later period.

The transition also changes how later growth is calculated. Ping An reports assets of 11,583,417 at the end of 2023. Subtracting its restated 2022 comparative gives a reported-balance increase of $11,583,417 - 11,009,940 = 573,477$. Subtracting the original 2022 amount instead gives 446,249, mixing the old and new bases and understating that comparable movement by 127,228 (Ping An Insurance (Group) Company of China, Ltd., 2024, p. 141). The first difference is a movement in reported asset balances; it is not a cash-flow measure or an explanation of every economic cause of growth.

An original-source query for assets at 31 December 2022, explicitly selecting the 2022 report, therefore returns 11,137,168. A query selecting the 2023 report's restated comparison for that date returns 11,009,940. Each assertion retains its report identity, consolidated scope, RMB-million scale, accounting basis and economic date. A link records that the later assertion restates the comparison; it does not delete the original. A request for a comparable annual movement must reject the mixed-basis pair unless an explicit reconciliation has first aligned it. Because the basis and disclosure version differ, these two amounts are not a same-context source conflict.

The information date remains a separate question. The 2023 report says its board authorised the statements on 21 March 2024 (Ping An Insurance (Group) Company of China, Ltd., 2024, Note I, p. 154). Neither that authorisation nor the 31 December reporting date establishes when the exact retained document became public. The corpus has archive-date leads but no verified publication date or historical byte vintage for either selected report. A query for information available to investors at 31 December 2023 must therefore retain an unresolved availability result for this source selection; it cannot silently import the later restated balance. The corpus's October 2026 revalidation timestamps describe local evidence handling, not investor knowledge in 2023.

Finally, Note IV.34 reports financial-asset reclassification, including use of the classification overlay for relevant assets derecognised during 2022. It also states that some insurance groups used modified-retrospective or fair-value transition because retrospective application was impracticable. The headline bridge consequently cannot be reconstructed by applying one percentage to every insurance liability. The example establishes how to retain and compare disclosed histories; reconstructing the transition itself still requires the affected groups, asset classifications and underlying measurement inputs (Ping An Insurance (Group) Company of China, Ltd., 2024, Note IV.34, p. 192).

## 12.8 Interfaces, exclusions and acceptance cases

Insurance events are group_recognize, premium_settle, service_release, claim_incur, claim_settle, assumption_change, onerous_change and treaty_change, each bound through the event-basis, authorization and commit contracts to the exact group/version, evidence and information view. Journals and component roll-forwards must reconcile independently; a component reallocation is not always a net liability change. Invalid grouping, missing rate/law, unsupported option valuation, ambiguous contract boundary or unknown service units produces the corresponding unresolved/outside-scope result. Failed numerical paths retain their coverage/mass, without successful-only normalization. Modified contracts need the applicable derecognition analysis; no new ID can erase unfavorable old-group obligations.

Initial transition requires an actual method choice and eligibility for full retrospective, modified-retrospective or fair-value transition under AASB 17 Appendix C, with acquisition, grouping, past service and historical-rate records. This collection does not claim an executed transition or a general VFA valuation. The nonzero-rate general case and onerous allocation through expiry above are finite specification examples. Required later acceptance still includes a priced guarantee, acquisition/reinsurance interaction, transition and actuarial backtesting. These retained obligations prevent the two-year illustration from being described as full insurance-domain readiness.

# Chapter 13

# Implementation contracts

## 13.1 Reference implementation and component boundaries

The proposed reference implementation uses Python for orchestration and a deterministic reference evaluator, exact decimal/rational arithmetic for booked amounts, SQLite for transactional metadata and versioned records, and columnar files for large scenario outputs. DuckDB is an optional analytical reader. These are convenience choices justified by inspectability and modest initial scale; they do not establish production performance. The IR and conformance tests allow a later compiled executor without changing accounting meaning. Committed journals, economic deltas, receipts and branch heads belong to the one transactional store, including for simulation branches. Columnar files hold derived analytical exports, never independently writable committed books.

```
accdsl/
  evidence/       documents, spans, filing contexts, extraction versions
  model/          typed IR, entities, books, symbols, contracts
  language/       grammar, parser, resolver, type checker, lowering
  constraints/    propagation, solver adapters, bounds and conflict cores
  accounting/     rule selection, posting validation, closing, consolidation
  economics/      contract and behavioural plugin interfaces
  execution/      scheduler, branch state, stochastic streams, run records
  reporting/      statements, notes, movement schedules, explanations
  storage/        versioned records, migrations, result readers
  interfaces/     Python API, CLI, structured import/export
  packages/       reviewed demonstration and accounting modules
```

The first service boundary is local and synchronous. A job abstraction can be added for extraction and simulation without requiring distributed execution. The model compiler returns an immutable execution plan; the executor receives that plan and run inputs. Neither should retrieve live market data implicitly. Input datasets and rule packages are selected and versioned before execution.

## 13.2 Persistent records

The table identifies logical storage relations and their obligations; exact versioned record schemas determine their payloads. Candidate calculations and diagnostics remain separate from the committed accounting relations.

| Relation | Key information and constraints |
|---|---|
| `document` | Hash, original path/URI, filing ID, entity, publication and availability times, media type, revision links. Original bytes are retained. |
| `source_span` | Document hash, page/element locator, bounding box, text/table representation, extraction run ID. |
| `observation` | Context ID, value or missingness tag, precision domain, source-span IDs, semantic-reading ID, review status. |
| `context` | Entity/perimeter, concept, basis, currency/unit, instant or interval, dimensions and taxonomy version. Contexts are immutable. |
| `assertion_version` | Logical assertion ID, payload hash, economic validity, ingestion time, public availability, superseded version and reason. |
| `model_node` | IR node ID, type, serialised payload, dependency IDs, package/model version; no executable arbitrary text. |
| `constraint` | Expression, scope, source/assumption references, supported solver fragment, applicable alternative condition. |
| `alternative` | Parent model, added conditions, rationale, evidence, coverage declaration and rejection history. |
| `economic_event` | Stable entity/book/branch/business-event key; versioned event, contract/cohort, dates, quantities and dependencies. |
| `branch` | Branch key and immutable definition reference, including entity, book and required synthetic or real book mode. |
| `branch_head` | Branch owner, current state reference and generation; updated only by the expected-head transaction. |
| `posting_authorization` | Exact basis, posting profile, method, authority, findings, permission and historical view. A permission record alone does not establish a committed event. |
| `journal_batch` | Committed balanced postings with event key, book/basis, date, method and measurement references. Candidate journals are separate; membership here requires the complete accounting transaction. |
| `economic_delta` | Committed contractual and physical state change, bound to the same basis and authorization as its journal. |
| `committed_receipt` | Unique stable event key, full request fingerprint, original request, opening/closing state, journal and delta references. Published atomically with those records and the advanced head. |
| `review_view` | Immutable parent view, decisions and dependency selection revisions; preserves historical interpretation. |
| `dependency` | Typed source and derived-record edges, including candidate-set membership needed for transitive invalidation. |
| `search_checkpoint` | Semantic manifest, partition, explored/unresolved regions, frontier, used budget and result references, committed together. |

| Relation | Key information and constraints |
|---|---|
| `run` | Evidence and model hashes, cutoff, scenarios, branch selection, weight model, solver/numerical settings, seeds and environment. |
| `report_cell` | Run, branch/scenario, context, amount or result family, dependency references and completeness status. |

Persistent identifiers are opaque; display labels can change without changing identity. References use foreign keys or validated content-addressed links. The full logical key of a reported fact includes its dimensions, not merely company, label and year. Duplicate assertions are stored as distinct source claims until a documented duplicate-resolution operation relates them.

Version records are append-only. A derived transaction-time interval can be closed by the next version event without deleting the original payload. Validity intervals can overlap when assertions conflict. Such overlap is diagnosed and resolved by the evidence view, not prohibited by a database constraint that would erase one assertion. Scenario states use structural sharing of immutable historical nodes and branch-local changes.

Large simulation output is indexed by run, explanation, parameter draw, scenario, innovation path, entity, book, date and account. Reducing this to means across all dimensions would destroy the product's central feature. Online summaries may compress draws within a declared distribution, while retaining conditional summaries and representative replay identifiers. A retention policy that removes individual traces must state which explanations can still be reconstructed.

## 13.3 Public API and result types

The following unversioned names describe the user workflow. They are conceptual shorthand, not the portable operation registry. The exact versioned requests are specified below; Python, command-line and any service binding must exchange the same typed values:

```
ingest(manifest, extractor_config) -> EvidenceVersion
compile(source_or_ir, evidence_view, package_lock) -> ModelPlan | Diagnostics
reconcile(plan, cutoff, solver_config) -> HistoryFamily | Diagnostics
condition(family, assumptions) -> ConditionalFamily | Infeasible
simulate(family, scenario_set, run_config) -> RunResult
render(run_id, report_request) -> ReportBundle
explain(node_id, evidence_view_or_run_id) -> ExplanationGraph
compare(run_a, run_b, alignment_spec) -> Comparison
```

`HistoryFamily` contains a model-scope declaration, shared variables, constraints, finite choices, feasibility status and solver-completeness status. It may contain witness histories without claiming that those witnesses enumerate the family. `RunResult` is a tagged union of concrete outputs, conditional samples, certified bounds, explored ranges and unresolved outputs. Probability-bearing results reference a probability-model record. Results without that reference cannot be presented as probabilities.

Comparison requires an alignment specification: entity/perimeter, dates, reporting basis, scenarios and shared innovations where applicable. Comparing IFRS and US GAAP books holds economic events fixed and changes accounting rules. Comparing management responses holds the specified scenario and initial information fixed and changes the behavioural model. These are different questions and must not be combined into an unexplained difference.

## 13.4 Versioned requests and independent results

The workflow names above become portable operations only when their arguments, input versions and effects are fixed. Each request names an exact operation, profile and signature; an immutable view; typed arguments; execution settings; and the expected head for a mutation. The body is closed:

```
OperationRequest {
  operation, request_id, workspace_key,
  profile_ref, signature_ref, view_ref, arguments,
  execution: {mode, budget, environment_ref},
  expected_head: null | {owner_key, state_ref, generation}
}
```

All fields are present. The mode distinguishes a specification example from a runtime request. Budget states operation-count and elapsed-time limits, or a reason either is inapplicable. The expected head appears only in this envelope. For accounting its owner is the entity/book/branch; for review it is the review namespace; for probability selection it is the family/evidence/time owner. Resolving a mutable "current" alias happens before constructing the request, and a mutation compares the resolved revision again.

The fingerprint covers the complete request, including its request identifier, expected generation and execution settings. In accounting the lookup namespace is still the business-event tuple in Section 8.5, not request ID. Changing the request ID for a committed event changes the fingerprint and conflicts. Source and review mutations use their own operation/workspace/request namespace. A transport correlation identifier and an economic event are not interchangeable identities.

The result preserves the submission even when decoding fails. Once exact-byte retention succeeds, it has the following fields:

```
OperationResult {
  submission_ref, request_ref, operation, input_view_ref,
  disposition, statuses, payload, diagnostics, coverage,
  unresolved, continuation_ref, commit_observation,
  receipt_ref, evidence_mode
}
```

The validated request reference remains null until all required input checks, including bindings and authorization, pass. Rejected input has null payload. If submission retention itself fails, the caller receives a transport failure with its original correlation, not a fabricated stored submission or a known commit outcome. A read can inspect an earlier committed receipt while its own commit observation says no new commit occurred.

Completed, partial, rejected, unsupported, cancelled, internal failure, unknown commit and replay are operation dispositions. They do not replace the separate structural, mathematical, computational, empirical, accounting, forecast, product and human statuses. Computing a zero normalizer can complete while conditioning on that event is undefined. A valid symbolic family can have only partly explored regions. A supported insolvency scenario can be fully evaluated while its company defaults. These distinctions prevent an infrastructure error from erasing an economically important path.

For example, let an admitted law assign mass 3/5 to a path with cash 120 and 2/5 to an unavailable cash calculation. The output retains both regions and their masses. If separate evidence bounds the unavailable cash by $[80, 110]$, the unconditional mean is bounded by

$$\tfrac{3}{5}(120) + \tfrac{2}{5}[80, 110] = [104, 116].$$

Without those bounds, even this mean interval is unsupported. Reporting a mean of 120 would silently condition on computational success.

Portable-only validation proceeds through decoding, field shape, reference integrity, cross-record binding and time context, capability and admission, calculation, transaction comparison and publication. Native interfaces preserve their own explicit stage order; the native accounting sequence checks recognizable dependencies before shape and source versions before bindings. At the earliest failed stage, return every independently decidable diagnostic in canonical field order. A malformed parent prevents its dependent checks, but does not hide unrelated sibling errors. Later stages are marked unevaluated.

Appendix A gives the exact operation names and argument meanings. Financial method names such as delivery or claim settlement are versioned members of a package; they are not invented new platform endpoints. In particular, continuous inference and causal methods need their own admitted extension signatures before the finite/exogenous initial profile can execute them.

## 13.5 Correction changes a view without rewriting its history

A correction can affect a result that never explicitly selected the corrected source. Suppose a query selected one available report because it was the only eligible candidate. Admitting a second report may create a conflict. The query therefore depends on candidate-set membership as well as on its selected record. Invalidation must follow both dependencies transitively.

Each correction creates an immutable new view with parent and changed decisions. Old frozen outputs retain their original views. A current query must check ancestor dependencies even for descendants created after an invalidation scan; otherwise delayed computation could revive a stale reading. Unrelated closures remain usable. Self-supersession, incompatible replacement scope and a missing required predecessor reject before mutation.

The mathematical correction and the audit of affected work are separate records. A `CorrectionBasis` depends on the unaffected base law, replaced reading, corrected source, target admission and replacement method. An `InvalidationReport` additionally depends on the inspected store snapshot and lists affected descendants. Computing another optional descendant changes the report, not the corrected law. This also keeps the content-reference graph acyclic.

Probability calculation, admission and selection are three effects. A selected law belongs to the owner consisting of family, target evidence view, target time and purpose. Correcting a source does not mutate the old owner's selected pointer or automatically select the new candidate posterior. In the new owner, an inherited stale selection returns selection superseded with an explicit readmission and reselection obligation. With no inherited selection it returns selection required. A historical query naming the old view remains valid in that frozen scope.

## 13.6 Durability and migration have observable obligations

One logical transactional store owns records, dependency edges, review views, branch heads, journals, economic deltas, receipts and search checkpoints. Original documents reside in immutable byte storage. Their exact bytes must be durable and verified before a metadata transaction refers to them. A crash between blob creation and metadata commit may leave an unreferenced blob; it must not leave an accepted reference to absent bytes. Analytical tables, caches and PDFs derive from a named snapshot and do not become separate writable versions of truth.

The proposed local store serializes commits even when several clients submit requests. A single

writer still needs expected-head comparisons. An event-log implementation is also possible if its commit marker, projection visibility and replay obey the same observable rules. Neither a document trace nor a choice of SQLite establishes physical-media durability or multiwriter throughput. Those require crash, reopen and fault-injection evidence in the selected environment.

A search checkpoint commits the semantic manifest, partition identity, explored and unresolved regions, frontier, used budget and result references together. A cell is not complete before its result and frontier transition are durable. Resume names that exact checkpoint and manifest. Increasing a scheduling budget can continue it; changing evidence, methods, scenario, family or traversal meaning requires a new manifest and an explicit reuse relation. Cancellation preserves the last committed frontier.

Export first stages a closed manifest, verifies every required record and blob, then publishes the manifest atomically. An incomplete export remains explicitly incomplete. Import verifies required bytes before committing metadata and grants no new financial admission. A failed report export cannot undo a journal or justify posting it again.

A representation-only migration retains old and new bytes and IDs plus a field/meaning mapping. Its input names the old and new schemas, migration method, source snapshot and declared semantic relation; the common request envelope supplies the expected head for its transaction. It must preserve typed quantities, context, uncertainty category, reference closure, admission, event identity and supported reference query results. Changing a rounding policy, calendar, recognition rule or probability law is a new model, not a format migration. A failed comparison leaves old heads and objects usable.

Exact reruns compare values on the same evaluated domain. An elapsed-time cap can visit different search cells on different machines, so equal partial coverage cannot be presumed. Reproducible prefix comparisons need a pinned traversal and operation-count budget; resume uses the saved frontier. Approximate methods use their declared financial error comparison, and random streams name economic owner, date, shock kind and draw index independently of branch order.

## 13.7 Plugin interfaces

Every plugin declares a version, supported types, input information dates, determinism, numerical measure and capabilities. A contract plugin implements `schedule`, `observe`, `transition` and `payoff`; it may return future conditional events. A behaviour plugin implements `decide` and declares which state and scenario information it uses. A measurement plugin implements `measure` and returns an amount or vector with a basis and numerical status. A reporting plugin maps classified journals and balances to output contexts.

The plugin response must include all inputs that materially affect results. An external calculator's global evaluation date, market-curve cache, calendar version or random seed cannot remain invisible. For example, QuantLib's floating coupon interface makes fixing-date conventions and pricing objects relevant inputs; the adapter records them rather than relying on process-global state. ACTUS risk-factor observations receive explicit dates and identifiers.

The constraint interface accepts a typed formula and a requested operation: feasibility, witness, enumeration of declared finite choices, or bound. It returns `sat`, `unsat` or `unknown`, plus evidence appropriate to the operation. Timeouts and unsupported theories return `unknown`. The history layer exposes that solver outcome as `SolverIncomplete`, retains the affected region and marks its coverage unresolved. These are layer-specific descriptions of incomplete computation, not infeasibility. A witness proves existence within the formula, not completeness of a recovered historical family.

## 13.8 Human review and model-assisted extraction

The review interface should put the source passage beside the proposed observation, context and reconciliation impact. An analyst can accept a mapping, add an alternative, reject an extraction, define a symbolic unknown or request more evidence. These actions create structured changes with reasons. The useful unit of review is a coherent disclosure and its links, not an isolated number detached from its note.

A language model can identify candidate tables, draft concept mappings, propose contract terms and suggest alternative readings. Its output must reference source spans and a schema. Deterministic checks reject fabricated source references, invalid units, inconsistent periods and unbalanced postings. A model-generated interpretation that passes those checks remains a proposal until its evidentiary status is established. Token probabilities and self-reported confidence are never automatically used as weights over economic histories.

For routine local research use, a lightweight interface and immutable source versions are sufficient. Multi-user approval hierarchies, cryptographic attestations and deployment security systems are outside this initial design. They can be added for a concrete operational requirement without changing the scientific representation.

## 13.9 Failure recovery and reproducibility

Extraction jobs and simulation jobs write explicit progress records and atomic results. Restarting a failed document conversion reuses its document hash and creates either the same idempotent extraction result or a new version under changed settings. Restarting a simulation resumes only from a committed state whose model and input hashes match the run. Numerical failures retain the inputs needed for a focused diagnostic.

A run manifest records code revision when available, package lock, source hashes, cutoff rule, model hash, scenario data, priors or weights, seed/stream scheme, solver settings, decimal and rounding conventions, calendars, environment and result locations. Absence of a usable Git revision is recorded as unavailable with content hashes, rather than inventing a commit.

## 13.10 Review decisions have consequences for later results

A proposed reading starts in `Proposed`. Acceptance records a reviewer, date, reason, verified locator and context, together with the reconciliation result. Acceptance means that the stated layer has been reviewed: accepting an extraction of a company's sentence does not accept its accounting treatment. Competing defensible readings enter `AcceptedAlternative` under a choice group. Rejection preserves the source and reason. A correction creates a new reading; replacing an accepted reading requires an explicit `Supersede` action. A prior reading therefore remains reproducible without entering the latest model.

Suppose OCR reads Northstar's original 120 as 720. The source span supports 120; correcting extraction creates a new reading linked to the erroneous one. The later revenue correction from 120 to 110 is different: it changes the company's reported assertion. The original extraction remains faithful to its document. Finally, a reviewer may accept that two notes conflict without accepting either as a unique model constraint. The view then contains a conflict set requiring alternatives or further evidence.

A changed reading or mapping invalidates descendant constraints, families, measurements, reports and cached runs for a new view. Unaffected dependencies can be reused. Old frozen

runs retain their original decisions and evidence IDs; they are marked stale for current use, not overwritten. A journal derived under the old view is never silently attached to a new source. The local workflow needs no multi-user approval hierarchy to enforce these rules.

## 13.11 Identity must survive harmless reordering

The portable serialisation in Section 7.3 preserves decoded Unicode code points without global NFC normalisation. It uses sorted ASCII keys and exact rational financial quantities, while integer metadata remains integer. Only schema-declared sets are sorted. Source identity hashes original bytes. The old Core 0 encoding survives only for its historical fixtures through an explicit legacy import; it does not define new portable digests. Neither encoding proves arbitrary economic equivalence.

SHA-256 with a domain/version prefix identifies three different closures. Evidence identity includes source bytes, locations, extraction and review versions. Model identity includes typed IR, assumptions, rules, calculator locks and resolution, while excluding formatting and source byte offsets. Run identity adds the evidence view, horizon, scenarios, outputs, calendars, numerical policy, random-stream scheme and environment. A corrected source can change evidence and run identity without changing the numerical model; a new rate assumption changes the model as well.

Each economic business event has a stable identity within its entity, book and branch. Its lookup key is the four-tuple in Section 8.5; event version, payload, expected head and request settings belong to the full fingerprint. Exact replay returns the original receipt before a new-event freshness check. Changed inputs under that key conflict. A new run label or version cannot make the event post again. Independent events may reorder only when read/write sets are disjoint or a package proves commutation, including funding constraints. Alphabetical ordering cannot settle an economic timing ambiguity.

Stochastic keys contain seed, coupling group, entity, contract/cohort, economic date, shock kind and draw index. Branch enumeration order is absent. Shared coupling groups may use the same uniform draw with different marginal transforms. The reference fixture maps the first 64 bits of a SHA-256 key digest, interpreted as unsigned big-endian integer $v$, to $(v + 1/2)/2^{64}$. This fixes example replay; it does not validate a production Monte Carlo generator. Research runs select an assessed counter-based generator and record transforms and numerical environment. Inner measurement draws have different shock domains from outer realised defaults.

Schema migration preserves old and new payloads and a field/meaning mapping. A representation-only migration must demonstrate equivalent supported semantics; a changed recognition rule creates a new model. Restart resumes only a committed state whose locks and identities match. SQLite owns source, review, accounting and run metadata transactions; content-addressed files hold immutable bytes and JSON, while DuckDB reads versioned analytical exports. Each mutable transition has one owner.

## 13.12 Adapters exchange typed records

The portable interfaces described in this chapter govern new exchanges. The original `spec/core-contract.md` and `spec/core-ir.schema.json` retain the historical Core 0 examples. A named decoder and comparison link those examples to the portable records; their older fields do not silently extend a new closed signature. Every method returns a success record or a diagnostic with code, source IDs, entity/period, dependency path and retry status. The principal sequence

is `ingest`, `review`, `compile`, `reconstruct`, `simulate`, `report`, with `bound`, `explain` and `replay` over frozen identities.

Arelle supplies candidate facts with QName, entity, context, period, dimensions, unit, accuracy, nil status and locator. Duplicates survive for review. Its archived `ModelInstanceObject` interface exposes these properties; validation success does not establish economic truth. Docling supplies page/table proposals, locations and conversion status. Missing headers and failed pages remain unresolved. Z3 receives named rational-linear assertions and finite choices; witnesses are checked against exact inputs and unknown results retain their reasons. The reference checks exercise Z3; the other selected adapters remain interface designs grounded in inspected official source snapshots.

ACTUS receives contract terms, calendars and dated risk factors and returns schedules, transitions and payoffs. AccDSL applies accounting separately. QuantLib receives an explicit evaluation date, instrument, curves, fixings, calendar/day-count and measure; it returns dated measurements with numerical status. No process-global valuation date may leak between branches. Beancount exports concrete journals for later differential checks; it does not own symbolic families. Import failures retain request ID and source hash so retries are idempotent. Licence/version evidence remains in the software ledger; integration acceptance requires pinned releases or source digests.

## 13.13 A successful extraction can still lose the fact

Consider an Inline XBRL amount whose literal is `1,234`, whose admitted numeric transform treats the comma as grouping, whose scale is $-3$, and whose context and unit identify a particular dollar observation. The resulting amount is $1234 \times 10^{-3} = 1.234$. Scaling that normalized value again gives 0.001234 and is wrong. An adapter must retain the literal, transform, scale, context, unit, nil status and original occurrence so that this error can be found.

The same principle applies to dates and dimensions. A lexical instant of 31 December remains distinct from the exclusive 1 January boundary used for an interval. Publication and ingestion are separate timestamps. Two equal numbers with different contexts remain separate occurrences. A nil fact is unknown, not zero. Dropping a typed dimension or context reference yields a preservation failure for that fact even if the resulting Markdown looks clear.

The inspected Arelle interfaces support native XBRL contexts, dimensions, transformations and validation. Its internal object IDs do not replace stable source identity. The inspected Docling source offers useful layout and table proposals, but its examined XBRL graph path does not retain every required fact/context/unit/typed-dimension field. That path needs a preservation sidecar or must report unsupported conversion for affected records. These are source audit findings, not measured accuracy rankings of the tools.

The retained Apple annual filing makes the dependency problem concrete. Its issuer archive includes the instance, the issuer's schema and its calculation, definition, label and presentation linkbases. Those files also refer to US GAAP and SEC Reporting Taxonomy definitions, SEC schemas and XBRL definitions held elsewhere. The issuer archive alone therefore cannot supply every definition needed to interpret its tagged facts. For this particular archive, a static inspection now resolves the recorded imports, link references and typed-domain references to retained files, checks target namespaces and fragment identifiers, and checks that instance facts have declared concepts, contexts and units. This establishes a source dependency map. XML Schema and XBRL validation, the issuer's interpretation of each concept, and the completeness of the wider disclosure package remain separate questions.

An offline parser needs more than a map of intended filenames. At the inspected Arelle source

version, the cache lookup can return a local pathname when the file is absent; subsequent loading records an `IOerror`. The proposed adapter must check each mapped file and its hash before loading, retain the missing dependency and affected records, and return an unresolved-dependency result. It must not infer successful retrieval from a nonempty pathname or silently substitute a newer taxonomy. Original schemas remain unchanged; an external resolver or isolated cache associates their declared addresses with the retained bytes. Any built-in cache fallback must belong to the same checked source set (Arelle contributors, 2026).

Nor is a successful API return necessarily a validation verdict. In the same source version, the controller's return value takes accumulated validation errors into account only when `validationExitCode` is enabled. The proposed adapter sets it explicitly, requests validation, disables automatic formula execution, and examines the returned status, diagnostic messages and model errors together. It keeps the model open while copying the necessary facts, contexts, units and original occurrence references, then closes the session. One session runs at a time in its owning thread within a process; the inspected API uses shared package and plugin state. These are source-grounded interface requirements for a later implementation, not results from an executed Arelle adapter (Arelle contributors, 2026).

Each adapter therefore pins code, dependency, model and taxonomy versions, licenses, input and output schemas, transformations and field-loss behavior. Preparation preserves bytes and receipts before parsing, inventories the package, attaches occurrences and contexts, records normalization, checks required fields independently and submits candidates for review. Partial conversion retains both successful units and the exact failures or unexplored frontier. Conversion counts cannot accept a package or an accounting treatment.

The proposed extraction comparison uses the same retained files, roots, budgets and taxonomy resources. Manual located source units and transparent native extraction provide the reference; applicable pdfplumber/Arelle configurations, plain Docling and explicit context/footnote repairs form the candidate ladder. Transcription, context, table structure, mapping, source support, retention, historical admission and cost are separate outcomes. Lost hashes or locators, look-ahead, silent omissions and untracked downloads veto the affected exchange. The current 31 development annotations cannot establish a heldout accuracy ranking, and cell counts within one report are not independent observations.

Reuse also has distribution conditions. Arelle's Apache and Docling's MIT code licenses do not by themselves clear every model, taxonomy or dataset. ACTUS technical-specification reuse retains its share-alike obligations; QuantLib, Z3 and ledger tools retain their own code and dependency terms. The Beancount and hledger comparison needs an explicit distribution boundary for copyleft components. License admission records actual versions, notices, models and data, rather than treating a repository's headline license as universal.

The actual taxonomy notices show why that distinction changes the design. The inspected FASB notices permit specified financial reporting and unchanged incorporation in works that explain or assist implementation, subject to their notice and third-party conditions. They do not provide a general permission to edit taxonomy definitions. Keeping the original files and recording location mappings separately respects that distinction. The notices retrieved for this inspection are unversioned; their relationship to the terms attached to the 2024 files still needs to be established before a distribution decision. XBRL's public-document notice separately directs software and schemas to a Software Notice, so permission for an explanatory document cannot simply be applied to a schema bundle. Exact dependency licences, component-specific terms and the intended distribution remain explicit preparation tasks (Financial Accounting Foundation, 2026b,a; XBRL International, 2026).

# Chapter 14

# Acceptance and implementation sequence

The reference cases now make several of the acceptance obligations executable as design checks. Northstar provides full statements and a known compatible history; Apple supplies real source contexts and a bounded marginal stress; bank, insurance, consolidation and inventory fixtures test the core's broader state requirements. They do not constitute a production implementation.

## 14.1 Three different questions require different evidence

Engineering correctness asks whether the implementation follows the specified types, transitions and arithmetic. Accounting validity asks whether the selected rules faithfully implement the applicable standards and disclosed policies. Scientific validity asks whether the economic assumptions and uncertainty representation support the intended forecasts or comparisons. A balanced journal answers only part of the first question.

The implementation must therefore maintain separate validation results. Compilation and deterministic examples cannot establish forecasting quality. Agreement with a public report cannot identify an undisclosed contract term. A lower extraction error on a document benchmark cannot establish that the engine preserved every materially different scenario response.

## 14.2 Core conformance cases

| Test | Required case | Pass condition |
|---|---|---|
| T01 | Exact debit/credit batches, including reversals and period closing | Every committed book conserves signed balances; imbalance rejected atomically. |
| T02 | Money, quantities, currencies, instants and interval flows | Invalid additions and context combinations rejected; explicit conversions traced. |
| T03 | Original filing, later correction and as-known cutoff | Earlier query reproduces the original evidence view; later view shows revision. |

| Test | Required case | Pass condition |
|---|---|---|
| T04 | Annual and interim cumulative flows with revised comparatives | Quarter derivation only across compatible contexts and vintages. |
| T05 | Fixed/floating borrowing with identical historical postings | Both explanations survive; next-year expense is 5 and 7 under the stated shock. |
| T06 | Continuous fixed share $\alpha \in [0, 1]$ | Exact model returns $I = 7 - 2\alpha$ and extrema 5, 7 without assigning probabilities. |
| T07 | Duplicated encoding of one history | Canonical duplicates preserve results and declared weights; no extra probability mass from an additional encoding. |
| T08 | Linked interpretation across debt classification and payment treatment | Impossible combinations rejected through shared constraints. |
| T09 | Rounded totals, duplicate facts and contradictory notes | Correct interval compatibility; no automatic zero or balancing plug. |
| T10 | Infeasible or unsupported constraint problem | Conflicting sources exposed; timeout/unsupported case reported as unknown. |
| T11 | Zero-coupon bond and ordinary equity examples | Carrying amounts, payments, share quantities and report mappings match derivations. |
| T12 | Foreign-currency asset, translation and cash movement | No mixed-unit balance test; translation effects separated from cash flows. |
| T13 | Acquisition and intragroup balance elimination | Perimeter and adjustment book reproduce group totals without inventing capex. |
| T14 | Conflicting recognition rules and unknown guards | Explicit conflict or symbolic partition; file order has no effect. |
| T15 | Insufficient cash with constrained funding | Funding failure/default retained as an outcome; no hidden borrowing. |
| T16 | Expected-loss measurement before future default is realised | Measurement uses only information allowed at that date. |
| T17 | Repeated/reordered independent events and interrupted execution | Idempotent replay; branch isolation; identical committed outputs. |

| Test | Required case | Pass condition |
|---|---|---|
| T18 | Conditional distributions and optional weighted mixture | Branch distributions retained; mixture agrees with declared weights and total probability; unweighted family never labelled probabilistic. |

For synthetic cases, expected histories, excluded histories and scenario consequences are known by construction. Tests should include counterexamples that balance but have the wrong economic effect, such as treating a non-cash acquisition movement as operating cash. Such tests discriminate between arithmetic validity and the intended semantics.

Differential tests against an existing ledger check only the shared supported subset. ACTUS schedule comparisons check contractual events, and QuantLib comparisons check selected valuation calculations under matched conventions. Disagreements must be classified as different conventions, unsupported cases or implementation errors before one tool is treated as an oracle.

## 14.3 Extraction and reconstruction evaluation

A company corpus must include different standards, industries, languages, filing formats, extensions, interim calendars and restatements. Company and time splits are required so repeated report templates or revised versions do not leak into evaluation. Public benchmark tables are useful development material, while acceptance uses complete disclosure packages that include notes and their relationships.

Reviewers annotate source facts, contexts, mappings, required relationships, known ambiguities and unsupported assertions. Relevant measures include correct source-linked values and contexts, unresolved extraction volume, coverage of note obligations, reconciliation failures and invented facts. Report each separately. A single overall score can hide a serious error in a small but consequential financing note.

Historical reconstruction should test whether the known synthetic history remains feasible, whether known incompatible histories are rejected, and whether economically distinct alternatives are preserved. On real companies, public data generally cannot establish the true journal. Review instead assesses evidential compatibility, unsupported commitments and conditional economic consequences. Later disclosures may provide discriminating evidence for previously ambiguous terms.

No universal numerical threshold for semantic coverage or extraction accuracy is fixed in this document. Before a release evaluation, an accounting-reviewed test corpus and task-specific acceptance thresholds must be recorded in an evidence contract. Hard conformance failures, fabricated citations, silent discarding of valid alternatives and undocumented accounting adjustments veto acceptance regardless of an aggregate score.

## 14.4 Evaluate the DSL against the strongest practical baseline

The comparison uses the same IR, accounting examples and tasks. One group of models is authored through the typed Python API; another through the DSL. Both receive equivalent documentation and examples. Tasks include encoding a movement schedule, adding an alternative

interpretation, changing an effective date and tracing a simulated amount. Measure correctness, material omissions, time to a reviewed result, ease of explaining a change and maintenance effort.

This comparison tests whether the notation and compiler help users; it is not a comparison between careful DSL code and unstructured spreadsheets. If the DSL offers little advantage, retain the shared IR and library and narrow the textual language. Small pilots nominate improvements. A claim of superiority requires a predeclared comparison and uncertainty evidence, not a few favourable demonstrations.

## 14.5 Forecasting evaluation comes later

The proposed experiment asks whether a conditional company method adds out-of-time predictive information while preserving statement coherence and unresolved exposure. The comparator ladder uses the same available sources and targets: last-observation and seasonal naive forecasts; contractual runoff and accounting-identity forecasts; tuned classical statistical and conventional financial models; the structural model with one selected interpretation; the same structural model preserving alternatives; and declared extensions. This separates changes in economics from changes in the representation of ambiguity. Tuning stays within training windows. Test issuers or time blocks remain untouched, and unavailable predictions stay in the reported denominator.

For probabilities, predeclare proper scores, calibration, interval coverage, sharpness and dependence-sensitive joint checks. Unweighted feasible sets have coverage and width under a stated set interpretation, not probability intervals. Promotion needs a declared score difference with uncertainty evidence on matched eligible targets and no accounting or information-time veto. Issuer and time dependence require a justified paired resampling or equivalent inference design; overlapping reports are not independent replications. Runtime, a few losses and small-seed tail differences remain descriptive without that support. No such forecast experiment has been run for this proposal.

Look-ahead, broken conservation, a changed target or missing scope can invalidate an experiment. A valid comparison in which the proposed predictor performs poorly instead rejects that candidate and motivates its stated repair. The two outcomes must not be conflated when choosing the next experiment.

Forecast evaluation uses historical information cutoffs, fixed data vintages, declared economic scenarios and proper separation of training and testing periods. Every level of the comparator ladder above uses that same information boundary, so extra information cannot silently account for an apparent benefit.

Construct the inexpensive comparators before choosing a complex forecasting method. A no-growth operating forecast with scheduled contractual financing tests whether the model adds value beyond existing obligations. A constant margin and turnover model tests whether detailed operations improve on familiar financial ratios. A trailing-average growth model tests whether fitted dynamics improve on recent experience. A fixed-exposure contractual stress model tests whether estimated management responses help beyond repricing the existing book. Each comparator must generate coherent accounts through the same reporting rules and use the same historical information. In synthetic cases, the known hidden state also supplies an oracle conditional on the stipulated scenario.

Evaluate these comparisons separately under stable conditions, large rate resets, refinancing pressure, credit deterioration and changes in reporting perimeter. A pooled average can conceal failure in the situation the product is intended to explain. Supported underperformance against

a cheap comparator in a declared critical situation vetoes promotion of the complex forecasting method and must lead the result discussion. These checks are falsification tests, not training targets or evidence of superiority when merely passed.

Evaluate point forecasts where they are a stated target, distributional calibration when probabilities are justified, and decision-relevant robustness when only feasible sets are available. A wide feasible set is not automatically a good prediction, and an unweighted set cannot be assessed as if it were a 95% predictive interval. In multi-scenario stress tests, scenario severity and probability remain separate inputs.

Any serious numerical comparison must follow the repository's research plan, assumption audit, baseline, veto, uncertainty and result-record policies. The present document runs only deterministic checks of its own examples. It neither selects a scientific forecasting default nor reports comparative performance.

## 14.6 Capability contracts for sector packages

| Package | Required state and operations | Minimum acceptance evidence |
|---|---|---|
| Core corporate | Working-capital bridges, lots/assets, ordinary debt and equity, period closing and cash-flow mappings | Exact synthetic cases, reviewed public filing reconstruction, all statement reconciliations. |
| Financial instruments | Terms, calendars, resets, options, measurement categories, modifications and hedges | Contract schedule comparisons, valuation checks, paragraph-based posting cases and transition cases. |
| Banking | Credit cohorts, deposits, collateral, liquidity, regulatory capital, management actions | Allowance roll-forwards, repricing and runoff stresses, capital reconciliation and default/funding-failure cases. |
| Insurance | Policy groups, future cash flows, service and finance components, reinsurance and asset interaction | Cohort movement reconciliation, assumption change cases, service recognition and reviewed actuarial calculations. |
| Multi-basis reporting | Versioned IFRS/US GAAP/ASBE rules, entity elections and transition treatment | Matched economic-event cases with explained basis differences and independent accounting review. |

Every package declares capabilities at operation level. "Supports bonds" is too vague; "supports fixed-rate, single-currency, non-callable amortised-cost issuance and redemption under this reviewed rule set" is implementable and testable. Unimplemented operations retain their source and contract representation while producing an explicit capability error when execution requires them.

## 14.7 Implementation milestones and dependency order

| Stage | Concrete implementation deliverable | Exit evidence |
|---|---|---|
| P0 | Portable records and contextual types; source/review/time selection; typed constructors and versioned IR; separate posting authorization and atomic journal/delta/state/receipt transactions for the running cases | T01–T04, T09 and T17; source inspection and exact receipt replay. |
| P1 | Admitted history templates, finite choices, shared symbols, linear solver adapter and historical bridges; conditional simulation and report explanations | T05–T10, T14 and T18; synthetic family preserved. |
| P2 | Matched debt text frontend with identical portable IR; connected corporate methods through the typed authoring API; document/XBRL import adapters | Parser/API equivalence; T11–T13; reviewed three-date public-company case. |
| P3 | Contract and valuation adapters, dated standard packages, richer financing and tax/funding policies | Contract/valuation differential cases; accounting review; T15 and transition cases. |
| P4 | Bank package, allowance models, regulatory reconciliation and management response alternatives | T16; banking-specific reviewed histories and stresses. |
| P5 | Insurance cohort package, reinsurance and actuarial measurement adapters | Reviewed cohort roll-forwards, information-date and service/finance tests. |
| P6 | Optional probabilistic inference, model-weight sensitivity and broader scenario research | Explicit targets, sampler/numerical validity, prior sensitivity and downstream evaluation. |

These stages specify dependency order, not dates or staffing estimates. Exit checks are cumulative: a case can recur when a later stage changes the inputs or execution path that it tests. Banking and insurance representation requirements influence P0 types, although their measurement engines arrive later. Probability weighting is optional; P1 already produces useful conditional cases and bounds. A failure of a chosen extractor or inference algorithm triggers a local replacement or repair, rather than abandoning the product's representation requirements. The parser/API comparison in P2 covers the matched debt fragment. Complete textual authoring of the corporate profile is a further frontend task; its absence does not remove corporate declarations from the typed API.

## 14.8 Decisions and replaceable assumptions

The requirements fixed by this design are preservation of evidence and alternatives, explicit time and basis, balanced and admissible posting, economic state beyond journals, and distinct set/probability semantics. Replaceable choices include the storage engines, parser generator, solver, numerical libraries, aggregation resolution and user interface. Every scientifically material aggregation, behavioural parameter, prior or truncation rule must record provenance, purpose,

failure mode and a diagnostic before it becomes a default.

The immediate work is to finish and review the governing proposal, including its real-source, authority, domain and research obligations. Product implementation follows completion of that work and explicit implementation authorization. In the table above, P0 establishes portable records, review/time selection, durable receipts and typed authoring; P1 builds admitted histories; P2 connects the corporate methods and matched textual frontend. These stages are a design sequence, not an already running product campaign.

## 14.9 Concrete conformance and corpus protocol

The fixture directory contains independently entered inputs and expected outputs for Northstar, rounded borrowing, evidence views and sector cases. `check_repair_conformance.py` recomputes exact journals and statements, constructs the joint linear bridges in Z3, checks strict bounds, and exercises replay, atomic rejection, finite Bayes updating and funding failure. `check_core_contract.py` checks grammar/schema examples, canonical identities, stable streams, information restrictions and source inventory. These scripts are reference oracles for portions of the design. They do not test an unimplemented full DSL compiler, all adapters or standards compliance.

Every conformance case must specify input files, the operation requested, expected values or typed errors, comparison convention and supported scope. Exact money, finite probabilities, linear witnesses and example journals compare as rationals. Approximate valuations require an error bound or a justified tolerance tied to the valuation scale. A missing expected result cannot become a passing test simply because the operation returned something. An implementation release must pass all applicable positive and negative fixtures and preserve unresolved operations; the arithmetic oracle alone does not release it.

The retained panel contains all 192 target annual/interim reports for 2018–2025 across nine issuers and three evidenced accounting families, with eight consecutive complete years per issuer. Apple, Caterpillar and JPMorgan Chase each contribute one annual and three quarterly reports per year; the other six issuers each contribute one annual and one half-year report. Thus $3 \times 8 \times 4 + 6 \times 8 \times 2 = 192$. This is source acquisition, not semantic closure or a representative random sample. The 244 retained original versions include historical alternatives to the active panel. Apple remains a detailed development package, with Caterpillar supplying a contrasting inventory policy; bank, insurer, IFRS and ASBE reports are also retained.

Publication dates remain unverified for 136 of the 192 accepted reports; 56 have filing-date metadata. A historical forecasting population needs the actual byte-availability evidence described in Section 3.12, not imputed publication dates. The 31 provisional annotations across four reports are exposed development material, not heldout answers. Policies, note links, incorporated documents, changed comparatives and adjudicated meanings still require task-specific preparation. Synthetic revisions provide controlled cases without substituting for a required real conflict. The corpus protocol and active panel preserve their distinct populations and versions.

Annotators first record literal/context readings without seeing the model's predictions, then mark source conflicts, mapping relationships, narrative qualifications and admissible alternatives. A second accounting-aware reviewer adjudicates material disagreements while preserving both initial annotations. Reports from the same issuer family and duplicated tables cannot be split between development and heldout evaluation. Forecast tests freeze the actual publication cutoff; later revisions are separately labelled outcomes and may not enter predictor inputs. Extractor upgrades rerun the same frozen corpus.

For the initial supported release, loss of a source span, wrong entity/unit/ period/basis, invented evidence, hidden financing, dropped feasible branch, incorrect cash reconciliation or unreported unsupported treatment blocks acceptance. All selected critical facts and operations must have exact reviewed answers or the expected unresolved response. These zero-tolerance conditions follow from the product's meaning-preservation requirements. Non-critical extraction quality is reported by stratum and error category; no arbitrary overall accuracy threshold certifies whole-report understanding. An extraction benchmark needs its own sample size and uncertainty protocol before making statistical accuracy claims.

The DSL and typed-library comparison uses identical packages, evidence, schemas and runtime. Users perform the same tasks: correct a context, add a shared uncertainty, represent a restatement, add a supported event and explain a forecast difference. Task order is counterbalanced and experience recorded. Both frontends receive the same validation and error messages. Measure successful completion, material semantic errors, time and maintenance effort with uncertainty appropriate to the participants; report no winner from the current syntax checks. A textual language earns its maintenance cost only if that comparison supports its use.

## 14.10 Workload limits and release evidence

The first integration workload is one local company package, up to 10,000 observations, 100 declared aggregate event slots per period, 64 materialised finite-choice cells and a 30-second budget per solver operation. These are provisional engineering limits for bounded failure behaviour, not measured capacity or scientific defaults. They must be configurable and recorded. Reaching a limit yields partial coverage with the unvisited formula/cells, never an exhaustive-history claim. Peak memory and latency are measured only after an implementation exists; a slower but semantically correct result does not become a scientific failure.

Engineering release needs compiler/API equivalence, transactional replay and adapter subset checks. Accounting release additionally needs paragraph-level authority, dated applicability, exact fixtures and independent domain review for every promoted operation. Forecast promotion additionally needs preserved information sets, the stated baseline ladder, statistical uncertainty and critical-regime vetoes. Human comprehension review remains separate from all three. This version supplies a richer design candidate and reference checks while preserving these outstanding acceptance decisions.

# Appendix A

# Portable operation reference

These proposed versioned operations share the request and result envelope in Section 13.4. Required inputs are explicit even when a list is empty. An expected head mentioned below refers to the common envelope, never a second argument. The registry is a lookup reference after the financial and transaction meanings have been explained in the body. These are specified exchanges, not deployed endpoints.

## A.1 source.prepare@1

package manifest, source snapshot, adapter/config lock, prior review, traversal budget and optional continuation → ExtractionPreparation

Acquires/preserves sources and constructs candidates under the source-preparation method; no semantic admission

## A.2 review.propose@1

target Ref, layer, decision kind, evidence Refs, actor, rationale, replacement Ref or null → ReviewProposal

Candidate construction; accepts no layer

## A.3 review.commit@1

ReviewProposal, expected review view/revision → immutable ReviewDecision and new view

Review transaction under the review and storage rules; admissible action/layer required

## A.4 time.select@1

package/view, investor/system/exploratory mode, cutoffs, revision and unknown-time policy → selected assertions, conflicts and temporal obligations

Pure selection under the time-admission rules; no invented ingestion or publication time

## A.5 model.compile@1

the versioned CompileRequest → candidate Module, source map, diagnostics and target admission/capability report

Candidate construction; no solver or posting

## A.6 history.construct@1

admitted Module, opening, scope, template registry, query/horizon, resource budget → HistoryFamily, partition, witnesses and unresolved formulas

History-construction contract; witnesses are examples within the family, not its definition

## A.7 history.refine@1

frozen family, additional admitted constraint or refinement, dependency view, budget → child family and inclusion relation

Candidate construction; record whether correction replaces constraints instead of intersecting them

## A.8 history.query@1

family, typed joint observable, bound/feasibility request, capability manifest → symbolic/partial/exact answer, coverage and endpoint attainability

Pure; unsupported expressions remain explicit

## A.9 uncertainty.evaluate@1

exact native uncertainty request, law/attachment/admission and operation-specific event or observation block → native UncertaintyResult and candidate derived bundle

Native admission/attachment/chaining rules; never selects its result automatically

## A.10 uncertainty.admit@1

candidate bundle, target time/evidence/purpose, actor, rationale, requested assumption scope, expected selection revision → admitted attachment and decision

Admission transaction; author-stipulated measures keep that label

## A.11 uncertainty.select@1

admitted attachment/receipt, rationale and expected selection revision → new selected-view revision

Selection transaction; no probability recomputation

## A.12 scenario.project@1

family, exogenous scenario, policy/method lock, horizon/calendar, explicit information policy, resource budget → conditional event/journal/state proposals by history and scenario

Candidate construction; budgets preserve unresolved regions; no historical-book commit

## A.13 accounting.assess@1

native assessment request, event, opening, context/time, authority/method, evidence/assumptions and admissions → TreatmentAssessment

Pure accounting assessment; inherited `assess_only` profile cannot post

## A.14 accounting.authorize@1

AccountingEventBasis and posting-profile Ref → PostingAuthorization

Pure evaluation under a separately supported posting profile; an assess-only result cannot grant posting permission

## A.15 accounting.propose@1

PostingAuthorization, AccountingEventBasis and posting/rounding method → JournalProposal and independent EconomicDelta

Candidate construction; revalidate bindings; unsupported method gives no journal

## A.16 accounting.commit@1

AccountingCommitArguments/v1 and envelope expected branch head/generation → StateTransitionReceipt

Single atomic transaction under the event-commit contract; the separate commit schema, separate from native assess-only

## A.17 report.build@1

frozen branch/family result, statement-map version, interval/view, typed requested statements and presentation policy → joint statement family, reconciliation and coverage

Pure; derive from retained state/journals, no balancing plug or hidden branch selection

## A.18 record.explain@1

output Ref, its frozen view, requested lineage depth → source/method/assumption/decision paths and unresolved dependencies

Pure; depth truncation is explicit; cannot substitute today's view

## A.19 work.resume@1

checkpoint Ref, original semantic manifest, scheduling budget → operation-specific resumed result

Resume the original operation; checkpoint ownership/atomicity under the durability contract; no model change hidden as resume

## A.20 store.export@1

snapshot/root Refs, format/decoder, destination → closed manifest or explicit incomplete export

Publish a verified export manifest atomically under Section 13.6; no domain mutation

## A.21 store.import@1

closed manifest, exact blobs/records, decoder and target namespace → import receipt and original-to-local mappings

Verify before metadata transaction; import alone grants no financial admission

## A.22 store.migrate@1

migration request under Section 13.6 → preserved originals, new representation, comparison/mapping and receipt

Transaction only after required semantic comparisons; no changed-model-as-format claim

## A.23 Closed accounting records

The following field lists are exact for the named versions. Every reference uses the portable kind/schema/digest type; plural references are schema-declared sets. Unknown fields reject. Source and admission sets can be empty only when the method's admitted assumption scope explicitly permits and explains that.

```
AccountingEventBasis/v1 {
  identity: {entity_id, book_id, branch_key, business_event_id},
  event_ref, opening_ref, context_ref, branch_ref, time_ref,
  chart_ref, authority_refs, method_ref, evidence_refs,
  admission_refs, assumptions_ref, view_ref
}
PostingAuthorization/v1 {
  basis_ref, posting_profile_ref, method_ref, authority_refs,
  status, findings, posting_permitted, evidence_level, view_ref
}
AccountingCommitArguments/v1 {
  basis_ref, authorization_ref, journal_ref, delta_ref
}
```

Identity fields are nonempty ASCII names. Each singular reference is nonnull. Authority references identify applicable support or a specifically typed synthetic authority for an explicitly

synthetic example. Authorization status is eligible, unresolved, ineligible or outside scope; posting permission is a Boolean and can be true only for eligible status under a supporting profile. A branch's immutable book mode is synthetic or real, and a profile names its supported modes. Synthetic evidence cannot authorize a real-book posting. Journal and delta both identify the same basis and authorization. Their domain payloads obey the package's pinned journal and state schemas. No earlier record refers forward to the final receipt.

## A.24 Method and extension definitions

A versioned financial method supplies the following nonempty definitions:

```
MethodDefinition {
  method_id, version, event_kind,
  state_schema, event_schema, journal_schema, delta_schema,
  output_schema, required_contexts, authority_requirements,
  preconditions, algorithm, account_roles, report_mappings,
  numerical_capability, examples, unsupported_cases
}
ExtensionPackage {
  package_key, version, record_schemas, operation_signatures,
  supported_contexts, state_requirements, authority_refs,
  method_refs, capability_requirements, migration_refs,
  example_refs
}
```

Each precondition has an identity, typed predicate, required input/admission references and separate false/unknown consequences. Domain state and event fields are specified with the financial methods in the body. Schema names and package members resolve through immutable locks. An empty authority list makes no accounting-authority claim. Registering a schema does not establish the method's entity, instrument, reporting-period or probabilistic applicability. The full machine-oriented contracts and finite exchanges accompany the proposal in the architecture and methods directories; their exact versions are pinned by the manuscript's dependency index.

# Appendix B

# Core grammar, records and execution details

## B.1 Preserved Core 0 teaching grammar

The grammar governing the preserved Core 0 teaching examples is reproduced from `spec/core-0.lark`. It does not enlarge the first matched text frontend beyond the portable profile described in the body. Names are case-sensitive ASCII identifiers; strings are UTF-8 with JSON escaping. All literal words in the grammar are reserved. The companion contract specifies valid dates, exact rational literals, argument arity, type/context checks and package resolution. A syntax parse is not semantic validation.

```
// Normative Core 0 syntax, Lark EBNF. UTF-8 input; ASCII identifiers.
// Longest token wins; keywords cannot be declaration identifiers.
start: "module" NAME "version" STRING ";" declaration*
?declaration: use | source | entity | book | account | observation | symbol
            | choice | binding | constraint | alternative | contract | event
            | rule | scenario | report | opening
use: "use" NAME "version" STRING ";"
source: "source" NAME "=" expr ";"
entity: "entity" NAME "functional" NAME ";"
book: "book" NAME "entity" NAME "basis" NAME "version" STRING ";"
account: "account" NAME ":" NAME "book" NAME ";"
observation: "observation" NAME ":" type time_spec "=" expr "from" NAME ";"
time_spec: "instant" DATE | "period" DATE "to" DATE
symbol: "symbol" NAME ":" type "in" domain "because" STRING ";"
domain: "[" expr "," expr "]" | "{" expr ("," expr)* "}"
choice: "choice" NAME "in" "{" NAME ("," NAME)* "}" "because" STRING ";"
binding: "let" NAME [":" type] "=" expr ";"
constraint: "constraint" NAME ":" expr ";"
alternative: "alternative" NAME "{" assumption* "}"
assumption: "assume" expr ";"
contract: "contract" NAME ":" NAME "book" NAME "{" field* "}"
field: NAME "=" expr ";"
event: "event" NAME "(" [parameters] ")" "book" NAME ";"
parameters: NAME ":" type ("," NAME ":" type)*
rule: "rule" NAME "for" NAME "as" NAME ["overrides" NAME ("," NAME)*] "when" expr "{" rule_statement* "}"
?rule_statement: binding | require | post
require: "require" expr ";"
post: "post" "{" posting+ "}"
posting: SIDE NAME expr ";"
scenario: "scenario" NAME "{" setting* "}"
setting: "set" NAME "=" expr ";"
report: "report" NAME "for" NAME "{" field* "}"
opening: "opening" NAME "for" NAME "from" expr ";"
type: NAME ["<" type ("," type)* ">"]
?expr: disjunction
?disjunction: conjunction (OR conjunction)*
?conjunction: comparison (AND comparison)*
```

```
?comparison: sum [COMPARE sum]
?sum: product (ADD product)*
?product: unary (MUL unary)*
?unary: UNARY unary | postfix
?postfix: atom ("." NAME)*
?atom: number | STRING | DATE | BOOL | call | NAME | "(" expr ")" | array | record
call: NAME "(" [expr ("," expr)*] ")"
number: NUMBER [NAME | "per" "year"]
array: "[" [expr ("," expr)*] "]"
record: "{" [record_field ("," record_field)*] "}"
record_field: NAME ":" expr
SIDE: "debit" | "credit"
OR: "or"
AND: "and"
UNARY: "-" | "not"
COMPARE: "==" | "!=" | "<=" | ">=" | "<" | ">"
ADD: "+" | "-"
MUL: "*" | "/"
BOOL.2: "true" | "false"
DATE: /@[0-9]{4}-[0-9]{2}-[0-9]{2}/
NAME: /[A-Za-z_][A-Za-z0-9_]*/
NUMBER: /(?:0|[1-9][0-9]*)(?:\.[0-9]+)?/
STRING: /"(?:[^"\\\x00-\x1f]|\\(?:["\\/bfnrt]|u[0-9a-fA-F]{4}))*"/
%import common.WS
%ignore WS
%ignore /\/\/[^\n]*/
```

## B.2 Required core record definitions

The historical closed machine-readable definitions remain in the original design directory at `spec/core-ir.schema.json`. They define contexts, source spans, observations, narratives, coverage units, mappings, partitions, entity/control records, books, accounts, symbols, choices, expressions, constraints, assumptions, alternatives, contracts, events, rules, packages, measurements, states, journals, scenarios, information views, review decisions, evidence views, model scopes, historical families, probability models, report mappings, result cells, branches, runs and diagnostics. For those historical records the schema fixes field spelling and structural shape, and its companion contract fixes reference resolution and operations. New portable records instead obey the exact versioned contracts introduced in Chapters 7, 8 and 13. No historical field list grants new posting authority.

For example, these selected records reproduce the schema's field names. Each field is required; a nullable link is represented by explicit null.

```
EvidenceView {
  id, cutoff, system_as_of, mode,
  documents[], observations[], decisions[],
  conflict_sets[][], transforms[]
}
Context {
  id, entity, perimeter, concept, basis, measurement,
  time, unit, currency, taxonomy_version, dimensions[]
}
Symbol {
  id, type, context, domain, uncertainty_kind,
  scope, dependencies[], admission, provenance[]
}
Journal {
  id, event, book, date, accrual_period,
  postings[], rule, measurements[], assumptions[],
  precondition_hash, explanation
}
ResultCell {
  id, context, history_condition, scenario, path, kind,
  value, expression, lower, upper,
  lower_attained, upper_attained, sample_ref,
  certificate, reason, dependencies[]
}
```

A value is a reduced rational, or a typed symbolic/domain reference. A posting is nonnegative with its side carrying the sign, in the book's functional currency. Original units and the valuation reference survive conversion. Monetary contexts require currency; a period flow requires a calendar and exclusive end boundary. Historical zero postings may preserve traceability but establish no economic event. New portable proposals omit zero lines and retain their explanation in the event and findings instead. The amount may have finer precision before the declared posting-rounding operation.

Schema validation rejects unknown fields but cannot prove that a treatment is permitted, a reference points to the correct kind of record, or two periods are comparable. Those semantic checks precede solver or runtime dispatch. The structural fixture and complete demonstration source/package are supplied separately so that a future implementation can check both layers explicitly. `fixtures/core-ir-demo.json` supplies an authored expected lowering; the reference checker evaluates its selected debt expressions. A general source-to-IR compiler and complete external-registry validation remain to be implemented. Domain records retain disclosed display text and rounding rules alongside their exact bounds.

## B.3 Typing and partial-information rules

For expressions $x$ and $y$ in the same currency $C$, the money addition rule is

$$\frac{x : \mathrm{Money}\langle C\rangle \quad y : \mathrm{Money}\langle C\rangle}{x + y : \mathrm{Money}\langle C\rangle}.$$

Context compatibility is a separate premise for observations. There is no analogous implicit rule for different currencies. Explicit conversion uses a dated measurement and records its rate and purpose. A rate-times-principal calculation is typed through a declared accrual convention; the example's annual simple rate and year fraction produce money in the principal's currency.

Known expressions evaluate normally. Shared symbols evaluate to symbolic expressions subject to a joint constraint store. Missing source values cannot enter arithmetic until the model supplies a symbol or a conditional assumption. Boolean guards over symbols produce formulas, not the value false. This avoids the common error in which "unknown whether impaired" silently means "not impaired." Division requires a proven nonzero denominator or a partitioned domain that excludes zero; otherwise the expression remains unsupported for concrete execution.

Logical queries first require a nonempty feasible family. An empty family returns `Infeasible`; it must not produce vacuous answers to substantive questions. For a nonempty family, constraint evaluation has three results. `entailed` means every state in the declared feasible family satisfies the proposition. `refuted` means none does. `undetermined` means both remain feasible or the solver has not decided; the response distinguishes those causes. Probabilistic truth is a separate query requiring a measure. A language-level Boolean value must never be inferred from an arbitrary probability threshold.

## B.4 Operational rule selection

Let $V(e, h)$ be the applicable rules for event $e$ in concrete history $h$, after package/date selection and guard evaluation. Let $r \prec r'$ mean that $r'$ validly overrides $r$ through the declared acyclic graph. Define

$$\mathrm{Max}(V) = \{r \in V : \nexists r' \in V \text{ with } r \prec r'\}.$$

For one required recognition operation, a singleton selects that rule; the empty set yields `UnsupportedTreatment`; a larger set yields `RuleConflict`. Multiple distinct operations, such as measurement and recognition, are evaluated through separate declared interfaces. A rule that intentionally produces no postings must belong to a supported non-recognition event operation, whose economic state effect is still recorded.

Given a selected rule, the evaluator computes all local bindings and preconditions, emits proposed postings, verifies Equation (6.2), validates account/context restrictions and commits the event transaction. Failure returns the original state plus a diagnostic. This defines the atomicity requirement independently of the storage technology.

## B.5 Example run request and response

The following request executes two named conditions under one scenario. It does not request an aggregate probability distribution.

```
{
  "model": "BorrowingExample@1",
  "evidence_view": "synthetic-2025-exact",
  "starting_state": "after-2025-close",
  "horizon": {"start": "2026-01-01", "end_exclusive": "2027-01-01"},
  "conditions": ["allFixed", "allFloating"],
  "scenarios": ["rateShock"],
  "mode": "conditional",
  "weight_model": null,
  "reports": ["balance_sheet", "income", "cash_flow", "equity", "journals"],
  "package_lock": "demonstration-packages-v1"
}
```

The response contains two separately labelled results with interest expense 5 and 7, cash 90 and 88, debt 100 in both, and equity $-10$ and $-12$. A request for the parent symbolic family returns the formula $7 - 2\alpha$ and, under exact linear bounds, $[5, 7]$ before monetary quantisation. The

API rejects an unconditional mean request without a weight model. A user can instead request an arbitrary clearly labelled representative case, which does not stand in for the feasible family.

## B.6 Requirement traceability

| Requirement | Primary implementation area | Conformance cases |
|---|---|---|
| R1 | Evidence registry and extraction adapters | T03, T09 |
| R2 | Context/version model and cutoff queries | T03, T04 |
| R3 | Types, mappings and package selection | T02, T12, T13 |
| R4 | Symbols, choices and constraint graph | T05–T10, T14 |
| R5 | Posting, closing and consolidation | T01, T11–T13 |
| R6 | Contract and behavioural interfaces | T05, T11, T15, T16 |
| R7 | Conditional output and probability types | T06, T07, T18 |
| R8 | Explanation graph and report dependencies | T09, T11–T13 |
| R9 | Capabilities and versioned modules | T02, T14–T16 |
| R10 | Run records, scheduler and replay | T03, T17, T18 |

# Appendix C

# Research basis and outstanding review

## C.1 Source collection and reading scope

The companion library contains 32 verified paper PDFs: 25 retrieved from external sources and seven preserved copies from the original collection. Filenames use the title, first author's surname and year. Publication years are used where an archived working version underlies a later publication; the catalogue identifies the actual version read. The original collection has not been renamed or modified. Standards pages, contract specifications, software source snapshots and metadata are stored separately from papers.

The survey considered accounting algebra and DSLs, corporate and bank simulation, incomplete and temporal databases, provenance, inverse problems, partial identification, model averaging, causal scenarios, contract languages, probabilistic programming, document extraction and financial reasoning. OpenAlex and Crossref supplied dated metadata, while primary publisher, institutional and author copies supplied technical support. Public-index results were checked against the downloaded title pages; HTTP success alone was not accepted as a valid paper download.

Technical reading was targeted to the mechanisms used in this design. The full PDF is archived even where only specified sections support the argument. No claim is made that every theorem, appendix or empirical result in every archived paper was independently replicated. Representative support anchors are listed below; the source-support ledger contains the per-source scope and qualifications.

| Source | Inspected technical support | Design use |
|---|---|---|
| Nehmer and Robinson | Sections 2–5: account states, transactions and aggregation | Conservation plus admissibility. |
| Doshi | Chapters 4, 7–9 of the 2024 thesis | Dates, logs, corrections and DSL scope. |
| Shahnazarian | Sections 2.3–2.8; relevant Chapter 3 equations; data and evaluation sections; Appendix B | Accounting/behaviour separation; limits of data transfer. |
| Pierson | Sections 3–4 and reporting structures; supplementary Vensim equations | Accrual/payment bridges and reporting accumulators. |

| Source | Inspected technical support | Design use |
|---|---|---|
| Libkin | Sections 2–5: incomplete states, conditions and certain answers | Shared unknowns and joint alternatives. |
| Green et al. | Definitions 3.1–3.2; Section 4 | Composable relational provenance; limits outside positive operations. |
| Stuart | Section 2.2, observation equation and prior/regularisation discussion | Inverse-problem formulation, not imported convergence guarantees. |
| Giacomini and Kitagawa | Sections 2.1–2.6 and Theorem 1 in CWP12/20 | Prior sensitivity and classes of posterior beliefs. |
| Hoeting et al. | Equations (1)–(3); prior and predictive-evaluation discussion | Optional model-weighted predictions. |
| Merigoux et al. | Sections 3–4, default calculus and translation figures | Scopes, exceptions, empty/conflict outcomes. |
| Peyton Jones et al. | Sections 3–5, combinators and valuation semantics | Contract composition separate from valuation assumptions. |
| FinQA / TAT-QA | Dataset operations, annotation, model and evaluation sections; FinQA evaluation source | Evidence-linked local arithmetic; benchmark boundaries. |
| DocFinQA / FinBen | Context/retrieval and financial-task evaluations | Long-document difficulty and task coverage. |
| PubTables / GTE | Canonicalisation, dataset construction, methods and evaluation | Table structure and financial-domain extraction. |
| Cosentino et al. | Sections 3.1–3.2, 4.6 and 5 | Persistent RAG provenance; scope of auditability evidence. |

## C.2 Version and access qualifications

Several discrepancies mattered. Doshi's thesis is dated 2024 despite the original filename. The archived Mernik text is the 2003 CWI report underlying the 2005 article. Giacomini and Kitagawa's copy is the April 2020 working paper underlying the 2021 article. The Docling v5 copy lists Auer as first author. The probabilistic-programming tutorial is an arXiv version revised in 2021, first submitted in 2018; a merged index DOI was not used. The Libkin author PDF has a footer DOI that resolves to another paper; title/author metadata from the indexes resolves the cited work to DOI `10.1145/2594538.2594561`. These discrepancies remain documented rather than being hidden by a cleaned bibliography.

McCarthy (1982), Imieliński and Lipski (1984), Tamer (2010), and Caiani and co-authors (2016) are retained as important blocked candidates. Access failures and rejected wrong-content downloads are recorded. The design does not use their uninspected technical claims. More specialised work on accounting measurement, process mining, multi-company equilibrium, actuarial inference and learned forecasting is recorded as expansion work for the corresponding implementation stage. The survey supports the core architecture without pretending that those later modules have a complete literature or rule base.

OpenAlex retraction flags and available Crossref correction/version relations were inspected

alongside local versions. No reviewed source was excluded for a discovered retraction. Missing correction metadata is marked unavailable; absence of a discovered notice does not certify that no notice exists. Citation counts are dated index metadata, and venue rankings were not available in this campaign. Neither is used as evidence of correctness.

## C.3 Companion records and status

The six research ledgers are `source-support.json`, `citation-venue.json`, `backward-snowball.json`, `forward-snowball.json`, `claim-support.json`, and `omissions.json`. They record technical anchors, metadata sources, considered leads, omitted or blocked candidates and the distinction between source claims and this document's derivations. Citation leads with checked seed references are distinguished from cross-survey additions and topic expansion leads. Forward searches retain the most-cited and most-recent results returned by the recorded bounded queries; they do not establish exhaustive citation coverage. The separate `tools.json` records software evidence. Raw retrievals and text extracts permit a later reviewer to revisit the decisions.

The research library includes Mack's reserving paper, with its conditional-mean/variance assumptions, proofs, aggregate-error dependence and numerical comparisons inspected. It also archives official AASB texts incorporating IAS 2, IAS 32 and IFRS 17, distinguishing their Australian additions and applicable editions. The corporate filings are evidence of reported amounts and policies, rather than substitutes for every underlying standard. The ASBE 1 reproduction has not been verified against an accessible official current text, and official FASB/ASC access remains incomplete. These limitations are recorded at the specific operations they affect.

Every one of the 179 unique candidates in the acquired forward-query files has an individual title/metadata relevance decision. This includes previously omitted discovery files associated with inaccessible seed papers. The generation script requires an explicit decision and stops on an unseen candidate. Screening is still distinct from technical reading: deferred provenance, temporal-query, bank-optimisation, extraction and actuarial follow-ups remain identified expansion risks. An omitted-paper reason does not make that paper technically reviewed or make the survey exhaustive.

This document specifies the product; it does not report a completed engine. The companion review records cover deterministic example arithmetic, bibliography/build checks, rendered-page inspection and author review. Human comprehension review, independent accounting review and implementation conformance remain separate future evidence. They are not replaced by a clean LaTeX build or by the quantity of literature collected.

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